



GUIDE:

Advanced Therapy Medicinal Products (ATMPs)

Validation Methods and Controls Throughout the Cell and Gene Therapy Product Lifecycle

Phase Appropriate

Risk-based Approach

Analytical

Cleaning

Manufacturing
Operations

Supply Chain





GUIDE:

Advanced Therapy Medicinal Products (ATMPs)

Validation Methods and Controls Throughout the Cell and Gene Therapy Product Lifecycle

Disclaimer:

This Guide provides guidance on validation methods and controls throughout the cell and gene therapy product lifecycle. This Guide is created and solely owned by ISPE. It is not a regulation, standard or regulatory guideline document. ISPE cannot ensure and does not warrant that a system managed in accordance with this Guide will be acceptable to regulatory authorities. Further, this Guide does not replace the need for hiring professional engineers, technicians, or consultants.

Limitation of Liability

In no event shall ISPE or any of its affiliates, or the officers, directors, employees, members, or agents of each of them, or the authors, be liable for any damages of any kind, including without limitation any special, incidental, indirect, or consequential damages, whether or not advised of the possibility of such damages, and on any theory of liability whatsoever, arising out of or in connection with the use of this information.

© 2025 ISPE. All rights reserved, including rights for text and data mining and training of artificial intelligence technologies or similar technologies.

No part of this document may be reproduced or copied in any form or by any means – graphic, electronic, or mechanical, including photocopying, taping, or information storage and retrieval systems – without written permission of ISPE.

All trademarks used are acknowledged.

ISBN 978-1-946964-96-0

For individual use only. © 2025 ISPE. All rights reserved.

Preface

Advanced Therapy Medicinal Products (ATMPs) represent a revolutionary frontier in medicine, offering life-changing treatments for conditions previously deemed incurable. However, the intricate nature of these therapies, from their unique raw materials to complex manufacturing and logistics, presents significant challenges and inherent risks. This *ISPE Guide: Advanced Therapy Medicinal Products (ATMPs) – Validation Methods and Controls Throughout the Cell and Gene Therapy Product Lifecycle* serves as a resource designed to help navigate these complexities effectively. It provides an in-depth analysis of validation activities across the entire ATMP lifecycle, including analytical methods, equipment, processes, cleaning processes, and supply chain logistics.

This Guide provides approaches to identify, assess, and mitigate risks unique to ATMPs, enabling more implementation of risk-based validation strategies. It provides essential insights into regulatory considerations that can help expedite commercialization and fosters innovation by promoting flexible, science-driven development approaches. Ultimately, this Guide aims to empower stakeholders to make informed decisions that not only support regulatory compliance but also accelerate patient access to these transformative therapies.

Acknowledgments

The Guide was produced by an international Task Team, under the leadership and direction of:

Biana Torres (Lead)	Encoded Therapeutics	USA
---------------------	----------------------	-----

The work was supported by the ISPE Advanced Therapy Medicinal Products (ATMPs) Community of Practice (CoP).

Core Team

The following individuals took lead roles in the preparation of this Guide:

Maria Amaya, PhD	Genentech, a Member of the Roche Group	USA
Jose Caraballo-Oramas	Kite, a Gilead Company	USA
Pinar Cicalese, PhD	Immatics	USA
Jean-François Duliere (Co-Lead)		France
Ryan Murray	Valsource Inc.	USA
Nessiem Samuel	GxP Impact Consulting LLC	USA
Peter Walters	CRB	USA

Other Contributors

The Core Team wishes to thank the following individuals for their contribution to the document.

Zhimei Du	BlueSphere Bio	USA
Judith Koliwer, PhD	Koerber Pharma Software Inc.	USA
Robert Lechich	Global OT/IT Quality Services	USA
John McGrath	WuXi Biologics	Ireland
Dylan Stolp		USA

Special Thanks

The Team expresses their particular thanks to Erich H. Bozenhardt (United Therapeutics, USA) for supporting this guide as a Mentor and reviewer.

The Team thanks ISPE for technical writing, editing, and project management support by Nina Wang, Heather Watson, and Jeanne Perez (ISPE Guidance Documents Technical Editors) and production support by Lynda Goldbach (ISPE Senior Publications Manager).

The Team would like to express their grateful thanks to the many individuals and companies from around the world who reviewed and provided comments during the preparation of this Guide; although they are too numerous to list here, their input is greatly appreciated.

Company affiliations are as of the final draft of the Guide.

Cover photo: www.istockphoto.com.



3001 N. Rocky Point Dr. E., Suite 200-242, Tampa, Florida 33607 USA
Tel: +1-813-960-2105, Fax: +1-813-264-2816

www.ISPE.org

For individual use only. © 2025 ISPE. All rights reserved.

Table of Contents

1	Introduction	7
1.1	Background.....	7
1.2	Purpose and Objectives.....	8
1.3	Scope.....	9
1.4	Guide Structure.....	10
2	Advanced Therapy Medicinal Products	11
2.1	Overview of ATMPs.....	11
2.2	Challenges and Strategies in Developing ATMPs.....	12
2.3	Phase-Appropriate Considerations	12
2.4	Accelerated Timelines.....	13
3	Regulatory Framework for ATMPs.....	15
3.1	Introduction	15
3.2	Regulations Concerning ATMP Validation.....	15
3.3	Clarification for ATMP GMP in European Union.....	17
3.4	Inspection Documents	17
3.5	Key Regulatory Considerations for Validation.....	18
3.6	Risk Management.....	21
4	Quality by Design as a Risk Management Tool in ATMPs.....	23
4.1	Key Components in QbD	24
4.2	Lifecycle Management.....	31
5	Key Validation Differences and Challenges in ATMPs	33
5.1	Analytical.....	33
5.2	Process	34
5.3	Cleaning.....	35
5.4	Supply Chain.....	37
6	Analytical.....	39
6.1	Overview	39
6.2	Cell Therapy Testing	40
6.3	Gene Therapy Testing.....	43
6.4	Potency Testing for ATMPs	46
6.5	Rapid Microbial Testing for ATMPs	47
6.6	Analytical Method Validation for ATMPs	48
7	Manufacturing Operations	55
7.1	Introduction	55
7.2	Operations	55
7.3	Equipment.....	58
7.4	Process	65

8	Cleaning	73
8.1	Introduction	73
8.2	Cleaning Validation Program	73
8.3	Early-Stage ATMP Cleaning	73
8.4	Equipment and Facility Cleaning and Sanitization	74
8.5	Equipment and Process Platforms	74
8.6	Single-Use Equipment and Components	76
9	Supply Chain	77
9.1	Introduction	77
9.2	Cell Donor Collection	77
9.3	Other Raw Materials	77
9.4	Cryopreservation	79
9.5	Transportation, Logistics, and Shipping	80
9.6	Patient Administration	81
9.7	General Requirements	82
10	Phase Appropriate	85
10.1	Introduction	85
10.2	Phase-Appropriate Validation	85
11	Appendix 1 – References	87
12	Appendix 2 – Glossary	95
12.1	Acronyms and Abbreviations	95

1 Introduction

Advanced Therapy Medicinal Products (ATMPs) encompass a wide variety of innovative treatments, including gene therapy medicines, somatic cell therapies, and tissue-engineered products. These groundbreaking therapies hold significant promise as potential alternatives for patients with conditions that current treatments cannot cure. However, due to their intricate nature, ATMPs present both known and unforeseen risks unique to this category of products. Known risks include challenges with raw material variability, immunogenicity, vector shedding, and the complexities of manufacturing processes and cold chain logistics. Unforeseen risks may emerge from long-term safety profiles, unexpected host interactions, or the real-world performance of these highly individualized treatments.

Similar to traditional products, ATMPs undergo rigorously controlled preclinical and clinical studies. It is mandatory for Marketing Authorization Holders (MAHs) to implement a robust risk management system that identifies, characterizes, and mitigates these risks, in alignment with established principles of Quality Risk Management (QRM) such as those outlined in ICH Q9(R1) [1] which has been adopted by major health authorities including the US FDA [2] and the EMA [3]. Developing a comprehensive risk management plan is crucial during the development and manufacturing of ATMPs, given their complexity, technical specificity, and novel mechanisms of action. A proactive, risk-based approach is particularly vital for managing manufacturing changes and ensuring comparability for these products (as emphasized in guidance such as the FDA Draft Guidance for Industry – Manufacturing Changes and Comparability for Human Cellular and Gene Therapy Products [4]) to maintain consistent quality and safety throughout the product lifecycle. By identifying quality, non-clinical, clinical, and regulatory risks early through a risk-based approach, the approval process for ATMPs can be expedited, thereby increasing patient access to these innovative therapies.

This *ISPE Guide: Advanced Therapy Medicinal Products – Validation Methods and Controls Throughout the Cell and Gene Therapy Product Lifecycle* offers an in-depth analysis of validation activities specifically for ATMPs. This includes the validation of equipment, processes, analytical methods, cleaning procedures, and transportation logistics. Additionally, this Guide outlines regulatory considerations to help streamline the path to commercialization and provides essential insights into the lifecycle management of ATMPs.

1.1 Background

The development of ATMPs requires scientific innovation and manufacturing strategies that involve regulatory risks, such as the introduction of novel starting materials (human cells) and often small-scale manufacturing processes to supply a single batch for one patient. Due to these risks, it is important to employ a comprehensive approach for risk identification, evaluation, and mitigation. This comprehensive approach ensures that risks unique to ATMPs are addressed, mitigated, and minimized to safeguard patient safety.

The regulatory framework for ATMPs has been developed based on the risks associated with ATMPs and the establishment of mitigation measures. To create more flexibility concerning ATMPs while keeping the focus on product quality and patient safety, health authorities such as EMA [3] and organizations such as PIC/S [5] recommend the use of a risk-based approach during development. The purpose of this approach is to create a framework that encourages consideration of continually evolving science and technology in order to design a suitable ATMP development program. The EMA acknowledges that ATMPs are complex, at the forefront of innovation, and subject to advances in technology. For these reasons, a risk-based approach guided by QRM principles, such as those outlined in ICH Q9(R1) [1] and adopted by health authorities such as the FDA, is highly encouraged. This approach provides manufacturers with the flexibility necessary to adapt the best controls to the process while ensuring all requirements and critical aspects of the process are met. This is especially true for Investigational Medicinal Product (IMP) phases. Applying risk-based approaches can support companies to be more efficient in overcoming regulatory hurdles.

The objective is to enable a combination of guideline-based ATMP development and science-driven development strategies. Science-based, case-based programs have been defined considering different factors such as:

- The disease, targeted cells, vehicle, and route of administration
- The mechanism of action of the treatment and the potential side effects
- The level of risk acceptability that corresponds to the perceived degree of clinical benefit in a specific context

Critical risk factors that must be considered in the risk management plan for ATMPs include the Critical Quality Attributes (CQAs), Critical Process Parameters (CPPs), the manufacturing process, and the analytical techniques for characterization, release, and stability studies of these products. Although identifying risks is crucial for the development of ATMPs, equally important is the identification of control measures to address, mitigate, and minimize these risks. By employing an adaptive approach, manufacturers can better handle the specific risks associated with ATMPs and implement justifiable controls to ensure patient safety and product efficacy. Section 2 of EudraLex Volume 4 Part IV further elaborates on the risk-based approach for ATMPs, emphasizing the need for adaptable measures due to their innovative and complex nature [6].

The regulatory review process for investigational ATMPs requires a meticulous risk-benefit analysis tailored to the specific clinical indication under study. Establishing a Target Product Profile (TPP) and identifying the CQAs of the gene or cell therapy product under development are essential steps for conducting a thorough risk analysis. These elements not only guide the regulatory framework but also serve as invaluable tools for strategic development.

Adopting the risk-based approach is recommended for process, product, and equipment validation, allowing stakeholders to focus on assessing and mitigating the most significant risks associated with complex advanced therapies. In Process Validation (PV), this means identifying CPPs and CQAs that most impact product quality and patient safety and concentrating validation efforts—such as enhanced studies and controls—on these key areas. For product validation, including characterization and testing, the approach prioritizes tests and methods based on their ability to detect critical attributes and variations, rather than exhaustively testing every parameter. Equipment validation efforts are similarly focused on the most critical equipment and automated systems that directly affect product quality or data integrity, ensuring consistent and intended performance. By systematically identifying potential hazards throughout the product lifecycle, stakeholders can make informed decisions grounded in evolving scientific knowledge and regulatory expectations. Ultimately, risk-based validation enhances patient safety, supports robust product quality, and accelerates the delivery of safe and effective ATMPs.

1.2 Purpose and Objectives

This Guide provides a comprehensive framework for identifying, assessing, and mitigating risks throughout the lifecycle of ATMPs. It describes risk-based validation principles for development and manufacturing processes to ensure the safety, efficacy, and quality of ATMPs. The ultimate goal is to establish an effective control strategy that helps stakeholders navigate the complexities of ATMP validation, thereby facilitating regulatory compliance and fostering innovation in the field.

1.3 Scope

This Guide focuses on risk-based approaches for the validation of Cell-Based Medicinal Products (CBMPs) and gene therapy products. Tissue engineering products are out of scope. For the purposes of this Guide, the nomenclature ATMP is used to refer to cell and gene therapy products.

This Guide focuses primarily on the development of risk-based approaches for manufacturing operations (equipment and process), analytical, cleaning, and supply chain validation during the development and commercialization of ATMPs. It highlights the importance of applying Quality by Design (QbD) principles in the development of ATMPs and explains how the QbD framework has many implications to improve process validation, method validation, and product transfer.

Computerized systems software development and testing, as well as commissioning and qualification of facilities and utilities are out of scope.

Artificial Intelligence (AI) and manufacturing innovation, such as robotics, are important aspects to validation if they are used. As these are rapidly evolving during the writing of this Guide, these topics are out of scope. Refer to current guidelines and regulations on AI or other technologies that will need to be addressed as validation activities are conducted.

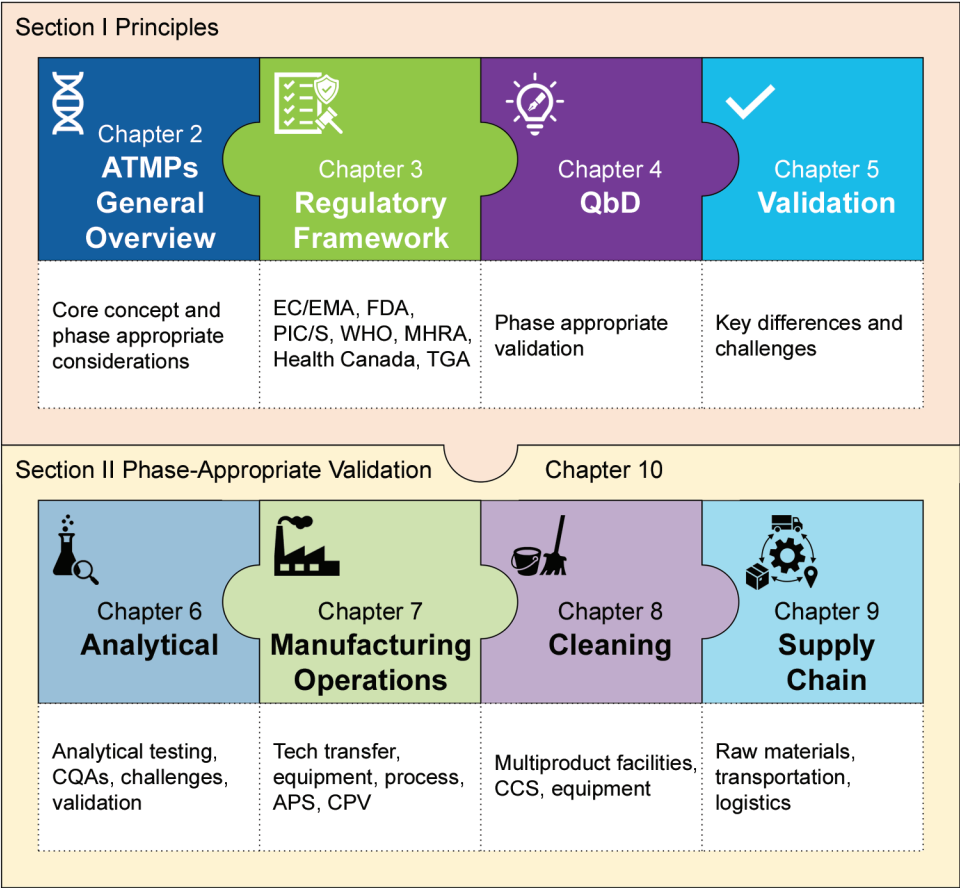
This Guide focuses on the unique aspects and validation for cell and gene therapies. Other validation considerations are covered within:

- *ISPE Guide: Advanced Therapy Medicinal Products – Autologous Cell Therapy* [7]
- *ISPE Guide: Advanced Therapy Medicinal Products – Allogeneic Cell Therapy* [8]
- *ISPE Good Practice Guide: Practical Application of the Lifecycle Approach to Process Validation* [9]
- *ISPE Guide: Cleaning Validation Lifecycle – Applications, Methods, and Controls* [10]
- *ISPE GAMP® 5: A Risk-Based Approach to Compliant GxP Computerized Systems (Second Edition)* [11]
- *ISPE Baseline® Pharmaceutical Engineering Guide, Volume 5 – Commissioning and Qualification (Second Edition)* [12]
- *ISPE Baseline® Guide: Volume 6 – Biopharmaceutical Manufacturing Facilities (Third Edition)* [13]
- *ISPE Good Practice Guide: Packaging, Labeling, and Warehousing Facilities* [14]
- *ISPE Good Practice Guide: Single-Use Technology* [15]
- *ISPE Good Practice Guide: Validation 4.0* [16]

1.4 Guide Structure

Figure 1.1 illustrates the structure of this Guide, which comprises two sections. Section I outlines the main principles underpinning the Guide, including a general overview of ATMPs, the regulatory framework, QbD, and key differences and challenges in validation approaches. Section II goes into the details of validation for analytical methods, manufacturing operations, cleaning, and the supply chain. Chapter 10 provides a summary of recommendations for implementation of the phase-appropriate validation approach throughout the product life cycle.

Figure 1.1: Structure of this Guide



2 Advanced Therapy Medicinal Products

2.1 Overview of ATMPs

ATMPs are based on gene supplementing and/or modifying therapies, cells, or tissues delivered to patients to provide a therapeutic benefit based on a specific target of interest. For ATMPs, the therapy is cells, engineered tissues, or cellular genetic components that modify, supplement, or manipulate the patient's genes. This is in contrast with traditional manufacturing processes for compounds that are synthetically derived (i.e., small molecule) or proteins or peptides expressed by cellular systems (i.e., large molecule biopharmaceuticals).

- **Cell-based advanced therapy products:** Living cells with or without *ex vivo* manipulations, sometimes using gene therapy methods that are introduced into the patient and act at the cellular level to treat disease or injury.
- **Gene therapy:** Rapidly growing field of medicine in which genes are introduced into the body to treat or prevent diseases. Gene therapy active substance contains recombinant nucleic acid that will regulate, repair, replace, add, or delete a genetic sequence in the patient. Although a variety of methods can be used to deliver the genetic materials into the target cells and tissues, modified viral vectors represent one of the more common delivery routes because of their transduction efficiency for therapeutic genes.
- **Tissue-engineered medicines:** “Contain cells or tissues that have been modified so they can be used to repair, regenerate or replace human tissue.” [3] Due to the further complexity of their nature, tissue-engineered therapies are out of scope for this Guide.

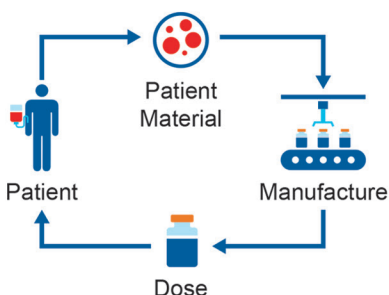
ATMPs encompass a wide range of product types including cell therapies, gene therapies, tissue engineering, and combination therapies. The complexity and variety of these products pose a significant challenge when establishing validation strategies for ATMPs.

Cell-based ATMPs can be classified into two main categories: autologous or allogeneic therapies. Figure 2.1 illustrates the end-to-end process, including manufacturing, for cell-based therapies.

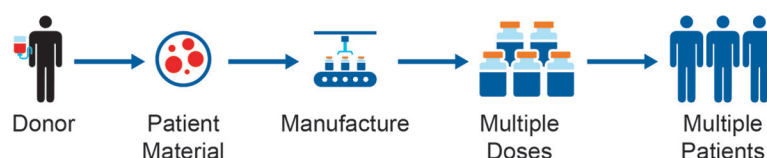
- **Autologous therapies:** Starting materials are derived from a patient and returned to the same patient after *in vitro* manipulation.
- **Allogeneic therapies:** Samples are derived from healthy donors, and the final therapy can be used to treat many different patients.

Figure 2.1: High-Level Illustration of Cell-based Autologous and Allogeneic Therapies

Autologous Therapies



Allogeneic Therapies



Additional information on the manufacturing facility and design for autologous cell therapies or allogeneic therapies for parenteral use is in the respective guides: *ISPE Guide: Advanced Therapy Medicinal Products – Autologous Cell Therapy* [7] and *ISPE Guide: Advanced Therapy Medicinal Products – Allogeneic Cell Therapy* [8].

2.2 Challenges and Strategies in Developing ATMPs

CBMPs present unique challenges due to the inherent high variability of their cellular starting material. This variability significantly impacts both the consistency of the manufacturing process and the quality of the final product. Additionally, the variability in manufacturing processes and raw materials used during clinical trials exacerbates these challenges. This is particularly true for autologous cell therapies, where each product is derived from an individual patient. As a result, broad final product specifications often emerge, increasing the risk of inconsistent efficacy and safety outcomes. Addressing these challenges requires establishing a clear link between CQAs and clinical performance to ensure product reliability and therapeutic success.

For autologous and allogeneic CBMPs alike, robust control of manufacturing processes and raw material quality is essential to mitigate variability. A systematic approach that defines CQAs and their relationship to therapeutic outcomes is critical. Leveraging advanced analytical tools and real-time process controls enables monitoring and making adjustments during production, thereby ensuring product quality. By integrating these quality considerations into the development lifecycle, developers can meet rigorous safety and efficacy standards while delivering consistent therapeutic benefits to patients.

Gene therapy products, such as viral vectors and plasmids, face equally complex manufacturing challenges. Although they share similarities with traditional biological medicinal products, their production involves specialized technologies requiring precise control. Key challenges include scaling production while maintaining product integrity, optimizing yields, and ensuring stringent purity and contamination controls, particularly for plasmid DNA, which is a critical starting material for many gene therapies.

To navigate these complexities, developers must implement robust control strategies, including defining CQAs, employing advanced analytical methods, and adopting innovative technologies for process monitoring and optimization. Regulatory requirements for gene therapies also continue to evolve, emphasizing the need for quality, safety, and efficacy. By applying QbD principles and innovative manufacturing approaches, developers can meet regulatory standards and ensure the reliable delivery of high-quality gene therapy products to patients.

2.3 Phase-Appropriate Considerations

A standard clinical trial program is often linear with progression through the phases. A Phase I clinical trial focuses on the safety of the investigational drug. In standard clinical trials, the participants in the trial are more likely to be healthy volunteers (typically around 20 to 80 participants). Once the Phase I trial indicates an acceptable safety profile, a Phase II clinical trial is started, which involves additional testing of a drug's safety and begins to test the drug's efficacy.

At this Phase II trial, various dosages are introduced and patients with the disease participate in the trial (typically involving a few hundred participants). Once the Phase II trial has confirmed the efficacy and safety profile, a Phase III clinical trial is started. A Phase III trial demonstrates whether or not a drug offers a treatment benefit to a specific population (typically involving 300 to 3,000 participants). Phase III trials compare the drug to the current treatment and ensures the safety profile for a longer duration. [6]

Clinical trial design approaches for ATMPs, however, are usually different from standard clinical trials for biologics, small molecules, etc. Currently, ATMPs tend to target disease indications that treat serious or life-threatening conditions or address unmet medical needs, and sometimes for orphan designation. For ATMPs, it is common to run a combined Phase I/Phase II clinical trial to establish the drug's safety and efficacy, and to not run trials with healthy patients.

Phase III trials continue the assessment of the safety and efficacy of the drug on a larger number of patients; however, the total participants are fewer than in a standard clinical trial design. It may also be common to see companies execute a Phase I trial, followed by a Phase II pivotal trial. The industry recognizes that the standard classical approaches to trial progression and execution may not always be appropriate for ATMPs, and with changing approaches, companies must take extra care to ensure that their validation program is commensurate.

Companies should consider the clinical trial design and strategy for their drugs when determining the validation efforts and requirements. Companies are encouraged to proactively reach out to regulatory agencies to ensure appropriate Chemistry, Manufacturing, and Control (CMC) strategies are aligned with their clinical goals.

2.4 Accelerated Timelines

Due to the high potential to offer transformative benefits for patients with a variety of diseases and conditions, ATMPs are often anticipated for FDA's Fast-Track, Breakthrough, or Regenerative Medicine Advanced Therapy (RMAT) designation or for the EMA's PRIME (PRiority MEDicines) designation [17–19]. Efforts to expedite the development and approval of ATMPs have led to a drive for accelerated timelines. This acceleration inevitably affects the validation process for these therapies. Consequently, data generated during the early stages of development can be leveraged to directly bolster the later, pivotal phases, streamlining the overall validation process and potentially expediting market access.

One potential impact of accelerated timelines is that there may be less time available for sequential validation of ATMP products and processes. Companies that receive an expedited drug development designation will need to pursue a more rapid validation program to accommodate the accelerated pace of the clinical program. Simultaneous development of validation strategies may be necessary. For example, the method validation may occur concurrently. A thorough plan that consists of risk assessments and scientific rationale is critical.

In summary, accelerated timelines in ATMP development can impact the validation process. It is important to maintain a balance between speed and rigor to ensure the safety and efficacy of these therapies.

3 Regulatory Framework for ATMPs

3.1 Introduction

Several national competent authorities around the globe include regulations for ATMPs. Many of those regulations are specific to ATMPs, whereas others inherently encompass ATMPs, and each country has specific regulations for requirements.

The objective of this chapter is to highlight some of the key regulations and guides pertinent to ATMP validation activities and risk management. This list is not fully exhaustive and is focused on the validation elements for analytics, process, manufacturing, cleaning, and supply chain. Because ATMPs are newer technologies, regulators are working with the goal of harmonizing regulations across the world and continue to release ATMP-specific regulations. The most recently released regulations (at the time of this Guide's publication) that are crucial for validation purposes include:

- PIC/S Annex 2A – Manufacture of Advanced Therapy Medicinal Products for Human Use [5]
- PIC/S Annex 1 – Manufacture of Sterile Medicinal Products [20]
- ICH Q9(R1) Quality Risk Management [1]

3.2 Regulations Concerning ATMP Validation

Table 3.1 outlines regulations that address ATMP validation. Some of these guidance documents apply directly to ATMPs and should be used in conjunction with other guidance documents that apply to Drug Products (DPs).

Table 3.1: Key Regulations and Guidance for ATMP Validation

Regulatory Authority or Organization	Regulation/ Guidance	Title	Year	Application Scope
PIC/S [21]	Annex 2A	Manufacture of Advanced Therapy Medicinal Products for Human Use	2023	ATMP
	Part 1	Guide to Good Manufacturing Practice for Medicinal Products	2023	General – GMP
	Annex 1	Manufacture of Sterile Medicinal Products	2023	General – sterile products
	Annex 15	Qualification and Validation	2023	General – qualification and validation
	PE 010-4	Good Practices for the Preparation of Medicinal Products in Healthcare Establishment	2014	General – for healthcare establishments (direct supply to patients)
	PE 011-1	Good Distribution Practice for Medicinal Products	2014	General – supply chain
EMA [18]	EudraLex Volume 4 Part IV	Guidelines on Good Manufacturing Practice Specific to Advanced Therapy Medicinal Products	2017	ATMP

Table 3.1: Key Regulations and Guidance for ATMP Validation (continued)

Regulatory Authority or Organization	Regulation/ Guidance	Title	Year	Application Scope
EMA [18] (continued)	Guideline	Guideline on Quality, Non-Clinical and Clinical Requirements for Investigational Advanced Therapy Medicinal Products in Clinical Trials	2025	ATMP
	Annex 1	Manufacture of Sterile Medicinal Products	2023	General – sterile products
European Commission (EC) [22]	2013/C 343/01	Guidelines on Good Distribution Practice of Medicinal Products for Human Use	2013	General – supply chain
FDA [2]	21 CFR Part 1271	Human Cells, Tissues, and Cellular and Tissue-Based Products	2023	ATMP
	21 CFR Part 210	Current Good Manufacturing Practice in Manufacturing, Processing, Packing, or Holding of Drugs; General	2017	General – GMP
	21 CFR Part 211	Current Good Manufacturing Practice for Finished Pharmaceuticals	2017	General – GMP
	Guidance for Industry	Process Validation: General Principles and Practices	2011	General – process validation
	Guidance for Industry	Analytical Procedures and Methods Validation for Drugs and Biologics	2015	General – analytical validation
	Guidance for Industry	Chemistry, Manufacturing, and Control (CMC) Information for Human Gene Therapy Investigational New Drug Applications (INDs)	2020	ATMP – IND requirements
	Draft Guidance for Industry	Manufacturing Changes and Comparability for Human Cellular and Gene Therapy Products	2023	ATMP – comparability
	Draft Guidance for Industry	Potency Assurance for Cellular and Gene Therapy Products	2023	ATMP – analytical
ICH [23]	Q9(R1)	Quality Risk Management	2023	General – QRM
	Q2(R2)	Validation of Analytical Procedures	2023	General – analytical
	Q14	Analytical Procedure Development	2023	General – analytical
WHO [24]	TRS 1019 – Annex 3	Good Manufacturing Practices: Guidelines on Validation	2019	General – validation
Health Canada [25]	GUI-0029	Guide to Validation – Drugs and Supporting Activities	2021	General – validation
	GUI-0028	Cleaning Validation Guide	2021	General – cleaning validation
TGA [26]	Guidance	Process Validation for Listed and Complementary Medicines (Technical Guidance on the Interpretation of the PIC/S Guide to GMP)	2019	General – process validation

Table 3.1: Key Regulations and Guidance for ATMP Validation (continued)

Regulatory Authority or Organization	Regulation/ Guidance	Title	Year	Application Scope
ISO [27]	21973	General Requirements for Transportation of Cells for Therapeutic Use	2020	Supply chain

3.3 Clarification for ATMP GMP in European Union

In Europe, there are discernible regulatory gaps, particularly highlighted by the development of PIC/S Annex 2A for ATMPs [5] and Annex 2B for biological products [28]. Notably, Annex 2A is closely linked to various Good Manufacturing Practice (GMP) documents and annexes, implying that for PIC/S, both Annex 1 (Manufacture of Sterile Medicinal Products) [20] and Annex 15 (Qualification and Validation) [29] apply comprehensively to ATMPs.

The European Commission/EMA [3] follows a distinct process wherein Annex 2 [103] pertains to biological products and ATMPs fall under EudraLex Volume 4 Part IV [6]. Unlike PIC/S Annex 2A, EudraLex Volume 4 Part IV stands alone and is not linked to other GMP parts and annexes. While PIC/S Annex 1 [20] focuses on sterilization, other aspects are addressed within EudraLex Volume 4 Part IV. The EMA has bridged EudraLex Volume 4 Part IV with the tables provided in the initial section of PIC/S Annex 2A through a Q&A document [30]. Notably, Annex 15 (Qualification and Validation) [29] does not apply to EudraLex Volume 4 Part IV, as Part IV contains its own chapter on qualification and validation tailored specifically for ATMPs.

3.4 Inspection Documents

Regulatory agencies ensure products are safe by conducting inspections. Understanding what regulators focus on can guide the implementation of risk-based validation strategies. Table 3.2 lists PIC/S [21] documentation related to ATMP validation activities for regulators inspecting facilities. PIC/S was selected as a good reference; however, other regulatory agencies may have similar guides for inspectors. These documents are helpful tools for the industry to use to clarify regulatory expectations on specific points.

Table 3.2: PIC/S Documents for Inspectors [21]

PIC/S Reference	Title
PI 006-3	Recommendation: Validation Master Plan, Installation and Operational Qualification, Non-Sterile Process Validation, Cleaning Validation
PI 007-6	Recommendation: Validation of Aseptic Processes
PI 023-2	Aide-Memoire: Inspection of Pharmaceutical Quality Control Laboratories
PI 024-3	Aide-Memoire: Inspection of Biotechnology Manufactures
PI 038-2	Aide-Memoire: Assessment of Quality Risk Management Implementation
PI 043-1	Aide-Memoire: Cross-Contamination in Shared Facilities
PI 044-1	Aide-Memoire: Inspection of Good Distribution Practice (GDP) for Medicinal Products in the Supply Chain
PI 054-1	Recommendation: How to Evaluate and Demonstrate the Effectiveness of a Pharmaceutical Quality System in relation to Risk based Change Management

3.5 Key Regulatory Considerations for Validation

This section highlights several key elements from various regulatory guidelines that are important for the phase-appropriate validation elements discussed within this Guide. This is not intended to be all encompassing, and it is recommended to review all relevant guidance and regulations depending on the product, intended geography, and appropriate phase of development or commercialization.

Although not in scope for this Guide, it is important to ensure that the facilities, equipment, and computer systems used during the validation activities described next are qualified and validated following the appropriate regulations and guidance.

3.5.1 Analytical Methods

The validation of analytical methods is crucial for ATMPs as reliable assays ensure the safety, identity, strength, purity, and quality of the product. Regulators advocate for the application of QRM in both the development and maintenance of these test methods. Validating analytical methods confirms that they are appropriate and effective for their intended use, providing confidence in the integrity of the ATMPs.

The use of QRM in the development of analytical methods is encouraged. ICH Q14 Analytical Procedure Development states: *“Risk assessment tools as described in ICH Q9 Annex 1 can be used to:*

- *identify analytical procedure parameters (factors and operational steps) with potential impact on its performance ...;*
- *assess the potential impact of analytical procedure parameters on the analytical procedure performance;*
- *identify and prioritize analytical procedure parameters to be investigated experimentally;*
- *inform the need and the extent of ongoing monitoring as part of risk review.”* [31]

A phase-appropriate approach to analytical method validation for ATMPs is recognized and encouraged by regulatory agencies. The FDA states in its guidance for gene therapy products that the *“validation of analytical procedures is usually not required for original IND submissions for Phase 1 studies; however, you should demonstrate that test methods are appropriately controlled.”* The FDA emphasizes that *“to ensure safety of gene therapy products, you should also qualify the assays used to determine dose (e.g., vector genome titer by qPCR, transducing units, plaque-forming units, transduced cells) prior to initiating clinical studies.”* [32]

The EMA states that for investigational ATMPs, *“during clinical development a gradual approach can be applied: First in-[human] and exploratory clinical trials: Sterility and microbial assays should be validated.”* [6] Safety assays should be validated, with increasing validation to pivotal studies (e.g., potency, batch release, and stability) and all methods validated for commercialization. [33]

The scientific principles described in the ICH Q2(R1) guidance [34] should be applied in a phase-appropriate manner to analytical procedures used during clinical development.

3.5.2 Process Validation

Given the diversity and unique characteristics of ATMPs, PV can pose a challenge. The PV principles described in EudraLex Volume 4 Part IV [6] aligns closely with some of the key elements of PIC/S Annex 15 [29], including clarification on the use of raw materials or starting materials for which some finished products are variable. Due to the limited availability of raw materials, concurrent validation can be used where needed and justified. *“The use of surrogate material may be acceptable when there is shortage of the starting materials (e.g., autologous ATMPs, allogeneic in a matched-donor scenario, allogeneic where there is no expansion of cells to [Master Cell Bank]).”* [6]

ATMP manufacturing knowledge and considerations are enhanced during progression from exploratory studies to clinical studies (Phase I–Phase III) to commercial manufacturing. *“Process validation studies are generally or typically not required for early stage manufacturing, and thus, most original IND submissions will not include process performance qualification. We recommend that you use early stage manufacturing experience to evaluate the need for process improvements and to support process validation studies in the future.”* [32]

EudraLex Volume 4 Part IV states *“the manufacturing process for investigational ATMPs is not expected to be validated but appropriate monitoring and control measures should be implemented to ensure compliance with the requirements in the clinical trial authorisation.”* [6] Additionally, PIC/S Annex 2A states *“ATMPs manufactured for exploratory, early phase clinical trials (phase I and phase I/II), are expected to be validated proportionately with the knowledge and the risk associated with the respective phase.”* [5] In order to appropriately implement phase-appropriate PV, it is important to perform risk assessments on the process.

The majority of ATMPs are manufactured using aseptic techniques and are not terminally sterilized. For all aseptic processes, Aseptic Process Simulations (APS) should be performed as part of the initial validation and repeated thereafter every six months in alignment with EudraLex Volume 4 Annex 1 [35] and Part IV [6]. All aseptic and sterilization processes, as well as virus inactivation or removal, for investigational and authorized ATMPs are expected to be validated. APS are expected to be performed for clinical Phase I programs and throughout to the commercial product as long as aseptic processing is used for manufacturing. This is a widely accepted regulatory expectation.

EudraLex Volume 4 Part IV also states *“where the same manufacturing platform is used for a number of closely related products (e.g., genetically modified cells where viral vectors are manufactured according to the same manufacturing process), the extent of validation work for each new product should be based on a justified and documented risk assessment of the process. This should take into account the extent of process knowledge, including existing relevant process validation work, for each significant step in the process. Thus, in so far as the other manufacturing steps remain the same, it may be possible to limit the validation to only the steps that are new to the process.”* [6]

The FDA recognizes the importance of leveraging platform programs to expedite bringing transformative ATMP therapies to market [36]. PV strategies should consider the organization of the facility and if the ATMP is manufactured in a multiproduct or multi-patient facility. Risk management and Contamination Control Strategies (CCS) should be assessed and implemented to mitigate the potential of contamination or cross-contamination [35]. The PV studies should be conducted in accordance with defined procedures. The results and conclusion should be recorded and include the following:

- ATMP manufactured for clinical trials or commercial.
- For aseptic processes, APS should be part of the initial validation, covering small batch production specificities.
- *“[I]n the case of infrequent production (i.e., if the interval between the production of two batches is more than six months), it is acceptable that the process simulation test is done just before the manufacturing of the next batch, provided that the results of the process simulation tests are available prior to the [start] of production. Nevertheless, in cases of long periods of inactivity (i.e., over one year), the validation prior to restart of production should be done with three runs.”* [6]
- The use of surrogate material during PV *“may be acceptable when there is shortage of the starting materials. ... Where possible, consideration should be given to complementing the use of surrogate materials with samples from the actual starting materials for key aspects of the manufacturing process.”* [6] When it is necessary to use surrogate materials, these materials should be assessed in parallel with samples of the initial raw material.

ATMPs are at the forefront of scientific innovation, with ongoing rapid technological change that will impact the manufacturing processes. As new manufacturing models emerge and industry knowledge evolves, it is imperative to apply QRM when evaluating the changes to manufacturing processes. If significant manufacturing changes are made, then comparability studies may be necessary to determine the impact of these changes on the safety, identity, strength, purity, and quality of the product. The extent of comparability testing depends on the manufacturing change, ability of analytical methods to detect changes in the product, and stage of clinical development.

For first-in-human studies, any differences between toxicology lots and clinical lots should be assessed for their impact on product safety. For later-phase studies, especially those designed to measure product efficacy, differences in clinical lots should be assessed for their impact on product safety and activity. Risk assessment should be performed for all types of manufacturing changes, regardless of the stage of product development. Based on that risk assessment, the *“appropriate validation and/or other clinical and/or nonclinical laboratory studies”* [4] *should be performed. This is to provide evidence that “each manufacturing change does not adversely affect product quality before distributing a product manufactured using the change.”* [4]

3.5.3 Cleaning

Cleaning validation should consider that ATMPs are often manufactured in the context of multiproduct and multi-patient processing. Another key consideration is that ATMP products are typically manufactured using aseptic manipulation throughout the entire process. Due to the nature of the mechanism of action, there is no terminal sterilization for ATMP products. Therefore, the cleaning program plays a critical role in ensuring product safety through preventing contamination and cross-contamination. *“You should clearly describe any controls for cleaning and change over as well as tracking and segregation procedures that are in place to prevent cross-contamination throughout the manufacturing process.”* [32]

It is important to recognize that equipment cleaning is only one of many measures that should be taken to control the risk of cross-contamination in a multiproduct facility or on equipment proposed to be shared. *“All potential sources of cross contamination should be assessed via a documented QRM process.”* [37] PIC/S Annex 15 states that *“for all cleaning processes, an assessment should be performed to determine the variable factors which influence cleaning effectiveness and performance,”* and that *“the risk presented by microbial and endotoxin contamination should be considered during the development of cleaning validation protocols.”* [29]

When developing a cleaning validation approach for ATMPs, consider the following guidance provided in PIC/S Annex 2A:

“Measures to prevent cross-contamination appropriate to the risks identified should be put in place. Measures that can be considered to prevent cross- contamination include, among others:

- (a) segregated premises,*
- (b) dedicating the entire manufacturing facility or a self-contained production area on a campaign basis (separation in time) followed by a cleaning process of validated effectiveness,*
- (c) adequate cleaning procedures:*
 - i. the cleaning procedure (technique, number of sanitation steps, etc.) should be adapted to the specific characteristics of the product and of the manufacturing process;*
 - ii. a risk-assessment should be used to determine the cleaning and decontamination procedures that are necessary, including the frequency thereof;*
 - iii. as a minimum, there should be appropriate cleaning and decontamination between each batch and*
 - iv. all cleaning and decontamination procedures should be validated.”* [5]

3.5.4 Supply Chain

Good distribution practices are essential for ATMPs, as they are a critical element in ensuring and maintaining product quality and viability. ATMP starting materials or end products often have unique shipping or shelf life requirements that increase supply chain complexity.

Currently (at the time of publication of this Guide), there are no specific regulations for good distribution practices specific to ATMPs, but general concepts and practices can apply. PIC/S Annex 2A states that *“application of QRM principles to the total supply chain is a critical part of the process to understand the risks to material quality.”* [5] Due to the sensitive nature of ATMPs, it is critical that traceability throughout the supply chain is established. Ensuring Chain of Identity (COI) and/or Chain of Custody (COC) is essential to enable a comprehensive supply chain.

Verification of transportation is critical to ensure that product quality is maintained from raw materials or starting material collection through administration of the product. Shipping containers, packaging materials, and transport routes should all be considered. A key element for verification of transportation is that *“a risk assessment should be performed to consider the impact of variables in the transportation process other than those conditions which are continuously controlled or monitored, e.g., delays during transportation, failure of monitoring devices, topping up liquid nitrogen, product susceptibility and any other relevant factors.”* [29]

When performing the validation of packaging, *“qualification of the equipment used for primary packing should be carried out at the minimum and maximum operating ranges defined for the critical process parameters such as temperature, machine speed and sealing pressure or for any other factors.”* [29] Additionally, EU guidelines state that *“the scope and extent of such qualification and/or validation activities (such as storage, pick and pack processes) should be determined using a documented risk assessment approach.”* [38]

3.6 Risk Management

3.6.1 Quality Risk Management

The importance applying risk management throughout the product lifecycle is well established in the pharmaceutical industry. In January 2023, ICH Q9(R1) [1] was released for industry and regulators to address challenges and improve the practical application of QRM principles. ICH Q9(R1) is commonly used by global regulators to review practical risk management implementation. *“FDA encourages the use of modern pharmaceutical development concepts, QRM, and quality systems at all stages of the manufacturing process lifecycle.”* [39]

The main principles of QRM remain key to implementing a scientific and practical approach to decision-making:

- *“The evaluation of the risk to quality should be based on scientific knowledge and ultimately link to protect the patient. (Note: Risk to quality includes situations where product availability may be impacted, leading to potential patient harm.)”* [1]
- *“The level of effort, formality, and documentation of the quality risk management process should be commensurate with the level of risk”* [1], the complexity of the process and uncertainty.

It is critical to ensure that the appropriate level of risk management is implemented in the various phases of the product lifecycle. ICH Q9(R1) [1] discusses that the level of formality within QRM is not a binary concept and can be influenced by many factors. There are three factors that are to be considered when determining the appropriate risk management activity:

- **Uncertainty** means the *“lack of knowledge about hazards, harms, and, consequently, their associated risks.”* [1] The amount of knowledge typically increases as the product moves through the clinical programs and into commercialization, thus decreasing uncertainty. For ATMPs, this level of uncertainty may remain higher even in commercial products, as they have been fast-tracked due to their transformative nature. Thus, it is important to consider not only the status, but also the available knowledge for the ATMP product.

- **Importance** is linked to how important that risk-based decision is with respect to product quality. For ATMPs, understanding the impact of the risk to product quality may be a challenge as these products are unique and often novel.
- **Complexity** refers to that of the process or subject area for the risk management activity. It is common for ATMPs to have aspects that are more complex.

Determining the level of formality for the risk assessments and risk-based decisions is key for facilitating informed decisions within ATMP validation programs. Regardless of how much formality is applied, a robust management of risk is the goal of the process.

3.6.2 QRM for ATMPs and Validation

Applying QRM for ATMPs is critical, as these products have varying characteristics and levels of complexity in the manufacturing process and analytical methods. *“The risk-based approach (“RBA”) is applicable to all type of ATMPs. It applies in an equal fashion to all type of settings. The quality, safety and efficacy attributes of the ATMPs and compliance with GMP should be ensured for all ATMPs, regardless of whether they are developed in a hospital, academic or industrial setting.”* [6]

A risk management plan should be developed to identify, characterize, and minimize risks associated with ATMP manufacture and use. Given the unique characteristics of ATMPs, outlining potential risks and mitigation strategies is especially important. Introducing RMPs in the early phases of development can be challenging because many parameters are still unknown and knowledge is gained throughout the lifecycle. For this reason, the RMP should be maintained as a living document that guides the risk-based strategy for elements including validation.

QRM is used in the context of validation as follows:

“To identify the scope and extent of verification, qualification and validation activities (e.g., analytical methods, processes, equipment and cleaning methods);

To determine the extent for follow-up activities (e.g., sampling, monitoring and revalidation);

To distinguish between critical and non-critical process steps to facilitate design of a validation study.” [1]

Additionally, the short shelf life of many ATMPs often necessitates product release before all testing results are available. Thus, a strong risk management approach is required to support product release based on continuous data verification, which is connected to PV. QRM is critical to developing control strategies across all stages of development to minimize process variability and to reduce contamination and cross-contamination.

The use and application of QRM within ATMP validation programs is required across the regulatory landscape. This can not only expedite these transformational medicines but also maintain the focus on product quality and patient safety.

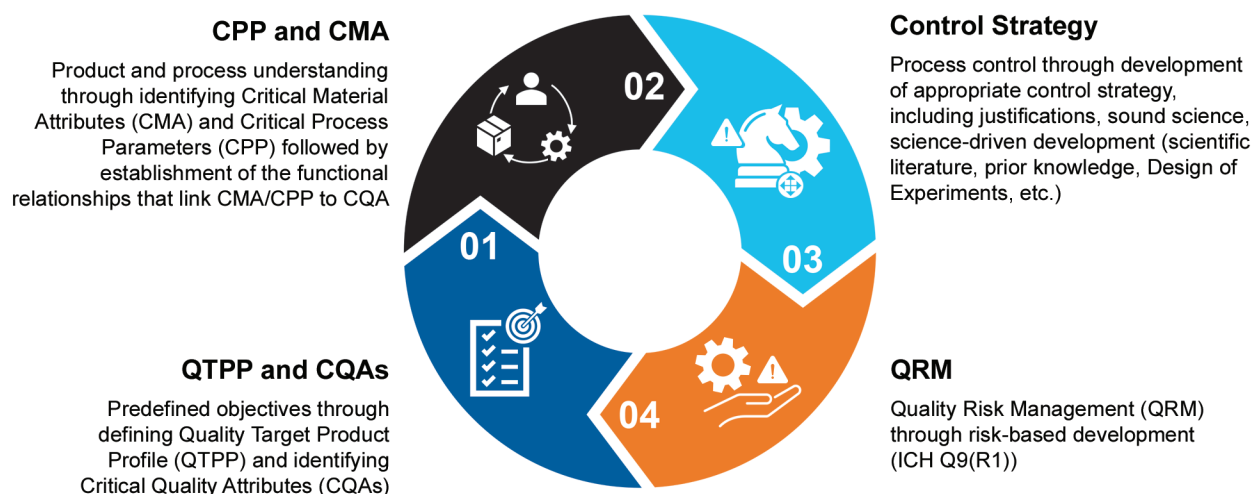
4 Quality by Design as a Risk Management Tool in ATMPs

In traditional drug development, the design of the product and processes is managed by distinct functional groups, with Research and Development (R&D) focusing on the product and CMC overseeing the process, each at different stages of development. However, this conventional workflow is suboptimal for emerging therapeutics such as ATMPs due to several factors [40]:

- Process is inherently tied to the product
- Limited understanding of both the product and the process
- High complexity
- Significant variability
- Accelerated development timelines

Consequently, it is crucial to implement QbD methodologies early in the development of ATMPs by integrating manufacturing and discovery activities into a comprehensive, holistic roadmap and strategy. The QbD approach is defined by ICH Q8(R2) as *“a systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management.”* [41] Figure 4.1 describes the tools used for the QbD approach.

Figure 4.1: Specific Tools Used for QbD



As outlined in ICH Q8(R2) [41], Q9(R1) [1], and Q10 [42], QbD facilitates a deeper understanding of the processes involved and promotes a more systematic and scientific approach to drug development. In ATMP development, where complexity and variability are notably higher than for traditional products, QbD provides a cohesive approach to integrating quality at the outset. This alignment between manufacturing and discovery processes ensures that accelerated development timelines for these therapies meet rigorous quality expectations. This integrated approach not only enhances the efficiency of the development process but also ensures that the therapies meet regulatory requirements and deliver the intended clinical benefits to patients.

4.1 Key Components in QbD

4.1.1 Quality Target Product Profile

The Quality Target Product Profile (QTPP), as a forward-looking framework of a DP's quality attributes, is instrumental in achieving the desired quality while adhering to safety and efficacy requirements outlined in ICH Q8(R2) [41]. Defining the QTPP is a pivotal early step in the development of ATMPs, as it provides a detailed and structured profile of the final product. This profile encompasses essential elements such as the product's intended purpose, route of administration, dosage form, strength, and CQAs. For ATMPs, the QTPP highlights key parameters such as potency, purity, identity, and stability, which are crucial for ensuring the product meets therapeutic goals and complies with regulatory standards.

Beyond serving as a foundational framework, the QTPP is an indispensable tool for developers, offering strategic guidance to identify the studies and data necessary for clinical trial entry and Marketing Authorization Applications (MAAs). When thoughtfully designed and regularly updated with emerging data, the QTPP provides a systematic and holistic understanding of both the product and the development process. By defining critical parameters such as potency, purity, identity, and stability, the QTPP ensures alignment with therapeutic objectives and regulatory expectations. This proactive approach streamlines the development pathway, facilitating efficient progression toward clinical success and commercial approval. Table 4.1 provides examples of QTPP elements. [43–45]

Table 4.1: Examples of QTPP Elements

QTPP Elements	Target
Starting and raw material attributes	Specifications for starting and raw materials, acceptance criteria, and limits based on product design and process validation
Manufacturing process	• Product design • Process platform • Batch scale • Production limits (e.g., Product Design Specifications (PDS)) • Identification of Critical Process Parameters (CPPs) and In-Process Controls (IPCs) • Consistency and Process Validation (PV) • Aseptic processing, GMP
Drug product quality attributes	• Identity • Purity/impurities • Potency/functionality • Genetic stability • Sterility
Analytical methods	• Assays for IPC, release, and stability testing • Assays for extended characterization (including PV and comparability testing) • Validation of release assays • Qualification of characterization assays • Setting specifications with acceptance limits
Stability	• Drug Substance (DS) and Drug Product (DP) stability and shelf life • In-use stability instructions for handling • Stability of intermediates and validation of hold steps
Container closure	Support product stability during transport and storage
Indication(s)	• Product fit for the disease(s) to be treated • Potency assay in line with mode of action • Correlation of potency versus efficacy

Table 4.1: Examples of QTPP Elements (continued)

QTPP Elements	Target
Dosage form	Target product form and formulation
Dosing	Single dose/repeated dosing versus manufacturing scale, DP strength
Route of administration	- Impact of route of administration on formulation and product profile/strength - Impact of additional substances (e.g., medical devices)
Pharmacokinetics	Persistency and functionality of DP versus product design
Safety	- Immunogenicity profile of the DP - Toxicity profile of the DP - Specification limits for impurities
Efficacy	- Specification limits for potency; correlation between potency and efficacy - Quality of batches used in clinical studies; comparability toward commercial batches

Developing ATMPs begins with understanding the disease and its pathophysiology. The unique starting materials and complex manufacturing processes directly impact the structural and functional characteristics of these therapies. This necessitates an integrated approach to CMC, non-clinical, and clinical development. Product design must account for known and potential risks associated with components and raw materials, starting with a comprehensive risk assessment of materials, processes, control strategies, and intended clinical use. Table 4.2 provides a summary of challenges in different ATMP categories.

Table 4.2: Challenges with ATMPs

Cell-Based Medicinal Products (CBMPs) - High variability in cellular starting material impacts product consistency and quality. - Autologous therapies are especially affected – each product is patient-specific. - Variability in raw materials and processes during clinical trials further complicates standardization. - Broad product specifications can lead to inconsistent efficacy and safety. - Difficulties in establishing predictable clinical performance. Strategies to Mitigate Variability - Define and link CQAs to clinical outcomes. - Implement robust manufacturing process controls and raw material quality management. - Utilize real-time process monitoring and advanced analytical tools for in-process adjustments.
Gene Therapy Products - Complex manufacturing (e.g., viral vectors, plasmids) demands specialized technologies. - Key issues: scalability, yield optimization, purity, and contamination control (especially for plasmid DNA). Enabling Product Reliability - Apply innovative control strategies: - Define CQAs early - Use cutting-edge analytics - Monitor and optimize processes in real time

A risk-based approach can be instrumental in early-stage development, helping identify and address known and unknown risks, similar to the principles outlined in the QTPP. During early research, testing product functionality and proof of principle is crucial, particularly when non-clinical models are unavailable. Patient cells or organoid stem cells can serve as substitutes for testing. Pilot-scale production runs further validate production capacity and product suitability, ensuring scalability and reproducibility.

The QTPP serves as a vital tool for ATMP development, providing a comprehensive framework for considering factors that influence the final product's quality, safety, and efficacy in both clinical and commercial contexts. Established during the research phase and continuously updated throughout development, the QTPP forms the foundation for the entire CMC module of the MAA. Properly constructed and maintained, the QTPP ensures a systematic and integrated approach to product development, aligning therapeutic goals with regulatory expectations and streamlining the path to commercialization.

4.1.2 Critical Quality Attributes

Aligned with ICH Q8(R2) guidelines [41], CQAs are defined as the physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality. Developing an analytical testing strategy for CQAs involves a systematic, risk-based evaluation to determine their significance across various categories. This includes their potential impact and associated uncertainties. This strategy ensures the prioritization of attributes critical to DP quality, such as physical properties, assay performance, content uniformity, degradation products, and microbial contamination. Early identification of CQAs during the development phase is vital for proactively managing risks, safeguarding the product's integrity, and maintaining compliance throughout its lifecycle.

The role of CQAs in ATMPs presents unique challenges due to the inherent complexity and variability of these products in comparison with traditional pharmaceuticals [32]. Companies are required to define CQAs that are specific to their distinctive manufacturing processes and product characteristics, which is especially crucial for autologous therapies where the starting materials are derived from individual patients. This individual sourcing introduces natural variability, with sterility being a primary concern because these materials are typically aseptically collected under clinical conditions rather than being inherently sterile.

Currently, regulatory guidance on categorizing and managing such materials is limited, necessitating the development of a common scientific understanding regarding their sterility status and the appropriate control strategies. The complexity is further compounded by the frequent use of manual aseptic processes in open systems and the restricted application of standard purification methods such as low pH inactivation or filtration. Thus, a comprehensive understanding of the manufacturing process is essential to establish meaningful CQAs and ensure the consistent quality of ATMPs throughout production. This thorough approach mitigates the absence of traditional safety measures, thereby maintaining reliability and homogeneity in ATMPs.

The intricate nature of ATMPs presents considerable hurdles in defining and monitoring CQAs. Although certain measurable attributes can be decisively identified and controlled, numerous critical performance attributes often remain elusive or challenging to quantify precisely. The industry's tendency to concentrate on measurable CQAs can inadvertently lead to insufficient attention to managing unknown variables. Effective specification setting for these therapies demands a deep understanding of the CQAs that directly impact safety and efficacy. Regulatory authorities are cognizant of these complexities and advocate for a systematic, phased approach to CQA development, often necessitating the early identification of CQAs prior to finalizing the QTPP. Key attributes such as the viability of donor material, cellular impurities, and comparability of allogeneic products are particularly crucial during initial assessments, highlighting the need for comprehensive strategies to manage quality and maintain consistency.

When a gene or cell therapy product fails to meet a defined CQA, its release is contingent upon comprehensive, risk-based analyses performed by expert teams from various disciplines, including manufacturing, regulatory compliance, clinical data interpretation, toxicity assessment, and efficacy evaluation. CQA limits are set conservatively, based on historical failure trends and the complexity of the manufacturing process. If a lot shows Out-of-Specification (OOS) results, a formal quality event is triggered, leading to investigations of laboratory and manufacturing practices. Each affected batch undergoes a detailed impact assessment, reviewed by safety and quality committees to contextualize the batch with the patient's clinical condition. If safety risks are identified, the product is not used; however, if the benefit-risk balance is favorable, the treating physician may be consulted to evaluate the use of the nonconforming product. Exceptional release might be pursued through regulatory pathways such as a Single-Subject Investigational New Drug application (sIND) or a Managed/Expanded Access Program (MAP/EAP). This adaptive approach ensures continuous evaluation, balancing patient safety, therapeutic benefit, and manufacturing realities throughout the ATMP product lifecycle. [46, 47]

4.1.2.1 Real-Time CQA Monitoring Strategies

Real-time monitoring of CQAs is essential for ensuring product quality and minimizing batch failures in rapid autologous cell and gene therapy workflows, where compressed timelines and patient-specific batches leave little margin for error. Although traditional offline assessments remain valuable, they often lack the immediacy required to detect and correct deviations during manufacturing. Process Analytical Technology (PAT) addresses this gap by enabling continuous, minimally invasive monitoring of key attributes such as cell viability, differentiation status, and metabolic activity. Using tools such as in-line glucose and lactate sensors, optical spectroscopy methods—including Raman and near-infrared—and automated imaging systems, PAT facilitates real-time feedback control and data-driven decision-making [48].

This approach reduces manual interventions, batch variability, and costs while improving product consistency. Robust and scalable manufacturing of ATMPs depends on innovation across three interconnected layers: closed and automated tools, robust and frequent process measurements, and advanced data analytics, which together enable well-characterized, adaptive production methods. Integrating these elements into cohesive systems demands a multidisciplinary approach that combines deep biological understanding with engineering expertise to identify relationships between CPPs and CQAs, consolidating process knowledge within digital twins.

Given the complexity of these challenges, innovation management, and cross-disciplinary collaboration are paramount. This requires experts in product engineering, cell biology, sensor development, data modeling, material sciences, end users, and regulatory bodies to work together. This collaborative effort is crucial for steering the industry toward meaningful adoption of Industry 4.0 concepts, creating scalable, reliable manufacturing platforms that can deliver novel therapeutic options to a growing patient population and advance the promise of ATMPs [49].

As the ATMP industry embraces Industry 4.0 principles, manufacturing is shifting from manual, paper-based workflows toward distributed, automated, and digital knowledge-driven systems. This transformation enhances process understanding, scalability, and the central role of real-time CQA monitoring in assuring quality and consistency. Within this advanced framework, risk-based decision models help determine when real-time monitoring is essential—particularly for CQAs closely linked to efficacy, safety, or high variability—and when offline analysis suffices for stable, well-characterized attributes. Integration of PAT with closed-loop feedback systems facilitates a proactive and adaptive quality strategy, ensuring robust and reproducible manufacturing even in the most time-constrained environments. For viral therapies, real-time monitoring is equally critical to optimize upstream and downstream processes by tracking viral particle characteristics—such as full versus empty capsids, aggregation, and impurities—to maintain product safety and efficacy, though current PAT adoption in viral vector manufacturing remains nascent and often relies on offline methods [48].

Despite the transformative potential of PAT, widespread implementation in ATMP manufacturing faces significant challenges. Identifying CPPs and correlating them with CQAs is complex, necessitating tools such as systems biology and bioinformatics. Upstream processes such as cell expansion and viral vector transfection require intensive monitoring of variables including cell density, viability, and metabolites, whereas downstream cell therapy processes pose unique PAT challenges due to their short duration and regulatory constraints. Additionally, technical limitations (such as a lack of suitable sensors capable of sensitive, accurate real-time measurements across broad dynamic ranges) and operational hurdles (such as the need for process redesign, regulatory uncertainty, and limited expertise) hinder adoption [50].

Autologous therapies add complexity due to patient variability, and many organizations lack awareness of QbD approaches and integrated PAT strategies involving automation and data analytics. Nonetheless, industry-wide collaboration through initiatives such as the Cell and Gene Therapy Catapult's PAT Consortium [51] are advancing novel technologies, including Near-Infrared Spectroscopy (NIR) sensors, microfluidics, and machine learning-enabled systems. The ultimate aim is achieving true real-time, in-line monitoring across upstream and downstream operations, empowering manufacturers to deliver consistent, efficient, and safe ATMPs.

4.1.3 Critical Material Attributes

Critical Material Attributes (CMAs) are fundamental characteristics of input materials that need to meet rigorous specifications to guarantee the consistent quality of ATMPs. These attributes include a wide range of components such as cellular starting materials harvested from patients or donors, human-derived culture media essential for cell growth, and specialized feeder cell lines used in certain treatments. The inherent variability in sourcing and handling of these materials poses significant challenges to the manufacturing process. For example, fluctuations in donor cell quality or inconsistencies in growth media can directly impact the manufacturing profile, affecting the potency, purity, and safety of the final product. Effective management of these raw materials requires a strategic approach to mitigate variability and maintain consistency throughout production.

Managing the quality of raw materials and patient-derived components is notoriously complex. Although variability in raw materials can be partly controlled through meticulous supplier selection and comprehensive testing protocols, patient-derived materials are more challenging due to their unique nature. Thus, ensuring process robustness becomes critical to counteract variability and achieve consistent results. Advanced strategies such as real-time monitoring, adaptive manufacturing techniques, and enhanced process controls are vital to overcoming these hurdles. According to European Pharmacopoeia (Ph. Eur.) 5.2.12 [52], maintaining the quality and consistency of these materials is crucial for product safety and efficacy. Exploring additional options to manage variability may involve advancements in raw material characterization, forming partnerships with suppliers for enhanced quality assurance, and developing innovative processing techniques to improve adaptability and consistency.

In addition to managing CMAs, a thorough understanding of ancillary materials, defined by United States Pharmacopeia (USP) <1043> Ancillary Materials for Cell, Gene and Tissue-Engineered Products [53], is crucial in ATMP manufacturing. Ancillary materials (such as fetal bovine serum), enzymatic digestion agents (such as collagenase and DNase), growth factors, cytokines, monoclonal antibodies, antibiotics, purification resins, cell-separation devices, and various media components all play a significant role in the production process despite not being part of the final therapy. They directly influence important attributes of the product, including cell growth rates, differentiation, viability, and therapeutic potency. The management of these materials involves addressing variability, ensuring consistent quality, and minimizing the risk of contamination. This requires stringent selection criteria, robust supplier qualification, and thorough testing protocols to evaluate purity, potency, and suitability for intended uses. Implementing comprehensive traceability and documentation systems also ensures compliance with regulatory standards.

4.1.4 Critical Process Parameters

The phrase “the process is the product” has long emphasized the critical relationship between manufacturing conditions and product quality. For ATMPs—which are biologically active, dynamic, and often not fully characterizable with current analytical methods—a robust and well-controlled process has been essential to ensuring consistency and safety. However, although originally protective, this mindset has increasingly become a barrier to innovation. Treating process changes as threats to product identity can discourage technological progress, particularly as ATMPs grow in complexity and personalization. Success in ATMPs requires not just replicating processes but continually improving them.

For example, manufacturing a single autologous Chimeric Antigen Receptor (CAR) T therapy involves thousands of variables, many of which go unmeasured or are poorly understood [54]. Advancements in analytics, automation, and data integration must shift validation from a backward-looking exercise into a proactive, real-time system that supports adaptive control and predictive quality. In this modern context, validation should enable innovation, not restrict it. Rather than holding on to rigid interpretations, the field should adopt a more forward-looking view: the product is the product, but deep process understanding and agile optimization are what will make the product accessible, affordable, and scalable for the patients who need it most.

Regulatory agencies emphasize the critical role of robust manufacturing process controls in the successful development of ATMPs. Central to this control is the identification and definition of CPPs, which form the foundation of a resilient and reproducible manufacturing control strategy. Furthermore, the importance of equipment in ATMP production is paramount. The relationship between equipment, process, and product quality in many ATMP products is intricately interconnected, highlighting the necessity of selecting, validating, and maintaining equipment that precisely meets the unique requirements of the therapy being developed. Collectively, these elements create a cohesive strategy aimed at ensuring ATMP products adhere to stringent quality, safety, and efficacy standards while supporting scalable and sustainable production processes.

Effective identification of CPPs in ATMP manufacturing involves a systematic approach that integrates several key steps. Initially, a thorough gap analysis is conducted across all process steps to pinpoint areas needing improvement. Following this, studies are performed to quantify the impact of each process parameter on CQAs, helping understand how changes in these parameters can affect product quality. The final step involves prioritizing CPPs based on their influence and variability, ensuring meticulous control over those with the greatest impact on product consistency and performance. Given the multitude of interconnected parameters in ATMP manufacturing, evaluating all of their effects is impractical, highlighting the necessity of prioritization to focus resources on the most critical aspects—such as maintaining precise cell culture temperatures and managing variability in donor cell characteristics. This systematic approach supports the development of a stable and reliable manufacturing process capable of consistently producing high-quality therapies.

Monitoring and establishing hold times are critical aspects of ATMP manufacturing due to the sensitivity of living cells and viral vectors to environmental conditions. Improperly controlled hold periods—such as those occurring after apheresis, post-transduction, or before cryopreservation—can significantly affect CQAs such as cell viability, transduction efficiency, and product potency. As such, hold times often represent CPPs that must be rigorously evaluated through time-course studies assessing the impact of various durations and storage conditions (e.g., temperature, agitation) on product quality.

These studies help define acceptable hold windows and are essential for ensuring product consistency, sterility, and efficacy. Regulatory agencies, including the FDA [2] and EMA [3], expect hold times to be data-driven, validated, and clearly integrated into a control strategy. As ATMP manufacturing moves toward more automated and distributed models, continuous monitoring technologies and robust data capture tools are increasingly vital to managing hold steps effectively and minimizing variability in these highly individualized therapies [55].

Common methods for identifying CPPs encompass leveraging scientific knowledge, historical data, and risk assessment tools. Statistical experimental design techniques, such as Design of Experiments (DOE), are typically employed to systematically vary process parameters and assess their impact on CQAs, thereby ensuring a comprehensive understanding of the process dynamics and enhancing control over product quality. Understanding the relationships between CPPs and CQAs is crucial for establishing optimal process conditions and defining acceptable ranges for process parameters to ensure the desired product quality. Therefore, CPPs are the parameters that must be precisely controlled within specified limits to maintain consistent product quality and achieve the intended clinical outcomes.

The inherent variability of starting materials sourced from patients presents significant challenges in the development and manufacturing of ATMP products. This variability complicates the identification of CQAs and the determination of CPPs, which are essential for controlling the manufacturing process and ensuring consistent product quality. It also impedes the establishment of robust acceptance criteria for release testing, increasing the risk of OOS results. To address these challenges, the identification and control of CPPs play a pivotal role in mitigating the impact of variability and maintaining a consistent product profile. Proactive strategies, including advanced process control, real-time monitoring, and robust analytical techniques, are critical to overcoming these complexities. By integrating these approaches into the development and manufacturing lifecycle, developers can enhance process reliability, meet regulatory expectations, and deliver safe, effective, and high-quality therapies to patients.

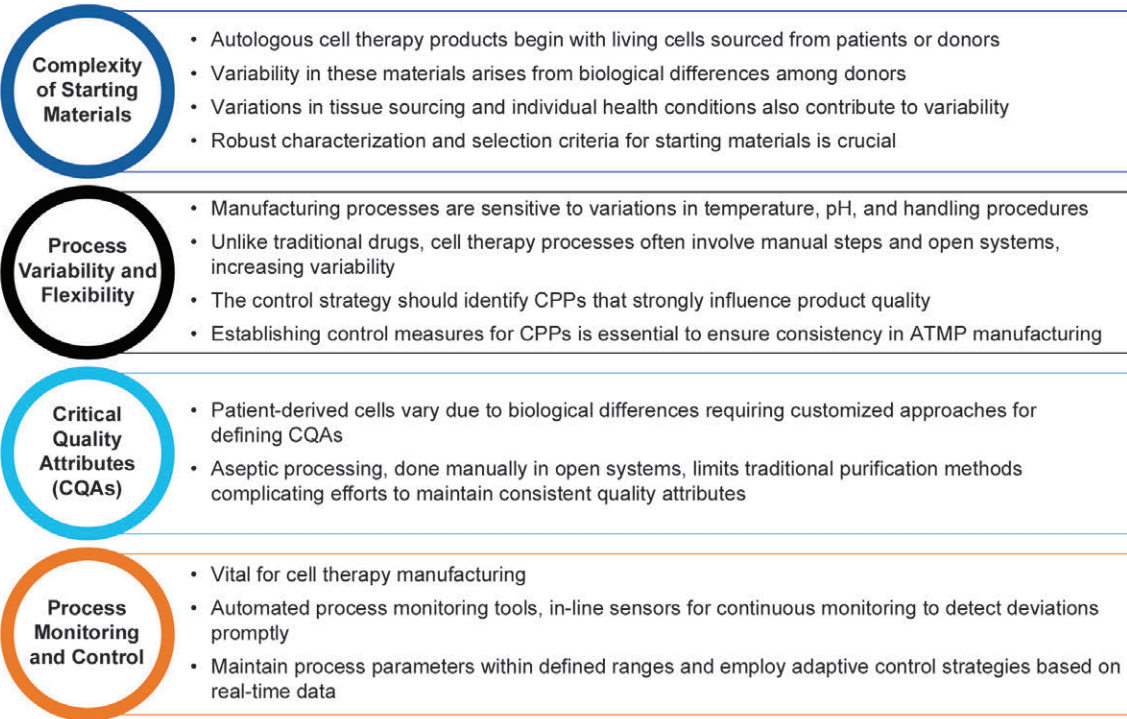
4.1.5 Manufacturing Control Strategy

Developing a control strategy within the QbD framework is fundamental to ensuring consistent product quality throughout the manufacturing of ATMPs. This strategy incorporates critical process controls, In-Process Controls (IPCs), and Quality Control (QC) tests, all meticulously designed to monitor and maintain product quality within predefined parameters. To enhance process robustness and accelerate development, a risk-based validation approach should leverage previous validation results through a comprehensive system that aligns manufacturing stages and digitizes historical data. This integration allows information gathered during early process development and validation phases to be reused in later stages, streamlining knowledge transfer and supporting dynamic risk assessment and decision-making. Establishing robust process controls that encompass equipment settings, operational conditions, raw material specifications, and validation protocols minimizes variability and enhances manufacturing consistency.

IPCs play a pivotal role in providing real-time insights into product quality, despite challenges presented by the complexity and variability of ATMPs. These systems have the potential to significantly improve manufacturing success rates and manage cost constraints. A robust control strategy also entails the development of appropriate QC tests aligned with CQAs and stringent final product release criteria. This comprehensive approach supports adherence to quality standards and addresses the unique challenges of ATMP products, such as variability in starting materials and complex manufacturing processes, ensuring high-quality outcomes and successful clinical applications.

Integral to the success of a control strategy is establishing a comprehensive documentation system that encompasses all aspects of the manufacturing process, including CPPs, acceptance criteria, IPCs, and QC tests. This documentation not only facilitates manufacturing oversight but also ensures regulatory compliance and enables continual improvement through periodic review and updates. The digitalization of process and validation data across the product lifecycle is key to supporting this continual improvement cycle, enabling knowledge sharing, reducing time to market, and fostering innovation. Despite the inherent challenges in understanding product and process characteristics specific to ATMPs, a proactive approach integrating evolving scientific knowledge, advanced analytical techniques, digital tools, and adaptive manufacturing processes is essential. Figure 4.2 outlines specific aspects to consider when designing a control strategy for autologous cell therapy products.

Figure 4.2: Aspects to Consider when Designing a Control Strategy for Autologous Cell Therapy Products



In essence, though the fundamental principles of establishing a robust control strategy apply universally to both conventional DPs and ATMPs, the distinct challenges faced in ATMPs—such as limited understanding of products and processes—directly impact the identification of appropriate CQAs and CPPs, leading to challenges in manufacturing control and higher failure rates. Therefore, adopting a proactive strategy that integrates evolving scientific knowledge, advanced analytical techniques, and adaptable manufacturing processes becomes essential.

4.2 Lifecycle Management

Continuously monitoring the performance and quality of ATMPs throughout their lifecycle is critical to their successful development, commercialization, and long-term reliability. This ongoing oversight enables continual improvement in understanding both the product and its manufacturing processes, supporting proactive management of changes through robust change control procedures. Effective lifecycle management requires an integrated approach that addresses scientific, regulatory, manufacturing, clinical, commercial, and ethical considerations.

Essential to this process is the implementation of rigorous process monitoring, which is a core requirement of validation lifecycle management. By ensuring consistent adherence to quality standards and enabling timely identification and resolution of potential issues, continuous process monitoring safeguards the safety, efficacy, and accessibility of these transformative therapies. Furthermore, the incorporation of innovative solutions, robust regulatory strategies, and comprehensive patient monitoring reinforces the commitment to delivering high-quality therapies that meet the evolving needs of patients and regulatory bodies alike.

The lifecycle management of ATMP products requires addressing complex supply chains, ensuring robust traceability systems, and navigating stringent regulatory environments. The high cost of failure underscores the importance of managing the ATMP product lifecycle correctly from the outset, with a clear understanding of regulatory requirements and proactive product development to meet these needs. Beginning with the procurement of materials and extending through clinical dose administration, stringent controls and robust traceability are essential to maintain safety and efficacy. This is especially critical for autologous therapies, where precise patient-specific modifications are crucial to avoid severe adverse reactions.

Securing the COC and employing precise analytics for cell validation are fundamental aspects of maintaining product integrity throughout the manufacturing process, ensuring that cellular materials remain segregated, safe, and well-identified from sourcing through patient administration. Continual improvement is a cornerstone of advancing ATMP products, driving innovation in both development and manufacturing processes. Integrating breakthroughs in gene editing technologies, vector design, and cell therapy techniques enables the creation of therapies with enhanced precision, efficacy, and safety. These advancements allow for more targeted and efficient treatments, addressing complex diseases while improving scalability and consistency in manufacturing. By continually refining these processes, the industry can deliver therapies that better meet patient needs and adhere to rigorous quality and regulatory standards.

Effective data management is critical in ATMPs, where vast amounts of data are generated across the product lifecycle—from research and clinical trials to manufacturing and post-market surveillance. To manage this complexity, an integrated digital system is essential. Platforms that unify data from disparate sources enable real-time visibility, traceability, and streamlined process optimization. Advanced informatics tools such as Machine Learning (ML), big data analytics, and cloud-based platforms are essential for organizing, analyzing, and interpreting this data to extract actionable insights. These technologies enable continuous process optimization, improved patient outcomes, and enhanced alignment with evolving scientific and regulatory expectations.

To further support the integration of AI into regulated areas, it is essential to establish effective risk management strategies, especially when dealing with dynamic online learning and ML operating models. By prioritizing risk mitigation activities that deliver the greatest impact and provide iterative feedback, organizations can improve both the quality of AI applications and the robustness of related processes. Taken together, this comprehensive, data-driven approach supports innovation, strengthens product reliability, and builds confidence among patients, regulators, and stakeholders [56].

Effective management of ATMPs requires addressing technical challenges alongside rigorous regulatory compliance and proactive engagement with regulatory bodies. Communication and a thorough understanding of global regulations are crucial to enable adherence to standards, from development through commercialization, with emphasis on traceability and control systems necessitating robust processes within ATMP companies. Additionally, navigating varying regulatory frameworks across regions demands comprehensive knowledge. Addressing potential challenges early in the development of ATMP products is essential to avoid costly setbacks during clinical development and commercialization.

A proactive and experienced approach anticipates scientific, regulatory, and commercial demands, including market scalability, manufacturing feasibility, and long-term supply chain needs. Integrating commercial requirements into the development process enhances the risk-based approach by aligning immediate and long-term objectives. Implementing integrated digital solutions rather than paper-based records streamlines validation activities, improves data integrity, enables real-time monitoring, and facilitates cross-functional collaboration. These capabilities reduce the potential for errors, support regulatory compliance and audit readiness, and ultimately control costs and accelerate the journey to market. Although this strategy may require greater initial effort and resources, it establishes a solid foundation for the successful commercialization of ATMP products, ensuring they are developed with a focus on safety, efficacy, and global accessibility.

5 Key Validation Differences and Challenges in ATMPs

The emerging and dynamic nature of the ATMP industry introduces additional risks and uncertainties that must be addressed during validation. These may include developing and progressing manufacturing processes, evolving regulatory requirements, emerging safety concerns, and promising technological advancements that outpace the existing validation frameworks. As such, validation practices within the ATMP community must remain agile and adaptable, incorporating ongoing updates and improvements to keep pace with the rapidly evolving landscape.

5.1 Analytical

In the evolving landscape of advanced therapies, the validation of quality analytical methods stands as a critical pillar in ensuring product quality, safety, and efficacy. Although the fundamentals of analytical method validation remain consistent across the life science industry, there are nuanced considerations specific to ATMPs that require a tailored approach.

ATMP analytical and QC validation practices must account for the unique nature of these therapies. Unlike traditional biologics, ATMPs are often living therapies that may involve personalized manufacturing processes tailored to individual patients. Consequently, analytical methods must be versatile and adaptable to accommodate the inherent variability associated with these conditions. This necessitates robust validation protocols that account for cell line and/or patient-specific factors without compromising on accuracy, reliability, and reproducibility, while also mitigating variability associated with patient-specific materials and ensuring product consistency.

One of the key distinctions in ATMP validation lies in the types of testing performed. Conventional biologics may rely on established assays and methodologies, whereas ATMPs may require novel approaches to assess CQAs specific to cell-based or gene-based therapies. This includes the validation of assays for cell identity, viable cell potency, purity, gene identity, and product stability, as well as transduction/transfection efficacy, vector replication competency, and other antigen-specific functionalities. These tests may involve innovative techniques such as flow cytometry, Polymerase Chain Reaction (PCR)-based assays, and next-generation sequencing.

The introduction of personalized manufacturing adds another layer of complexity to the validation landscape, demanding meticulous adherence to maintaining COI and COC throughout the manufacturing and control testing process. For patient-specific therapies, it is crucial to ensure testing results are correctly linked back to the products they represent, as each individual patient lot represents a unique dose. The controls associated with COI and COC must be integral to the handoffs, movements, testing, and documentation of these therapy samples. Such controls often include solutions such as digital track and trace.

Personalized therapies or ATMPs cannot be effectively scaled up and must be scaled out in production count. As such, facilities pursuing a high volume of annual batches will be subject to an equally scaled-out requirement for QC in-process and release testing. It should also be noted that for anything patient specific, there is a time criticality for quality release testing, as it will have a direct impact on the timing that the final dose(s) can reach and then be administered to the patient, often referred to as vein-to-vein time for autologous therapies.

In the pursuit of effective commercial solutions, rapid and efficient testing technologies are often sought, with ATMP laboratories looking to leverage emerging technologies to streamline the quality process. These new solutions include the use of novel technologies, automated handling and testing systems, densified testing systems, and robotics. Although these technologies hold promise for accelerating throughput, reducing laboratory footprint and staffing requirements, and possibly assisting in automating COI and COC, they also pose unique validation challenges. New, emerging technologies can often face issues regarding method robustness, reliability, and comparability with conventional assays. Therefore, development and validation efforts must strike a balance between innovation and regulatory compliance, ensuring that rapid test methods meet the same rigorous standards as traditional approaches.

Several agencies have published guidelines and/or requirements that can help provide structure to companies looking to validate their quality testing. ICH Q2(R2) [34], though high level, provides structured guidance on analytical procedure approaches. Section 12 of EudraLex Volume 4 Part IV [6] provides specific considerations that must be taken with regard to quality testing, sample control, and handling. Overall, the validation of analytical methods for ATMPs requires a nuanced and tailored approach that accounts for the unique challenges and complexities associated with personalized medicine and emerging technologies. By embracing innovation while upholding rigorous standards of quality and safety, stakeholders can ensure the continued advancement and success of ATMPs while meeting the quality expectations set forth by the industry.

5.2 Process

ATMP products target a wide variety of modalities and encompass a wide variety of manufacturing processes; thus, each has manufacturing challenges that are often complex and unique. There are clear expectations for the validation of processes (e.g., FDA Guidance for Industry: Process Validation – General Principles and Practices [39]), but these guidelines do not address ATMP-specific validation challenges. Given the transformative nature of ATMPs, the validation strategy must consider both product knowledge and patient needs.

As a manufacturing process is developed, it is important to ensure there is a focus on producing quality products by design and scalable processes that can advance rapidly if early clinical trials are promising. Often, the amount of product and process knowledge is limited due to the novel technology and techniques being used. In addition, there may be a lack of established analytical methods or equipment available for these novel approaches. These aspects contribute to challenges in ATMP PV approaches.

Both autologous and allogeneic ATMPs have the challenge of inherent differences in starting materials that result in variability within the process. For autologous therapies, each product is made from the starting material collected from an individual patient. These starting materials have extensive variability due to differences in the patients themselves, their disease state, and any prior and/or current treatment regimen.

For personalized cell therapies, it is recommended to use healthy donor cells (or surrogate material) when designing and implementing the validation strategy. For allogeneic therapies, a surrogate material may be composed of only certain components of the final formulation that have demonstrated equivalency to product and no impact to validation. However, the inherent variability between patients and the difference in patient versus surrogate could lead to difficulties in the validation approach and results. It is important to understand these differences and perform risk assessments to justify the appropriate material for the PV strategy.

ATMP products are often manufactured in individualized and smaller batch sizes. For autologous therapies, because they are a personalized medicine and are limited in starting material, a manufactured batch typically has just enough for the patient dose (whether limited by manufacturing complexity or sheer vein-to-vein time). For allogeneic therapies, although a manufactured batch can produce enough supply for several patients, the batch sizes are significantly smaller than typical large-scale antibody manufacturing, as an example. Some processes take months and are not produced very frequently, per clinical need. The amount of material required for extended characterization, stability, and Process Performance Qualification (PPQ) will often reduce or eliminate the ability to have a patient dose.

Often, a limited number of batches is produced prior to and during PPQ. It is encouraged to utilize process and product information from similar products when designing the validation strategy. Where it is infeasible to produce sufficient material under normal conditions, alternative schemes should be explored. Use of surrogate starting material, or approaches such as continuing to expand product cells after the patient dose has been withdrawn, can be considered when appropriate. Whenever available, leveraging platform approaches is highly encouraged. This will increase efficiency and support the accelerated timeline to commercialization.

When manufacturing personalized therapies, strict control of COI and COC throughout the entirety of the therapy lifecycle (from receipt of the patient material through delivery of the DP) must be maintained. These products are designed to treat a single patient, and controls must be in place to ensure against cross-contamination and manufacturing mix-ups. These controls must be integral to handoffs, movements, equipment use, storage locations, materials used, and all quality testing performed. Due to the criticality of these controls, digital track, trace, and gating are often used to capture where the process and product are at all times, and to guard against the use of incorrect equipment and materials.

Another challenge for ATMP PV is the limited knowledge and variability of product quality attributes. Due to the unique and novel elements of ATMPs, it is often a challenge to define CQAs. Poorly defined CQAs can be a hinderance to an effective PV strategy. Given the unique and novel nature of many ATMP manufacturing processes, it is also a challenge to determine the CPPs. Using a risk-based approach to understand the critical process steps will support an expedited and focused approach for PV of the critical unit operations.

The manufacturing of most cell-based ATMPs relies on manual aseptic practices, highlighting the importance of implementing robust risk assessment measures. Ensuring the adequacy of the aseptic validation program is paramount for each product, encompassing critical components such as aseptic operator qualification and comprehensive APS. This meticulous approach not only safeguards product integrity but also mitigates the inherent risks associated with manual aseptic techniques, thus upholding stringent standards of safety and efficacy in ATMP manufacturing.

The design of the PPQ is challenging for ATMPs. Companies must ensure that they continue to gain knowledge from the different elements that contribute to the sources of variability during process development and process characterization. Data from manufacturing for investigational drugs can continue to contribute to this knowledge, and understanding the variability and the risks will support efficient PV. Although it is acknowledged that there is inherent variability in the starting material, the PV should demonstrate that the final product can be produced through a consistent process and meets the quality characteristics.

Manufacturing changes for both investigational products and commercial products should have a risk assessment that indicates if this change can impact product quality. Based on the risk assessment and clinical stage of the product, the appropriate PV should be implemented to demonstrate comparability. The FDA recognizes that *“risk assessment for changes to the manufacturing of [cell and gene therapy] products may be more challenging than for other product types because the effects of manufacturing changes are often difficult to predict for these complex products.”* [4]

In summary, a risk-based approach is critical for the ultimate success of PV throughout the lifecycle of the ATMP and to support successful movement through the clinical phases to commercialization. The combination of variability from the starting material, the novel and complex manufacturing process, product quality attributes, and analytics pose a unique challenge for ATMP PV. Companies are highly encouraged to consider platform approaches, within both the investigational and commercial phases, to increase their efficiency and decrease time to market.

5.3 Cleaning

For ATMPs, the manufacturing process can include, or in some cases be predominantly composed of, open aseptic manipulations manually performed using a number of complex and critical materials. As facility capacity is expected to increase, the number of systems and operations taking place in a given space must be considered. Core to that assessment is the ability to routinely and reproducibly clean the equipment and surrounding environment to minimize any chance of contamination or cross-contamination. This risk becomes even greater when manufacturing involves personalized materials.

Especially with regard to personalized therapies, there is often a large number of kitted components and materials being transitioned to highly classified cleanrooms to support operations. ATMP manufacturing can include a variety of complex materials, require highly classified spaces and environmental equipment, and might require facilities to turn over and accommodate large numbers of batches per year in order to remain efficient and cost effective. As such, cleaning validation procedures of equipment and cleanrooms is integral to ensuring the integrity, safety, efficacy, and quality of these innovative therapies.

Although foundational principles of equipment and cleanroom validation remain consistent across the life science industry, ATMPs introduce unique considerations that demand an additionally rigorous approach. The variety of equipment typically found in ATMP manufacturing is sometimes atypical for the industry. Companies must plan for a multifaceted cleaning regime to handle critical infrastructure systems such as Biosafety Cabinets (BSCs), incubators, and isolators. Room cleaning performance must consider that oftentimes there may be single-use systems (e.g., cell culture flasks, bags, tube sets) conducting aseptic manufacturing steps still active within the space. Use of equipment-based cleaning and decontamination such as Vaporized Hydrogen Peroxide (VHP) must consider its compatibility with cell culture containers, which often have permeable membranes, screw top caps, or other surface complications.

For a high-batch-throughput facility, the building and operations program must facilitate the robust cleaning and decontamination of large numbers of kitted components from the warehouse all the way into the Grade A/ISO 5 aseptic operations environments. As the majority of these decontamination operations occur manually, companies must validate robust and highly repeatable Standard Operating Procedures (SOPs) for surface sanitizing of materials and kits.

ATMP manufacturing environments also necessitate stringent cleaning and disinfection protocols to prevent cross-contamination and maintain product quality. Unlike traditional biologics manufacturing, ATMPs often contain viruses, viral vectors and/or genetic elements, unique patient-specific or process-specific factors, and formulations that influence cleaning requirements and validation testing and that add layers of complexity to cleaning validation. Risk concerns could include biosafety and gene contamination that may be below the limit of detection to properly identify.

Companies need to demonstrate inactivation of genetic viral vector types used in their specific process, as well as a measurable standard of DNA destruction to ensure against cross-contamination and to limit the risk of genetic carryover between batches. When equipment surfaces are shared among products or patient-specific batches, it becomes crucial for companies to determine the acceptable limit of residual quantity that could impact a batch, especially when considering scaled production for individual patients. Additionally, assessing the risk associated with the detectability of residues in cleaning validation is essential. Given the heightened risk of cross-contamination and residue carryover in such scenarios, rigorous attention to cleaning protocols and validation methodologies is imperative to ensure product safety and quality standards are upheld.

For patient-specific manufacturing, a system or piece of equipment is often used and then cleaned and turned over to support the manufacture of the next patient lot. This high-throughput, patient-specific nature creates an increased emphasis on cleanliness and sterility to minimize the risk of contamination and cross-contamination and to ensure patient safety. Thorough validation of cleaning procedures—including surface disinfection, equipment sterilization, and Environmental Monitoring (EM) to meet the standards of cleanliness and sterility—is critical for aseptic operations conducted within environmental operations equipment such as BSCs or isolators. As systems may be required to process hundreds or even thousands of batches every year, downtime due to cleaning inefficiencies, confusing procedures, or qualification/requalification can cause direct implications for facility manufacturing capacities and the ability to serve patients in need.

Validation testing methods in ATMP cleaning validation may encompass a comprehensive array of approaches, including analytical techniques, surface sampling and swabbing, and microbial monitoring to assess the effectiveness of cleaning procedures. In some cases, companies use advanced analytical techniques such as next-generation sequencing to identify potential process residuals and cross-contamination. These methods must be sensitive enough to detect minute traces of contaminants while being compatible with the delicate nature of ATMPs. Additionally, validation studies should encompass worst-case scenarios to challenge the cleaning processes effectively and validate their robustness under varying and evolving conditions to properly reflect the manufacturing state.

Overall, companies should develop a comprehensive CCS, with cleaning and disinfection procedures and practices that are tailored to the unique challenges and complexities of the materials, potential patient-specific components, and processes performed. Developing this control strategy requires a deep understanding of the unique challenges posed by these innovative therapies and a commitment to implementing rigorous validation practices tailored to the intricacies of ATMP manufacturing. By implementing robust validation protocols and maintaining a commitment to cleanliness and sterility, stakeholders will be well situated to manage their manufacturing program.

5.4 Supply Chain

Supply chain validation demonstrates the consistent delivery of therapies to patients while maintaining safety, efficacy, and quality attributes. For ATMPs, supply chain validation is an important element from the collection of starting material (e.g., from patients, healthy donors, or cell banks) through the manufacturing processes, all the way to patient treatment. Ensuring a robust supply chain in ATMPs is crucial to ensuring timely and safe delivery of therapies to patients. Because ATMPs are often a transformative medicine, it is imperative that there is timely delivery of the drugs to patients who are often critically ill. In addition, there can be strict timelines for delivery of the drugs to the patients based on the treatment schedule or clinical protocol.

Personalized medicines in ATMPs have added complexity to the supply chain with the necessity to ensure COI and COC from patient enrollment through to treatment of the patient (vein-to-vein). Ensuring COI/COC throughout the supply chain is critical to ensuring patient safety in both autologous therapies and allogeneic therapies. For autologous therapies, where the medicine uses the person's own starting material, the supply chain needs to ensure the return of the medicine to the same patient. For allogeneic therapies, the starting materials come from a different donor, but there may be requirements related to the allowable types of donors to ensure patient safety. Given the criticality of maintaining COI from vein-to-vein and the complexity of the supply chain (often using multiple supply chain routes), it is recommended that companies consider automated systems to track along the supply chain. Companies should perform risk assessments for the COI/COC processes and ensure supply chain mitigations are in place, as these are critical to the safety of a patient.

Although the ATMP industry has grown significantly, a large portion of the reagents, materials, or equipment used continue to be single sourced. If there is a disruption in the supply, this could cause a manufacturing and supply issue to patients in need of the transformative medicines they require. Therefore, logistics plays an even more critical role in the supply of ATMPs.

Amongst the biggest challenges in the ATMP supply chain are the temperature sensitivity and short shelf life of both starting materials and products. Once starting material is collected from patients or healthy donors, there is a time sensitivity for shipment to the manufacturing site. Certain ATMPs have a short shelf life from when the product is manufactured to when it must be administered, even as short as six hours. The temperature conditions that are required can vary depending on the starting materials or product. In addition, there may be movement of materials within the manufacturing process from one location to another. When determining the supply chain validation requirements, the various temperature requirements and time requirements for the end-to-end supply should be considered.

It is critical for ATMPs to have a resilient supply chain that can prepare for and adapt to unexpected events, providing flexibility and operational agility. Companies should create a robust and risk-based plan that supports a resilient supply chain during clinical trials, which can easily be scaled up as they move into commercial operations.

6 Analytical

6.1 Overview

Analytical development in the field of ATMPs is particularly challenging due to the rapid pace of innovation and the necessity for new methods tailored to unique mechanisms and targets. For cell-based ATMPs, ensuring product characterization and QC is crucial to consistency and reproducibility, given the inherent variability in cell composition, proliferation ability, viability, and antigen recognition capabilities. These attributes are often influenced by the personalized nature of the treatments and the differing donor sources required for larger patient populations.

Multiple assays are developed throughout the process to measure diverse characteristics, with each batch requiring validated QC testing to demonstrate assay suitability, precision, and reproducibility. According to ICH Q6B [57] and Ph. Eur. [52] reference methods, typical QC release assays for biological products include product identity, purity, and impurities; product quantity and/or potency; and safety. Manufacturers continually refine and streamline these assays, improving the manufacturing process and product quality, and gaining deeper insights into the CQAs and process characteristics specific to ATMPs.

Clear guidelines and robust QC measures are essential to establishing standardized approaches that ensure the integrity and comparability of analytical testing across different stages of product development and regulatory approval. Table 6.1 summarizes the guidelines associated with analytical methods development and validation, providing essential guidance for a consistent and standardized approach needed in this evolving field.

Table 6.1: Guidelines Associated with Analytical Methods Development and Validation

Regulatory Authority or Organization	Document	Year
FDA [2]	Guidance for Industry: Chemistry, Manufacturing, and Control (CMC) Information for Human Gene Therapy Investigational New Drug Applications (INDs)	2020
	Guidance for Industry: Considerations for the Development of Chimeric Antigen Receptor (CAR) T Cell Products	2024
	Guidance for Industry: Content and Review of Chemistry, Manufacturing, and Control (CMC) Information for Human Somatic Cell Therapy Investigational New Drug Applications (INDs)	2008
	Guidance for Industry: Potency Tests for Cellular and Gene Therapy Products	2011
	Draft Guidance for Industry: Potency Assurance for Cellular and Gene Therapy Products	2023
	Guidance for Industry: Analytical Procedures and Methods Validation for Drugs and Biologics;	2014
EMA [3]	Guideline: Quality, Non-Clinical and Clinical Requirements for Investigational Advanced Therapy Medicinal Products in Clinical Trials	2025
	Guideline: Requirements for Quality Documentation Concerning Biological Investigational Medicinal Products in Clinical Trials	2022
	Guidelines on Good Manufacturing Practice Specific for Advanced Therapies (EudraLex Volume 4 Part IV)	2017
	Guideline: Quality, Non-Clinical and Clinical Aspects of Medicinal Products Containing Genetically Modified Cells	2021

Table 6.1: Guidelines Associated with Analytical Methods Development and Validation (continued)

Regulatory Authority or Organization	Document	Year
ICH [23]	Q2(R2) Validation of Analytical Procedures	2023
	Q14 Analytical Procedure Development	2023
USP [53]	<1220> Analytical Procedure Life Cycle	2022

Sponsors face significant challenges during the development and manufacturing of cell therapy products, particularly due to the limited availability of analytical tools for QC and process changes. The variability of patient samples, small lot sizes, potency requirements, and product stability further complicate the selection of analytical tests necessary for characterizing and releasing cell therapies. Defining the CQAs of cell-based ATMPs presents unique difficulties due to the complexity of live cells, which can differ significantly from one product to another.

Unlike biologics—which use multiple assays to characterize product identity, potency, purity, and quality—ATMPs require a more intricate approach to understand complex biological mechanisms and patient-specific variations. Consequently, the development of numerous assays throughout the process is essential to measure various physical and biological characteristics [58]. As manufacturers streamline the manufacturing process, enhance product quality, and deepen their understanding of the CQAs and process characteristics, they refine existing assays, introduce new assays, and streamline the test panel accordingly.

As cell-based methods evolve to support gene therapies and assess product potency through gene or protein expression, the growing complexity and variability in assay readouts raise important questions about best practices in method development and analysis. Ensuring the accuracy, reliability, and reproducibility of these assays becomes paramount as their sophistication increases. Stakeholders must navigate challenges such as selecting appropriate cell lines, optimizing culture conditions, standardizing assay protocols, and rigorously validating analytical methods.

Additionally, the dynamic nature of gene therapies necessitates the ongoing adaptation of these assays to accommodate new therapeutic modalities and technological advancements. Personalized therapies introduce an additional layer of complexity, requiring tailored analytical methods capable of addressing individual variations in patient responses and treatment outcomes. This shift necessitates a departure from the uniformity and mass production typical of conventional drug manufacturing processes.

6.2 Cell Therapy Testing

Developing CBMP involves navigating significant challenges due to the inherent variability and dynamic nature of cells. Analytical tests are pivotal throughout the development of both stem-cell-based and non-stem-cell-based therapies. They ensure comprehensive characterization and implementation of QC measures. These tests assess critical aspects such as cell viability, identity, potency, purity, and safety profiles, thereby ensuring the safety of patients and the effectiveness of treatments. Table 6.2 provides a summary of analytical tests used to characterize cell therapies [59–62].

Table 6.2: Analytical Testing for Characterization of Cell Therapies

Purpose	Cell Type	Test
Identity	Pluripotent Stem Cells	- Molecular: Short Tandem Repeat (STR) genotyping - Surface markers: SSEA-3, SSEA-4, TRA-1-60, TRA-1-81 - Transcription factors: Nanog, SOX-2, OCT-4 - Phenotypic: Colony formation assays, EB formation assays
	Multipotent Stem Cells	- Surface markers: Lin-, CD34+, CD38-, CD90+, CD45 (HPC); CD73+, CD90+, CD105+, CD34-, CD45-, CD11b-, CD14-, CD19-, CD79α- (MSC) - Functional assays: Colony formation, differentiation potential
	Unipotent Cells	- Cell-specific surface markers - Morphological evaluation - Additional tests: Karyotype analysis, SA-β-Gal assay
	Somatic Cell Therapies	- Surface marker determination - DNA fingerprinting (STR) - Species determination - Morphological examination - Bioassay - Biochemical marker analysis
Purity	Pluripotent Cells (iPSCs, ESCs)	- Fluorescently or magnetically activated cell sorting - Metabolic selection - Density gradient separation
	Multipotent Stem Cells (HPCs, MSCs)	- Surface marker-based sorting (e.g., CD34+, CD45+ exclusion) - Functional assays (e.g., differentiation potential)
	Unipotent Cells	- Transplants: Identification and exclusion of non-target cells in donor tissue - CAR T therapies: Antibody-based T cell enrichment (CD4+/CD8+ antibodies on bead conjugates)
Sterility		- In-process sterility testing during manufacturing to detect bacterial and fungal contaminants - Consideration of rapid testing methods due to lengthy standard incubation periods - Tests for HIV types 1 and 2, hepatitis B and C, human TSE, syphilis, and additional tests as necessary - Mycoplasma contamination monitoring and rapid testing strategies
Potency		- HPCs: Total nucleated cells, CD34+ cells, and cell viability - CD19+ T cell immunotherapy: Cell viability and CAR expression - Chondrocytes: Expression of Hyaline1 - Keratinocytes and fibroblasts: No known mechanism of action; cytokine assay - Postauricular fibroblasts: No known mechanism of action; cell number - CD54+ cell immunotherapy: Expression of CD54 on antigen-presenting cells after activation
Safety	Particulate Testing	- Visual inspection: > 100 µm - Light microscopy, light obscuration, flow imaging microscopy, Coulter principle, flow cytometry: 1 µm to 1 mm - Dynamic Light Scattering (DLS), static light scattering, Aggregate Ultracentrifugation (AUC): 1 nm to 1 µm - SDS-PAGE, SEC-HPLC: < 1 nm to 100 nm

Table 6.2: Analytical Testing for Characterization of Cell Therapies (continued)

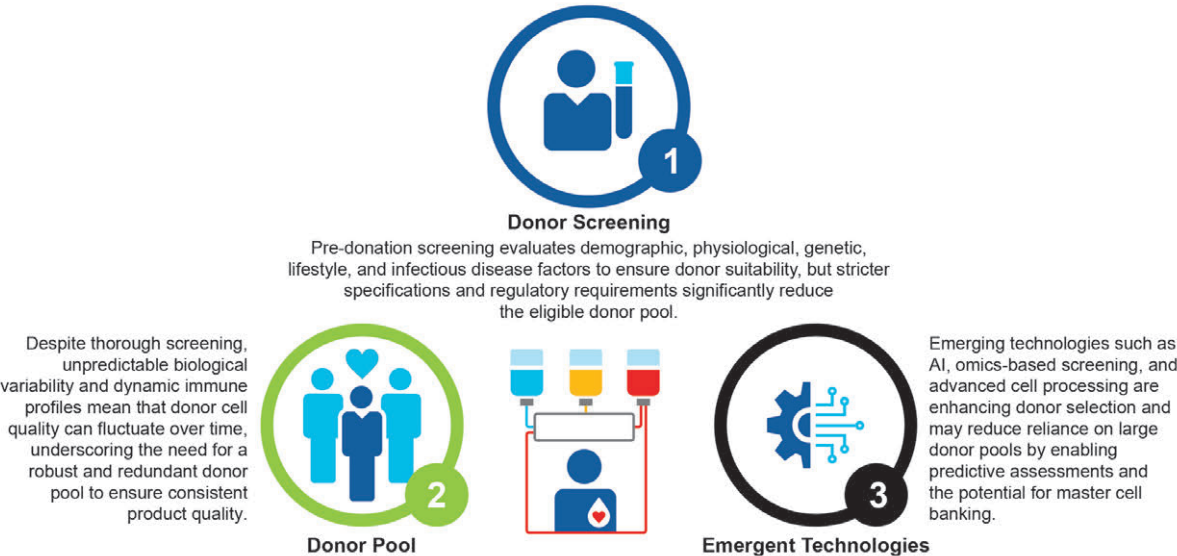
Purpose	Cell Type	Test
Safety (continued)	Viral Testing	- Cell-based assays: Assessing the ability of the vector to infect and transduce cells - PCR and sequencing: Identifying the presence and characteristics of the viral vector - Western blotting: Identifying the presence and composition of viral proteins - Replication Competent Lentivirus (RCL): The presence of RCL events in vector batches increases the risk of insertional mutagenesis due to potential ongoing viral replication and multiple integrations into patient's cellular DNA: the assays involve detection of an enzyme found in a retrovirus

6.2.1 Patient/Donor Traceability and Testing

In cell therapies, maintaining comprehensive records regarding patient/donor information, manufacturing processes, and processing facilities is paramount to ensure the safety and efficacy of treatments. Establishing a robust COI is integral for traceability throughout the entire supply chain. The COI system links essential identifiers and data to specific patients or donors and encompasses thorough testing of the donor's starting material and medical history [63]. Typically, donor material undergoes rigorous screening for infectious diseases and exposure to infectious agents at molecular, antigen, and antibody levels to mitigate the risk of disease transmission through the cell therapy product. Screening requirements for all allogeneic cells and tissues include testing for HIV types 1 and 2, hepatitis B and C, human TSE, syphilis, vaccinia virus, West Nile virus, and SARS [64]. These stringent measures ensure the safety and quality of cell therapy products, safeguarding patient health and treatment outcomes.

Donor optimization is crucial in cell therapy because the quality and characteristics of donor cells directly impact the safety, efficacy, and consistency of the final therapeutic product. Variability in donor health, genetics, and immune profiles can lead to differences in cell potency, expansion capacity, and functional performance, which in turn affect treatment outcomes and manufacturing reliability. Optimizing donor selection helps reduce the risk of immune rejection or adverse events, ensures reproducibility across batches, and improves manufacturing efficiency. In allogeneic therapies, choosing donors with favorable traits such as low immunogenicity also supports broader patient compatibility and commercial scalability. Challenges in donor optimization and emerging solutions are shown in Figure 6.1 [65].

Figure 6.1: Challenges in Donor Optimization and Emerging Solutions



6.3 Gene Therapy Testing

Gene therapy products, specifically viral vectors such as Adeno-Associated Virus (AAV), retrovirus/lentivirus, and adenovirus, require unique testing and characterization methods that cannot always be accomplished with traditional approaches. Sponsors use a combination of routine and customized methods to fully analyze the identity, potency, safety, and stability of their new therapies. However, full method development and validation can extend timelines. The viral vector is a critical component of gene therapies, and its manufacture requires in-process analysis.

Lentivirus and AAV, the most commonly used vectors, are well-characterized but their complexity and potential variability during production pose challenges for end-product and in-process analytics [32]. To produce safe, consistent, and high-quality gene therapy products, robust, scalable, and optimized production and purification strategies are needed. These goals cannot be achieved without similarly optimized analytics for monitoring key CQAs related to the product's physical integrity and therapeutic objectives.

An overview of analytical methods employed in viral vector manufacturing to assess various quality attributes essential for product safety and efficacy is included in Table 6.3 [66, 67].

Table 6.3: Analytical Methods Employed in Viral Vector Manufacturing

Quality	Attribute	Technique
Identity	Confirm presence and identity of viral vector	SDS-PAGE, MS, Western blotting, genome sequencing, and PCR to ensure the correct viral strain is present
Potency	Physical viral titer	DNA hybridization, real-time PCR, optical density measurements
	Functional viral titer	Plaque-forming assays, fluorescence foci assays, and TCID50
Purity	Process-related impurities	MS, chromatography, TEM
	Host cell-related Impurities	PicoGreen, DNA Threshold assay, qPCR
	Host Cell Proteins (HCPs)	ELISA, TEM
	Capsid content (empty: full capsids)	TEM, AUC
Safety	Sterility	Standard sterility tests (Ph. Eur. 2.6.1 [52], USP <71> [53])
	Endotoxin	LAL method (Ph. Eur. 2.6.14 [52], USP <85> [53]), rabbit pyrogen assay
	Mycoplasma	PCR, Cell-cultured-based assays
	Replication competent virus (presence of rep or cap sequences)	Southern blotting, qPCR, co-cultivation assay, Next-Generation Sequencing (NGS)
	Adventitious agents	<i>In vivo</i> and <i>in vitro</i> assays
Physical Testing	pH	Potentiometry
	Osmolality	Osmometry
	Aggregate formation	Light microscopy, DLS, SEC-MALS, TEM, AUC, FFF-MALS

6.3.1 Cell Line Characterization

Cell line characterization assays are essential to ensure that the cell lines used in manufacturing are well-defined, free from contaminants, and genetically stable. Comprehensive authentication of cell lines, involving phenotypic or genotypic identity testing, is necessary when producing gene therapies with viral vectors.

Cell line characterization is performed at various stages, including Master Cell Banks (MCBs), Working Cell Banks (WCBs), and End-of-Production Cells (EPCs) or Cells at the Limit (CAL) of *in vitro* cell age used for production. Current regulations require the consideration of several aspects during cell line characterization. These aspects include the origin and history of the cell line, cellular morphology and growth characteristics, cellular identity, purity of the cell line (including absence of contaminating cells, microbial contamination, and adventitious viruses), tumorigenicity and oncogenicity, and genetic stability.

6.3.2 Identity

The identity of AAV vectors must be periodically checked during development to ensure the correct serotype and product configuration are being used. Immunoassays such as Western blotting and Enzyme-Linked Immunosorbent Assay (ELISA) are commonly employed for identity confirmation. Western blotting is considered more specific for confirming protein identity, as it provides information on the size and presence of specific capsid proteins using serotype-specific antibodies. In contrast, ELISA is well-suited for quantitative measurements and high-throughput screening, making it useful for large-scale lot testing. Additionally, MS-based proteomics is emerging as a powerful tool for vector identity confirmation, offering high resolution and detailed characterization of capsid protein composition.

Verifying the identity of the Gene of Interest (GOI) is a crucial step in ensuring the safety and efficacy of gene therapies, particularly in AAV-based therapies, where the GOI is directly tied to the therapy's mechanism of action. Accurate identification confirms that the correct genetic sequence is present and that no unintended mutations have occurred during vector construction or production. Common methods for GOI identity testing include PCR, Sanger sequencing, and Next-Generation Sequencing (NGS). Sanger sequencing remains a gold standard due to its precision and widespread regulatory acceptance. However, NGS is increasingly favored for its speed, cost-effectiveness, and ability to detect low-frequency variants. PCR-based assays also play an important role, particularly for initial identity confirmation and GOI quantification [32].

6.3.3 Purity

Impurities in vector products can originate from both the host cell system and downstream purification steps. Host-cell-derived contaminants include residual proteins (HCPs) and nucleic acids, whereas process-related impurities may result from components of the culture medium or purification reagents. Defining acceptable levels of these impurities early in process development is crucial to ensure the production of high-quality vectors. A range of analytical tests is employed to monitor impurity levels, including assays for HCPs, host cell DNA, and common process contaminants. For preclinical studies, these assessments should be efficient and cost-effective, whereas clinical-grade materials require more extensive and rigorous testing to meet regulatory standards. Techniques commonly used to evaluate purity include SDS-PAGE, capillary electrophoresis, Quantitative PCR (qPCR) or digital PCR, ELISA, HPLC, and MS. Table 6.4 summarizes the analytical methods used to detect process impurities and product-related substances [68, 69].

Table 6.4: Tests Used for the Detection of Impurities for Gene Therapies

Impurity	Type of Impurity	Test
Partially filled capsid	Product-related substance	A260/280 ratio, TEM, AUC, qPCR
Empty/full/partial capsid	Product-related substance	Mass Photometry (MP)
Empty capsid	Product-related substance	TEM, AUC, qPCR
Overfilled capsid	Product-related substance	TEM, AUC, qPCR
Sequence variants	Product-related substance	DNA sequencing, NGS
Virus aggregates	Product-related substance	DLS, Size Exclusion Chromatography with Multi-Angle Light Scattering (SEC-MALS)
Replication-competent AAV	Product-related substance	PCR, co-culture, NGS
Protein impurities	Process impurity	SDS-PAGE- Silver stain
Residual HCP (HEK-293, Sf9)	Process impurity	ELISA
Residual host cell DNA	Process impurity	qPCR
Residual plasmid DNA	Process impurity	ddPCR, qPCR
Residual baculovirus DNA	Process impurity	ddPCR, qPCR
Residual Triton X100	Process impurity	HPLC
Residual Benzonase	Process impurity	ELISA, quantification of Benzonase's ability to digest DNA (MS, gel electrophoresis, qPCR)
Affinity ligand	Process impurity	ELISA

6.3.4 Safety

Safety testing aims to ensure that process intermediates or end products in gene therapy are free from impurities that could harm patients. Critical parameters such as sterility, endotoxin levels, mycoplasma contamination, adventitious viruses, and replication competent viruses are rigorously assessed to achieve this goal. Sterility testing follows USP [53] or Ph. Eur. [52] recommendations to detect bacterial or fungal contamination. Endotoxin levels are commonly determined using the Limulus Amebocyte Lysate (LAL) assay or the Recombinant Factor C (rFC) assay, the latter of which is increasingly preferred in regulatory settings for its ability to reduce animal use. The rabbit pyrogen test, once a standard, has been largely replaced by rFC due to ethical and reproducibility considerations [70]. Mycoplasma contamination is detected through PCR-based methods or cell culture assays, with testing performed at critical stages of manufacturing.

The presence of replication competent viruses is evaluated using cell-based assays, Southern blotting, or qPCR, aiming to reduce levels to below one replication competent AAV per 10⁸ vector genomes. Adventitious agent testing employs a holistic approach, including *in vitro* infectivity assays, nucleic acid-based amplification, and antibody production tests, with NGS replacing animal-based detection methods. Raw materials undergo proportional testing based on risk, with special attention to high-risk materials for *ex vivo* cell therapies.

6.4 Potency Testing for ATMPs

6.4.1 Cell Therapy Products

CBMPs are intricate, often containing multiple active ingredients, complicating the identification of components contributing to potency. Potency testing, which is crucial for product quality, consistency, and efficacy, requires multiple assays to account for potential ingredient interference or synergy. This testing begins in preclinical and early clinical stages by identifying the critical biological activity and formulating a potency determination approach. A potency assay matrix may include viable cell count, *in vitro* differentiation, target-specific cytotoxicity or cytokine release, surrogate functional biomarkers, and biological activity, all scientifically justified. Given production complexities and donor genetic variability, validating these assays before Phase III clinical trials is essential to manage variations and ensure therapeutic effectiveness.

Potency testing of cellular therapies presents several challenges [71]. First, the patient-specific nature of most cellular therapies means that there is often a limited amount of suitable source material, which presents a challenge for lot release and potency testing. Administering large quantities of the product for testing purposes can reduce the available dose and clinical effectiveness of the product, thus limiting the number of patients that can be treated. Therefore, it is essential to develop potency assays that are both sensitive and efficient. Also, the need for comprehensive testing needs to be balanced with the limited availability of source material to minimize the impact on patient outcomes while maintaining regulatory compliance.

Second, cellular therapies require immediate testing after production but before clinical use due to the sensitivity of living cells to environmental factors. This presents a challenge in terms of developing rapid and reliable potency assays to ensure the product's safety and efficacy. Additionally, careful handling and storage protocols must be implemented to minimize any negative impact on the potency of the final product. Some cellular therapies require administration within hours of production completion, leaving very little time for testing and QC, making the development of suitable potency assays all the more crucial.

Third, a major challenge in potency testing for cell therapies is the lack of a clear reference standard, making it hard to compare results across batches. To address this, well-characterized assay controls—ideally made from the same process and similar to the test sample—should be used to ensure the assay works reliably. These controls help separate product variability from assay variability and support consistent testing over time. Even without a standard, potency results should be as quantitative as possible to help define product specifications and meet regulatory expectations.

Finally, cellular therapy products can exhibit lot-to-lot variability due to differences in starting materials, production processes, and environmental conditions, as well as genetic polymorphisms in donors. This can complicate potency testing, making it important to carefully select starting materials and standardized manufacturing processes to minimize variability and ensure consistent potency.

A modern, risk-based approach to potency testing for ATMPs emphasizes the use of surrogate assays, which are typically faster, less variable, and more practical than traditional bioassays. Rather than capturing the full mechanism of action in a single test, surrogate assays measure molecular or biochemical indicators that are physiologically linked to the product's functional activity and that may serve as predictors of clinical outcomes. These assays are particularly useful when the mode of action is complex or not fully understood, or when time constraints—such as those during DP release—make functional assays unfeasible.

Surrogate measures can include protein expression, surface markers, or viable cell counts, and may not always be bioassays. The relationship between these surrogate assays and product performance should be confirmed through bridging studies, which help validate their relevance and ensure that product changes, such as degradation or impurity levels, do not negatively impact function. In some cases, a matrix of multiple assays may be necessary, using bioassays for characterization and stability testing and surrogate assays for lot release [72].

Potency testing strategies can also include both *in vivo* and *in vitro* models. *In vivo* animal studies can directly measure functional activity but often face limitations in relevance, scalability, and ethical concerns. In contrast, *in vitro* assays serve as practical surrogates by capturing cellular or biochemical responses. For example, flow cytometry is a widely used *in vitro* method to assess specific surface antigens, although it can be limited by factors such as antibody availability, detection thresholds, and sample quantity. Establishing a strong correlation between *in vitro* assay results and the intended biological activity is critical for assay validation, as it supports the biological relevance of the assay and ensures it meaningfully reflects the product's clinical function. In cases where the mode of action is unclear, simple metrics such as viable cell count may be used as a proxy for potency. However, due to the diverse nature of cellular therapies and the current lack of harmonized regulatory standards, potency assays must be tailored to each product, with careful consideration of the assay's purpose, feasibility, and predictive value [73].

6.4.2 Gene Therapy Products

The assessment of physical and infectious titers is paramount for QC and dosing of viral therapeutics. Physical titer quantifies the total number of viral particles per mL using methods such as DNA hybridization and real-time PCR-based assays, despite assay variability. Infectious titer, measuring viral particle infectivity, is crucial and typically lower than physical titer. This is expressed as transducing or plaque-forming units per mL and determined by assays such as viral plaque assays. For recombinant AAV (rAAV) vectors, quantifying genome and capsid titers is essential, with PCR-based methods standard for genome titer and antibody-based or optical methods for capsid titer.

In vivo gene therapy products rely on *in vitro* potency assays reflecting mechanism of action, measuring functional effect or activity of transgene sequences in dose-dependent manners. These assays assess restoration of cellular function, tumor cell cytotoxicity, cytokine production, and cellular activation, providing insights into therapeutic efficacy. *Ex vivo* gene-modified cell therapy products require robust potency assays, ensuring product strength and stability. These assays, including NGS and Droplet Digital PCR (ddPCR), assess various target functions such as cellular function restoration and gene editing efficiency, facilitating consistent product quality throughout clinical studies [73].

Potency assays for *in vivo* gene therapies involve *in vitro* assessments monitoring target gene or protein expression levels, gene copy number, and inhibition marker expression. These are crucial for therapeutic efficacy assessment in *in vivo* settings. These assays provide valuable insights into therapy impact on molecular pathways and cellular functions, aiding in therapeutic optimization and efficacy evaluation [73].

6.5 Rapid Microbial Testing for ATMPs

Manufacturers of ATMPs face persistent challenges in developing effective microbiological QC testing approaches, particularly for products with a short shelf life. These therapies require constant monitoring to detect contaminants that could compromise both product quality and patient safety. Addressing these challenges requires early evaluation of modern QC testing methods within the product development process, ensuring that innovative treatments are not only effective but also safe for patient use. The adoption of rapid QC techniques offers several advantages, including enhanced control over the manufacturing process. These methods provide faster results compared to traditional approaches, enabling prompt action in response to deviations from set specifications. Testing critical control points using rapid QC methods enables manufacturers to ensure that the final product meets required specifications, even when released under risk conditions. This ability is particularly valuable for short shelf life products, where timely release is essential to meet patient needs [74].

Rapid detection technologies for microbial QC include Adenosine Triphosphate (ATP) bioluminescence, nucleic acid amplification, respiration, and solid phase cytometry. While most contaminants are detected quickly, slow-growing organisms can still take several days, though this is significantly faster than compendial methods. To ensure both product safety and process efficiency, setting a comprehensive CCS is crucial. This involves not only rapid sterility testing of the final product but also implementing rapid bioburden testing throughout the manufacturing process, from raw materials to intermediates. Quantitative bioburden results provide early indications of contamination levels, enabling corrective actions before the final product is completed. If bioburden levels are consistently within acceptable limits, it may even justify validating rapid sterility tests for earlier product release [75].

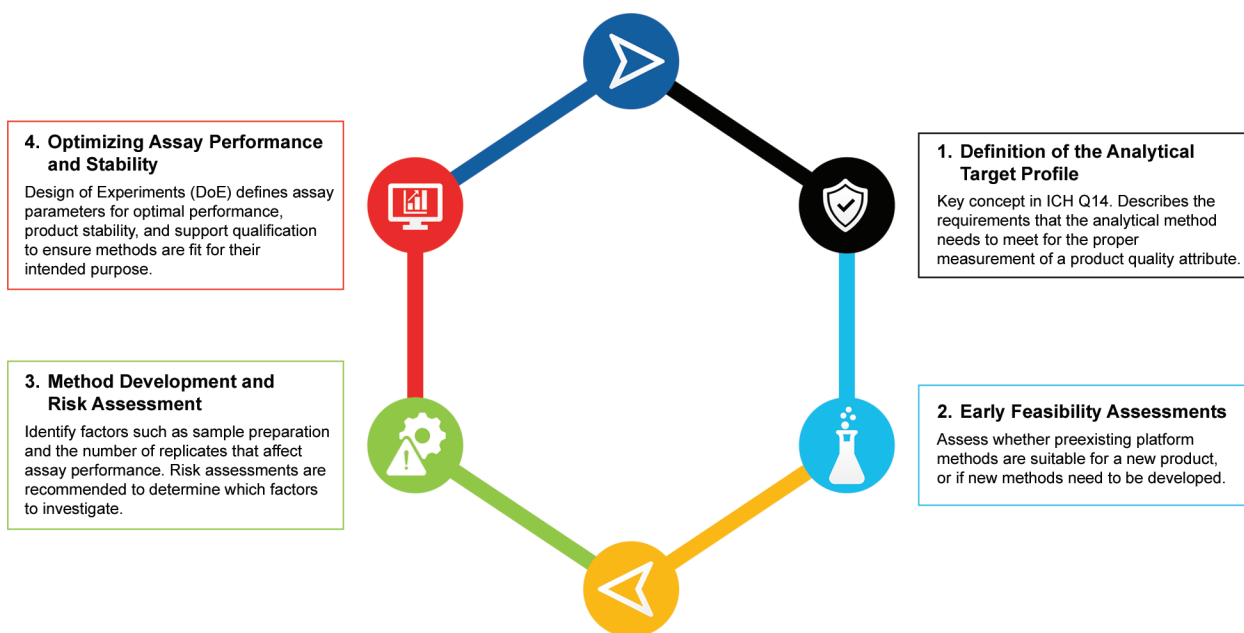
Rapid sterility testing emerges sometimes as a viable alternative to traditional methods outlined in USP <71> Sterility Tests [53] and also as a tool to facilitate conditional product release. Unlike conventional testing, which requires incubation periods of 14–18 days, rapid sterility testing can yield results in as little as 6 days, representing a significant reduction in testing time of 8–12 days. This expedited timeframe offers numerous advantages, including improved supply chain management, timely product release schedules, and quicker investigation of contamination events. Additionally, rapid sterility testing as covered in USP <1071> [53] supports conditional release strategies, allowing products to enter the market with a high level of confidence in sterility while awaiting final confirmation. Molecular methods further enhance this approach by providing results within hours, enabling near-real-time decision-making and immediate responses to potential sterility concerns. Through advanced technology, organizations can rapidly detect microbial growth, surpassing the capabilities of traditional visual methods.

6.6 Analytical Method Validation for ATMPs

Analytical method validation for ATMPs is essential due to their complexity and variability, requiring robust confidence in analytical measurements. As ATMPs progress to late-phase trials and regulatory approval, validation packages will face rigorous scrutiny, highlighting the importance of a comprehensive validation approach. Implementing a risk-based strategy early in the development process enables identifying and prioritizing the critical variables impacting assay performance, guided by existing knowledge, experience with similar methods, and input from development and operations teams.

Applying QbD principles offers a systematic framework emphasizing risk assessment, planning, scientific rationale, and empirical testing to build product and process understanding. This approach enhances assay variability understanding, ensures fit-for-purpose validation tailored to the assay's intended use, and establishes essential procedures for tracking performance. Integrating risk assessments and QbD principles allows manufacturers to control impactful variables, thus improving the robustness and reliability of ATMP analytical methods and facilitating successful regulatory approval [76]. Figure 6.2 describes a risk-based method lifecycle to ensure analytical methods are suitable.

Figure 6.2: Risk-Based Analytical Method Lifecycle



6.6.1 Assay Selection

The selection of test methods for cell therapies depends on their intended use and the data they aim to provide. Research methods may prioritize the relevance and availability of instruments, often requiring minimal assay development. Process development studies, internally controlled, may reduce the need for detailed assay characterization. However, product release assays demand specificity, sensitivity, quantitative capability, accuracy, precision, and robustness. As products near commercialization, practical considerations such as ease of use, throughput, hands-on time, and cost become essential. The extensive time and expense required for assay characterization and validation, essential for regulatory approval, necessitates careful assay selection [58].

The diversity in cell therapies, clinical indications, manufacturing processes, and assay technologies contributes to the absence of standard assays in the field. Analytical scientists need to consider what to measure and how to measure it based on intended use. They can select from various cellular properties, including physical, biochemical, and cellular responses. Platform choices, such as ELISA and flow cytometry, impact assay development complexity. Flow cytometry, common but complex, may require significant effort, especially for validation.

A strategy for assay selection involves identifying the simplest measure that provides necessary information. Consideration must be given to assay complexity and the requirements for validation. For instance, comparing the stability assessment of a cellular product using two analytical methods revealed different outcomes. Although cell viability displayed a decline after three days, cell functionality dropped after just one day. Despite the superior sensitivity of the functionality assay, its complexity and variability necessitate careful consideration for validation. The validation paths for simple and complex assays differ significantly, requiring analytical teams to determine the additional work needed for successful validation and control [77].

6.6.2 Considerations for Validation of Analytical Tests

Assay validation is the process of demonstrating the suitability of an analytical procedure for its intended application. This involves analyzing or verifying several assay parameters, including specificity, linearity, accuracy, and precision. A comprehensive validation involves extensive laboratory testing, but some of this work can be initiated during the assay's development and optimization phases. Trend data from positive controls and/or reference standards can be used for routine tests, but no trend data alone can sufficiently demonstrate the parameters of the assay. Several principles should be followed when proceeding toward an assay validation, including having qualified personnel, validated and qualified equipment, and knowledge of the assay's characteristics by the person designing the validation plan.

The validation of analytical tests for ATMPs, including gene and cell therapies, presents unique challenges stemming from the heterogeneity of cellular products, complexity of manufacturing processes, and evolving nature of regulatory expectations. As these therapies advance toward clinical trials and commercialization, establishing confidence in the accuracy, reliability, and consistency of analytical measurements becomes increasingly critical. However, the absence of standardized pharmacopeial methods and well-defined validation pathways—especially for novel cell and gene therapy assays—requires a comprehensive, risk-based approach. Traditional one-factor-at-a-time studies often fall short in capturing the full range of assay variability, underscoring the need for QbD principles that integrate risk assessment, empirical testing, and holistic planning.

Fit-for-purpose validation aligns the level of rigor with the assay's intended use, ensuring appropriate resource allocation without compromising data integrity. This approach becomes especially important as gene therapies near pivotal trials, where early-phase qualifications must evolve into full validations to meet regulatory expectations. Usability in QC laboratories, particularly for complex bioassays with long turnaround times, must also be considered, and orthogonal methods may offer practical alternatives to support method robustness [31, 34].

6.6.3 Evolution of Analytical Assays

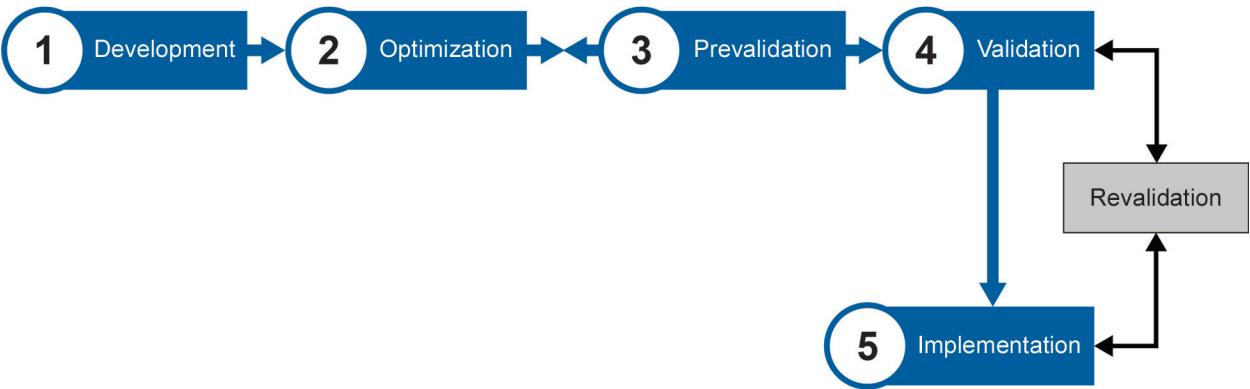
Assay development for a product should begin as soon as the product is available, and the requirements for assay performance change as the product development progresses. Although qualitative or semiquantitative methods may be acceptable for preclinical and Phase I studies, quantitative chromatographic methods are recommended for Phase II and Phase III. The extent and level of the validation package needed are determined by the product developmental phase. Biological assays—which measure the response of a living cell or organism induced by a drug or other stimulus—are particularly challenging to validate due to their laborious nature and large amounts of noise in the data obtained. Examples of bioassays include apoptosis, cell proliferation, cell migration, cell secretion, cell stimulation, gene expression, and cell function/inhibition assays.

Clinical trial design approaches for ATMPs, however, are often different from standard clinical trials for biologics and small molecules. ATMPs tend to target serious or life-threatening conditions or to address unmet medical needs, sometimes qualifying for orphan designation. It is common for ATMPs to combine Phase I/Phase II clinical trials to establish safety and efficacy and to avoid trials with healthy patients. Phase III trials continue the assessment of safety and efficacy on a larger number of patients, though they typically involve fewer participants than standard clinical trial designs. Companies may also execute a Phase I trial followed by a Phase II pivotal trial. The industry acknowledges that traditional trial progression and execution approaches may not always suit ATMPs, necessitating extra care in validation programs.

The revalidation of an assay is crucial once the initial validation is completed, as depicted in Figure 6.3. This process involves revisiting specific parts or the entire validation procedure due to scheduled timelines, changes in manufacturing, alterations in the product or analytical methods, or updates to equipment or software. Assays must undergo validation or revalidation before they are used routinely, especially if the conditions under which they were initially validated no longer apply, or when modifications exceed the original scope. Revalidation highlights the importance of maintaining accuracy and reliability in analytical testing, ensuring the assay continues to perform as intended under evolving conditions.

By periodically reassessing and confirming the validity of the assay, manufacturers can address any discrepancies or shifts in the testing environment promptly. This iterative process guarantees that analytical methods remain robust and dependable, adapting to changes in manufacturing practices, products, and analytical procedures, as well as technological advancements in equipment or software. Figure 6.3 illustrates this principle, emphasizing the necessity of revalidation for sustaining high standards of precision and reliability in assay processes post-initial validation.

Figure 6.3: Overview of Assay Revalidation



6.6.4 Challenges in the Validation of Analytical Assays

Analytical test methods for ATMPs present a multifaceted challenge due to their complexity and reliance on specialized techniques and instrumentation, often leading to high assay variability. Various factors contribute to this variability, including limited testing instances during validation and routine testing, outdated assay platforms, patient material variability, evolving manufacturing processes, and evolving regulatory guidance. To address these challenges, substantial resources in terms of material, time, and repetition are essential, alongside the application of empirical and statistical methods. Upgrading to modern assay platforms, implementing automation, and developing robust process control strategies can also play crucial roles in mitigating variability. However, regulatory expectations regarding CQA requirements and allowable variability in support of related methods are still evolving, adding another layer of complexity [58].

Another significant challenge stems from the unfamiliarity of ATMP assays and the limited data available to establish a comprehensive understanding of acceptable assay ranges and their correlation with clinical efficacy and safety. Furthermore, the unique nature of gene therapies may require novel R&D instrumentation to support both clinical and commercial release testing, necessitating further validation of instrument variability and assay robustness. In particular for gene therapies, a single analytical method may not be universally applicable to all AAV serotypes for a given CQA. Despite successful assay development, challenges persist in reliably associating assay results with clinical outcomes. To address these challenges, it is imperative to ensure early establishment and trending of reference standards or assay controls, adopt wider acceptance ranges during clinical phases, and implement redundancies to support *in vivo* assays. Moreover, proactive engagement with regulatory authorities is crucial to align expectations and ensure compliance when employing *in vivo* assays for commercial product support [78].

In light of the limited data available for cell and gene therapy assays and the unique challenges posed by gene therapy, it is important to adopt best practices to enhance the understanding of assay variability. This includes establishing reference standards or assay controls early in method development, using wider acceptance ranges during clinical phases, and maintaining frequent dialogue with regulatory agencies throughout the development process. Additionally, building a comprehensive analytics dataset encompassing current and historical therapies can provide valuable insights into assay performance and variability, aiding in the continual improvement of analytical methods and regulatory compliance.

6.6.4.1 Validation of Potency Assays

Potency assays play a critical role in assessing the activity of DPs, particularly in novel modalities, and their development should commence early in the drug development process. Given the unique nature of potency assays for ATMP products, regulatory agencies typically require a case-by-case evaluation, emphasizing the importance of initiating discussions with regulatory bodies at the outset of development. Challenges in ATMP potency assay development arise from limited availability of DP and reference standard material. In such cases, surrogate DP and assay controls can serve as viable alternatives. Retaining samples from different DP lots and critical assay reagents facilitates comparability and bridging studies, particularly when assay adjustments are necessary in later phases [79].

Setting validation acceptance criteria poses a significant challenge for ATMPs, often due to limited datasets resulting from a small number of manufactured lots or restricted cell availability during clinical phases. To address this challenge, it is advisable to develop assays early in the clinical phase and trend a reference standard, which may need to be internally generated, and assay control to generate a representative dataset for establishing acceptance criteria confidently. Appropriate statistical analysis is essential for setting acceptance criteria, with wider criteria potentially acceptable for functional assays, if justifications are provided. Given the profound implications of poorly developed potency assays on dosage determination, product stability evaluation, comparability studies, and overall product quality, it is imperative to meticulously develop and validate potency assays with appropriate acceptance criteria in place [80].

6.6.4.2 Validation of Rapid Microbiological Methods

The increasing complexity and urgency of advanced therapeutics, particularly gene therapies, highlight the necessity of rapid microbiological testing. To address this need, the USP [53] has introduced General Chapter <72> Respiration-Based Microbiological Methods for the Detection of Contamination in Short-Life Products, which takes effect in August 2025. This chapter offers a standardized framework for using faster, metabolism-based methods to evaluate product sterility, diverging from the traditional 14-day incubation period. General Chapter <72> focuses on rapid sterility testing for pharmaceuticals with a limited shelf life, such as cell and gene therapies and radiopharmaceuticals.

By leveraging microbial respiration, it detects contamination through gases such as CO₂ produced by growing microorganisms. Specialized instruments, using colorimetric or fluorometric sensors, allow for quicker results, facilitating timely product release. A key aspect of General Chapter <72> is method suitability testing, ensuring that chosen rapid methods are rigorously validated to reliably detect a low number of relevant microorganisms in specific product matrices. The chapter also introduces the concept of a "safety margin" for determining adequate incubation time, guaranteeing that even slow-growing contaminants can be detected. Though it does not mandate a specific technology, General Chapter <72> sets performance criteria for respiration-based systems, reflecting the industry's shift toward rapid microbiological methods to enhance patient safety and streamline manufacturing workflows [53].

Additionally, the USP [53] has released General Chapter <73> ATP Bioluminescence-Based Microbiological Methods for the Detection of Contamination in Short-Life Products, which also takes effect in August 2025. Chapter <73> provides a standardized framework for using rapid, ATP bioluminescence-based techniques to assess sterility, offering a significant advantage over traditional culture methods. It emphasizes method suitability validation, requiring laboratories to demonstrate that the ATP bioluminescence method can reliably detect low levels of relevant microorganisms, including slow-growing or product-specific isolates, within the product matrix.

This validation process includes specific experiments to confirm the method's appropriateness for its intended use under the test conditions. Furthermore, the chapter provides detailed guidance, including an equation for calculating the appropriate incubation time to ensure sufficient microbial growth for detection, even for challenging organisms, while maintaining rapid results. This rigorous validation framework described in USP <73> [53] ensures the reliability and accuracy of rapid methods, enhancing patient access to essential treatments while upholding high-quality safety standards for time-critical therapies. Its implementation signifies a broader shift in pharmaceutical manufacturing toward adopting innovative technologies that enable faster, more efficient QC processes.

When discussing the integration of rapid methods for ATMPs, it is crucial to understand that their implementation is not a simple off-the-shelf solution. These methods must be meticulously justified, fully validated, and appropriately filed with regulatory agencies. The approval process is not standardized; instead, it is handled on a case-by-case basis, requiring a comprehensive and well-documented approach to demonstrate the method's suitability and reliability for the specific ATMP application.

6.6.5 Challenges in Generating Reference Standards

Due to the inherently limited production scale of ATMPs, the amount of material available is often minimal—frequently only sufficient for a single patient dose. This limitation, combined with the novelty and complexity of ATMPs, often means that established reference standards are not available, complicating efforts to demonstrate that analytical procedures are suitable for their intended purpose. In such cases, the use of interim reference materials can be a practical solution to maintain confidence in analytical methods throughout the product lifecycle. These interim standards should be developed within the framework of the analytical lifecycle and refined over time. Ideally, reference standards should accurately reflect the manufacturing process and may require updates, supported by bridging studies, as process changes occur. When formal reference materials are not available, well-characterized assay controls can serve to monitor consistency and demonstrate method suitability during development [78].

Analytical data from reference standards and controls should be leveraged to define assay acceptance criteria. In early clinical phases, these criteria may be relatively broad, but they must be tightened as the product advances toward commercialization to ensure manufacturing consistency and product quality. Assay trending over time can validate established release specifications and reveal sources of variability. When such variability arises, studies should be conducted to identify and mitigate its causes. DOE, in alignment with ICH Q14 [31], offers a structured approach to investigating variability while minimizing the use of scarce materials.

Although commercial reference standards are available for certain biologics, options remain limited for AAV serotypes and are nearly nonexistent for autologous cell therapies such as CAR T, where patient-specific variability further complicates the establishment of representative materials. Unlike conventional biologics, which typically require new reference standards only after significant manufacturing changes, cellular therapies may necessitate more frequent updates due to the small batch sizes and variability of input material. Therefore, it is essential for developers and testing laboratories to maintain continual dialogue with regulatory authorities, proactively developing analytical strategies and communicating emerging challenges throughout the product lifecycle.

7 Manufacturing Operations

7.1 Introduction

The principles of QRM have been embedded within the pharmaceutical quality system since the adoption of ICH Q9 in 2005. These principles have been further codified with the latest revision—ICH Q9(R1) in 2023 [1]—with the addition of an added note to “Principles of QRM” to include availability risks as quality risks where there is the potential for patient harm. This sentiment is of particular importance in the ATMP environment. In autologous cell therapy, for example, a single batch of a DP often represents the entire availability of DP to the single patient population.

Given that the principles established in ICH Q9(R1) are equally applicable to ATMPs, the various methodologies defined within can be applied to include tasks such as risk identification, classification, assessment, mitigation/remediation, justification, and the formal acceptance of risk. However, it is recognized that some ATMP manufacturing platforms may require nontraditional approaches to risk management.

One of the goals of this Guide is to assist in identifying ATMP-specific mitigation and rationale to support the classification of product impact and critical risks. This chapter aims to serve as a starting point to risk management, to help break down a given manufacturing process into discrete steps in a formal documented process map, and to identify risks and each required risk assessment step. This approach is intended to help improve the reader's understanding of process risk while also ensuring operational readiness and support audit situations related to process and patient risk.

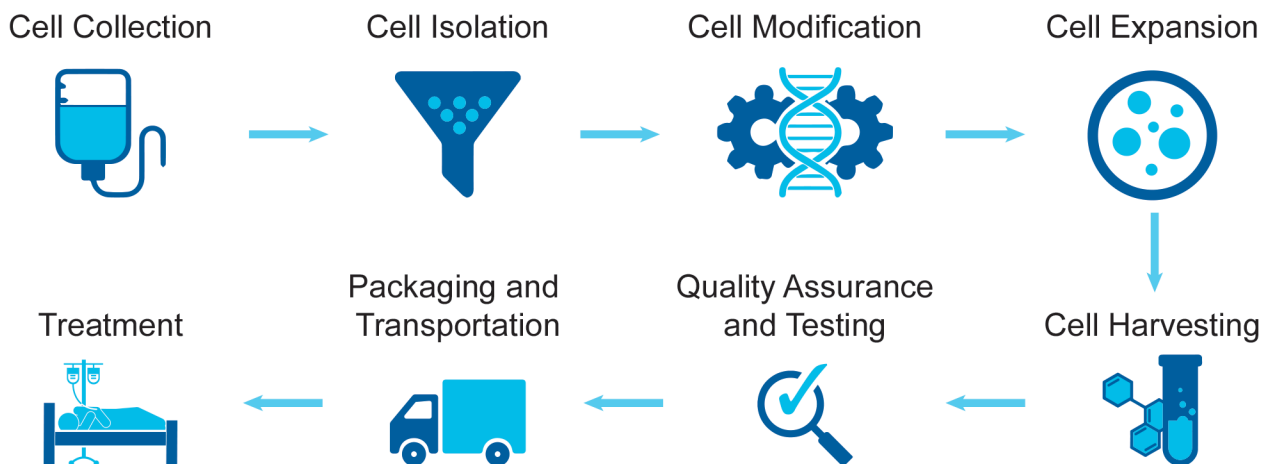
7.2 Operations

Manual interventions are common throughout ATMP processes. These manual interventions pose unique risks due to the inherent differences between autologous cell therapy, allogeneic cell therapy, and gene therapy. In autologous cell therapy, the donor and patient are one and the same. However, in allogeneic therapy, the donor provides the starting material (apheresis) that is then utilized to make multiple products that can be distributed to many patients. For both therapy types, capturing operational risks is paramount to ensuring a robust validation strategy that is commensurate with the level of risk.

As most products generated in ATMP manufacturing cannot be terminally sterilized, the manual operations of the process must be accounted for when developing a validation strategy to ensure aseptic technique is adequately controlled through all phases of the process. The overall approach to aseptic processing and its associated definitions are provided in both EU and FDA guidance extensively (e.g., [35, 81]) and as such are not covered in this Guide. Instead, this section highlights how to identify and mitigate risks to operational steps as applicable to ATMPs. In addition to defining mitigations to ensure a robust aseptic process is established, the operational risk assessment can also prove beneficial in assessing appropriate areas for closure improvements, implementing automated systems, establishing a robust training program for operators, and rationalizing a company's approach to an APS strategy.

7.2.1 Manual Operation Assessment

The meticulous management and execution of manual steps within the ATMP manufacturing process is central to product integrity. Although the focus of this section is on an autologous product, the principles also apply to allogeneic products that rely on *ex vivo* processing of cells. A typical autologous cell therapy process is used in this section as a base example for determining operational process risks. These principles also apply to allogeneic products that rely on *ex vivo* processing of cells. As shown in Figure 7.1, the typical steps of an autologous manufacturing process are as follows: cell collection, cell isolation/selection, cell modification (activation, transduction/transfection), cell expansion, cell harvesting, and finally cryopreservation.

Figure 7.1: Simplified Manufacturing Process – Autologous Cell Therapy (CAR T) [7]

The process initiates from the initial apheresis collection from the patient/donor, where cells are then transferred to the manufacturing facility for *ex vivo* processing across various days. Within each of these processing steps, there are several unit operations required for the addition of reagents, buffers, and cell transfers. Each of these operations poses potential risks to the integrity of the product. These risks are not only impactful from a microbial or particulate contamination standpoint, but also from an overall product quality standpoint (i.e., dose, purity, identity, and potency). To identify and mitigate these risks, a systemic evaluation of the manual interventions of the *ex vivo* processing steps should be captured.

Several common manual steps to consider in the assessment include the following:

- Sampling of cell culture in-process
- Formulation of single/multicomponent reagents
 - Collecting reagent
 - > Use of spike adapter
 - > Sterile reagent in vial
 - > Lyophilized reagent (addition of Water for Injection (WFI))
 - > Sterile reagent in bottle
 - Removal of reagent
 - > Use of needle from vial
 - > Pulling syringe on luer port of bag
 - Transferring reagents
 - > Pressing syringe on luer port of bag
 - > Bag-to-bag transfer via tube weld/gravity drain

- Centrifugation in single-use system
- Cell selection in single-use system
- Mixing of formulated reagents and/or final DP
- Aliquoting from formulated or in-process cell suspension
- Cryopreservation

Once the manual steps are appropriately identified for a given process, the complexity of each step should be considered.

7.2.2 Assessing Manual Operation Complexity

Complexity can vary between similar unit operations, for instance, the aspect of completing volume transfers. In this example, using sterile welds to transfer entire or partial contents from one culture bag to another is not as complicated as transferring components via a volumetric syringe.

- For a transfer through a gravity drain, typically additional equipment is used to facilitate the transfer and can be performed outside of a Grade A environment. For instance, a sterile connection between the transfer vessels is established through a welder. The target vessel is placed on a scale, and the contents can be readily drained from the source vessel. Upon completion, a sterile sealer is used to separate the vessels with an operator and verifier confirming final weight on the scale.
- For a transfer through a volumetric syringe, there is an increased effort in maintaining aseptic technique for both the target and source vessel within a Grade A/BSC environment. The volume to transfer is meticulously measured through graduation lines on the syringe. Motions are critical when transferring the filled syringe contents to the target vessel. Multiple connections and disconnections between the vessels and syringe are required to facilitate the transfer and increase the likelihood of microbial contamination.

As demonstrated in the previous example, complexity should not focus solely on the impact from microbial contamination due to the aseptic nature of ATMP manufacturing. Therefore, it is recommended that the manufacturer assess what actions are performed, pre, during, and post the manual operation to ensure aseptic techniques are maintained and that the process parameters (critical and noncritical) are satisfied. Likewise, the nature of the particular material being transferred should also be understood, as different materials might pose different effects on the assessment of transfer risk. In addition, the required attention from the executor, cognitive processes, and human performance (although not directly quantifiable) should be considered when determining the overall complexity of a manual step and in the design of the process simulation studies.

This can be viewed as a combination of motion elements (pickup, place, and reach) and mental functions (identify, locate, and decide), followed by the environmental condition in which the action is performed (i.e., under ISO 5 gowning versus ISO 8 gowning). Also, there may be a direct correlation with the higher number of motion elements, mental functions, and operating environment relating to a higher level of complexity. Ultimately, it is up to the manufacturer as to how to categorize the complexity of a manual step; however, this process will be of great assistance when the manufacturer wants to move toward closure solutions and automation to increase organization efficiency.

7.2.3 Repetition and Operation Oversight

With the number of manual steps assessed for each processing step and its associated complexity determined, two additional factors should be considered when determining the overall risk of the operational process: the repetitiveness of the manual step and the operational oversight provided.

- Most manual steps are commonly repeated in the ATMP manufacturing process. It is logical to conclude that if there are highly complex manual steps being executed multiple times in a manufacturing process, then there would inherently be a higher level of risk associated with that process. Appropriate mitigations, such as reducing the number of high-complexity steps, implementing automation solutions, or enhancing the training program for these steps, should be identified. These mitigations can be applied to all steps in the operational process that are repetitious, thus allowing for consistent, efficient process improvements to be rapidly applied to the manufacturing process.
- Operational oversight should also be considered when determining the risk of the operational steps. Depending on the criticality of the operation to product quality, or the calculated level of complexity, the appropriate level of oversight can be determined for the given manual intervention. For instance, highly repetitive and complex manual operations require a higher level of oversight to ensure product quality and patient safety. Conversely, less complex but still highly repetitive manual operations may allow for reduced oversight, relying on a robust training program.

7.2.4 Mitigations and Benefits

The repetition of an operational step and an increased level of oversight do not inherently reduce process risk. A holistic approach to the operational process is important to identify, classify, and categorize risks so that appropriate mitigations can be put in place. This includes capturing the number of manual steps in the process, understanding the complexity when executing the manual step, determining how many times the step is repeated within the process, and, finally, assessing the appropriate level of oversight.

This operational assessment will guide the decision of whether moving to complex automated equipment benefits the manufacturing process or introduces risks that outweigh the benefits. This assessment should also yield a more efficient training program that ensures operator readiness through highlighting highly complex or repetitive operations. Furthermore, grouping the operational steps will streamline the aseptic simulation strategy and process closure initiatives. This detailed assessment will identify which steps are repeated throughout the process and therefore do not need to be executed each time in an end-to-end aseptic simulation.

7.3 Equipment

The equipment used in ATMP manufacturing brings forth a set of considerations pivotal to a risk-based validation approach. Although the various manufacturing equipment used in the ATMP processes are described in the ISPE Guides on ATMPs [7, 8], this section focuses on additional factors to consider for a risk-based approach to qualification. These considerations range from the complexity of the manufacturing equipment and its direct contact with the product to the incorporation of advanced monitoring functionalities that allow for remote observation. These considerations all factor into the design of the equipment, which should be meticulously planned, qualified, and validated through a series of stages such as Factory Acceptance Testing (FAT), Site Acceptance Testing (SAT), Installation Qualification (IQ), Operational Qualification (OQ), and Performance Qualification (PQ). As part of a risk-based approach to qualification, leveraging of FAT and SAT data should be considered when appropriate.

7.3.1 Risk-Based Verification and Validation

A risk-based approach ensures that the equipment used within the process is fit for its intended purpose, operates consistently, and produces DPs that can reliably meet predefined quality attributes. The concept of verification becomes particularly important in ATMP manufacturing, where equipment is often sourced from vendors originally designing it for laboratory or hospital use rather than commercial GMP environments. Therefore, understanding equipment verification and how it integrates into the company's overall validation framework is essential for the effective implementation of a risk-based approach.

As emphasized in the FDA CBER OTP Town Hall on CMC Readiness for Gene Therapy Biologics License Application (BLAs) [82], a well-structured validation plan is expected to clearly define the strategy and justification for the qualification and verification of all critical equipment, including those used for closed-system processing, single-use technologies, and fill-finish operations. The validation plan serves as a foundational document to demonstrate how verification activities align with the overall control strategy, ensuring that equipment risks are appropriately mitigated and that the process consistently delivers quality product.

Verification can be viewed as “You built the system correctly.” For equipment, this consists of verifying the components were built within the drawing specifications or software was installed with the specified version, as well as the maximum controllable range of process parameters the equipment is capable of. Essentially, it is confirmation using objective evidence that the requirements have been fulfilled. In the context of a risk-based approach for Commissioning and Qualification (C&Q), this activity is the Design Qualification (DQ) that is completed prior to fabrication and verified post fabrication and installation.

Validation can be simplified as “You built the correct system.” Validation is the responsibility of the manufacturer producing the Drug Substance (DS)/DP. Therefore, although the equipment may not be designed with a specific manufacturer’s process in mind, it is the expectation that the manufacturer will provide the necessary evidence to demonstrate that equipment is the correct system. This means it is fit for the process and can produce consistent and reliable product.

Often, ATMP manufactures do not put an emphasis on the verification phase of the equipment, mainly because it is purchased as an Off-The-Shelf (OTS) commodity that fits into a manufacturing process. Poor verification of the equipment design could potentially impact validation by reducing the ability to track down and correct problems. Therefore, it is recommended that a manufacturer creates a traceability matrix early on that will capture design inputs (requirements) and establish a clear linkage to the design outputs, verifications, validations, and, most importantly, risk assessments. This should be further described in a validation plan.

Risk assessments, from a processing and PV perspective, will highlight the areas of the manufacturing process that are critical to product quality, and therefore should have rigorous test measures in place to adequately qualify. One such example is a centrifugation system that is responsible for removing process-related impurities. The manufacturer identifies the critical aspects of the equipment related to force and duration of centrifugation that have a direct impact on culture viability. Therefore, it would be beneficial to perform verification checks for the components within the equipment that facilitate this functionality. This could potentially lead to a reduced or rightsized validation effort required by the manufacturer if it can be demonstrated that components built within a specified range will manufacture products that consistently meet quality attributes. Before assessing the strategy for equipment verification and subsequent validation, additional factors should be accounted for as related to equipment used in ATMP manufacturing.

7.3.2 Equipment Factors for Consideration

Additional factors to consider when assessing the risk of equipment used in the manufacturing process include direct product contact, EM capabilities, and remote viewing functionalities. Equipment with direct product contact will require a greater focus on sterility and contamination control in the risk assessment process, resulting in enhanced cleaning validation procedures and sterilization methods. It can also potentially point to further material compatibility studies to ensure product integrity and safety are maintained throughout the manufacturing process.

7.3.2.1 Environmental Monitoring

EM is a crucial factor in maintaining the aseptic manufacturing environment required for ATMPs where the risk of microbial contamination could prevent the release of the DP. Factors to consider when assessing equipment with integrated EM functionality include the system’s capability of real-time monitoring of airborne nonviable particle counts and viable particulates, temperature, and potential microbial load. The continuous monitoring of these factors assists in risk-based validation by enabling the early detection of adverse trends or excursions that could lead to a breach in aseptic conditions if not addressed. The assessment of the equipment’s EM functionality to monitor the manufacturing environment accurately and consistently will allow for informed decision-making and prompt interventions that may prevent potential contamination of the product.

7.3.2.2 Direct Product Contact

Direct product contact surfaces inherently introduce risk of microbial, particulate, or chemical contamination that compromises product quality. Therefore, the design, operating conditions, and storage of single-use kits must be carefully selected and validated to minimize these risks. Single-use consumable kits are common and constructed for product equipment to help prevent cross-contamination between manufactured lots. The compatibility of the single-use material with the product should be well-documented to limit the impact of potential Extractables and Leachables (E&L) and to provide assurance of no animal origin or biological nonreactive material. QC analytical testing of these single-use systems can aid in the CCS. A manufacturer will typically require the following tests or assessments to be performed on these single-use systems:

- Sterility
- Endotoxin
- Bioburden
- Free of Transmissible Spongiform Encephalopathy (TSE)/Bovine Spongiform
- Encephalopathy (BSE)
- Particulates assessment
- E&L assessment

The integrity of these systems is paramount, as any compromise can directly impact the CQAs of the final product in ATMPs. Some single-use systems, especially for fully automated manufacturing systems, have rather complex designs. This leads to difficulty in developing reliable QC analytical testing methods that may prevent the manufacturer from testing the material in-house. As such, there may be a need to rely on the supplier methods in place on the Certificate of Analysis (CoA), through which the manufacturer should ensure robust controls are in place via supplier agreements and an auditing program that is commensurate with risk.

7.3.2.3 Remote Viewing Capabilities

The capability to remotely view manufacturing equipment is prevalent in assessing potential mitigation and detection mechanisms for identified process risks. Remote viewing provides significant advantages in operational flexibility and efficiency, but it also introduces potential challenges and risks related to data integrity and cybersecurity. Therefore, it is crucial to identify measures necessary to prevent unauthorized access and protect company manufacturing data. Some mitigations may include the implementation of enhanced data encryption, user access controls based on job type, and audit trails that can demonstrate maintenance of data integrity and comply with 21 CFR Part 11 [83] and EudraLex Volume 4 Annex 11 [84]. These data integrity safeguards must be considered in equipment verification. Additionally, the mechanisms for data transfer and integration and for additional capabilities such as the alarm interface must also be assessed.

Remote viewing capabilities of the equipment may aid in timely decision-making and intervention that ultimately reduce quality risks identified in the process, but it is vital that this technology is implemented in a manner that complements, instead of replaces, physical oversight. These factors can prove to be extremely beneficial in reducing risk to an ATMP process by enhancing the EM, maintaining an aseptic environment, and promptly alerting the user and necessary supporting teams remotely to mitigate on-the-floor issues rapidly. The culmination of the risk outcomes from these factors will highlight focused areas of training for the personnel involved in equipment operation. This is particularly identified as related to aseptic technique, EM capabilities, and data management.

7.3.2.4 Simple Versus Complex

When identifying risks for equipment that is used in the manufacturing process of ATMPs, it is crucial to consider unique factors that could influence product quality, process integrity, and, ultimately, patient safety. Manufacturing equipment used in ATMP manufacturing ranges from simple equipment instances to complex automated systems. Although GAMP guidelines [11] offer the principles of validating these systems, unique considerations regarding ATMP therapies must be accounted for to ensure product safety, efficacy, and quality. For instance, equipment directly impacting CQAs or CPPs would undergo more rigorous validation to ensure consistent performance compared to equipment that does not serve to meet any process attribute or parameters.

This is critical when these CPPs are maintained through parameterization or automated logic within the equipment and pose separate challenges in qualification. Simple systems that contain no software (e.g., simple mixers), though still potentially impacting a CPP, may have validation efforts focused on mechanical reliability and consistent performance as demonstrated through calibration, maintenance, and material compatibility. These non-software systems are accounted for in the risk-based approach to ensure the focus in qualification is centered on intended use in the process and the ability to demonstrate equipment robustness.

More complex equipment, with embedded software or configurable logic controllers, is expected to have additional risks regarding the software functionality, user access control, data integrity, and the ability to generate accurate and reproducible results. Typically, these complex systems can generate reports that contain process information that is required for CPPs and potentially for CQAs. These complex systems bring unique challenges of integrating with the existing manufacturing process, and therefore a company may require a higher degree of assurance that the report information is accurate and reported to the level of detail required for the process. Regardless of whether the equipment is simple or complex, it is important to incorporate these considerations into the equipment validation strategy to ensure it is appropriately validated to perform its intended function and that it ensures product quality and compliance with regulatory standards.

7.3.3 Equipment Qualification

As stated in ICH Q9(R1) [1], QRM principles should be applied to product equipment to determine the extent of qualification activities required. It is essential that the functionality of the equipment be directly linked to the CPPs, where possible, as defined through a formal risk assessment. By understanding the relationship between equipment functionality, CPPs, and CQAs, qualification efforts can be focused on those aspects of the equipment that are most critical to process performance and product quality.

QRM principles emphasize the identification of unacceptable risks to CPP control and the implementation of sufficient controls to mitigate those risks to an acceptable level. These controls may include a combination of engineering and equipment design features, integrated automation control strategies, data management systems that demonstrate process control over time, critical alarms that indicate loss of control, and clearly defined procedures for operational response. The appropriate combination of these controls—one or more depending on risk severity and process complexity—must be defined and implemented to establish and maintain a robust state of control.

In addition to supporting equipment qualification, this integrated risk assessment also informs the development of preventive maintenance strategies. For example, spare parts for components deemed critical to maintaining process functionality should be readily available, based on their impact to CPPs and the overall risk profile of the process.

7.3.3.1 Product Equipment Design

Primary to the equipment qualification process is establishing a robust User Requirement Specification (URS). This is developed based on a comprehensive understanding of the process, including the identification of CPPs and CQAs through risk assessments and process characterization studies. For ATMP manufacturing, these parameters often include operational settings that, when controlled within specified limits, ensure product quality. Examples include temperature control for incubators or sedimentation time and G-force settings for centrifuges, all of which can directly influence product attributes such as viability, purity, and potency.

A formal URS is typically required for custom-designed equipment, serving as a foundational document to guide the design and ensure that the equipment will meet the operational and quality needs of the process. The URS is a key input to DQ and subsequent qualification stages [11, 12, 85]. However, for OTS or catalog equipment, a traditional URS may not be applicable. In such cases, user requirements must still be clearly documented, even if the equipment has not been custom-built. These documented requirements should outline the intended use, functional expectations, critical operating parameters, and compatibility with the intended process. This documentation enables an effective DQ, an activity that verifies whether the selected equipment, as designed and supplied, is suitable for its intended purpose.

7.3.3.2 *Design Specification and Qualification*

Regardless of whether the equipment is procured from a third-party vendor or developed in-house by the ATMP manufacturer, the design and construction of the equipment must be clearly linked to the documented user requirements, either within a formal design specification or through supplemental technical documentation provided by the vendor. The design specification defines how the equipment has been built, including key functional and operational features, materials of construction, and control system architecture.

The DQ verifies that the equipment design, as described in the design specification, meets the user requirements and is suitable for its intended use. This includes confirming that all design features and control strategies identified through QRM have been appropriately implemented to mitigate any unacceptable risks to product quality or process performance. DQ should demonstrate clear traceability between user requirements, equipment specifications, and risk-based controls, ensuring that critical elements (e.g., alarms, automation logic, or process safeguards) are in place to maintain a state of control. Early stage alignment between the end user, the technical specifications, and applicable regulatory requirements is essential [11, 12, 85]. This alignment helps mitigate risks throughout the equipment's lifecycle and ultimately ensures that the equipment supports consistent, compliant, and robust manufacturing of ATMPs.

7.3.3.3 *Verification and Acceptance Testing*

Depending on the level of complexity and criticality of the equipment being procured, it may be necessary to perform FAT at the vendor's site, followed by SAT at the manufacturing facility. FAT is typically required for custom-designed equipment, as it provides initial documented evidence that the system has been constructed according to the design specification and can operate within the defined parameters. FAT also supports the risk management process by identifying potential issues early—such as design limitations, software malfunctions, or integration gaps—that may require mitigation or redesign before equipment deployment.

For OTS equipment, FAT may not be necessary if DQ has already confirmed that the selected unit meets the documented user and functional requirements. However, SAT may still be appropriate for complex or novel OTS equipment, particularly where environmental conditions, system integration, or automation interfaces may influence the equipment's performance at the manufacturing site.

Equally critical to FAT is the execution of SAT, which provides documented confirmation that the equipment has been delivered and installed, and it performs as intended in its actual operating environment. SAT verifies that installation and functional checks meet the acceptance criteria, ensuring that the system is fit for purpose and aligned with the URS. Acceptance testing should be applied selectively, based on a documented risk assessment that considers equipment type, criticality, and potential impact to product quality. Where applicable, the leveraging of FAT and SAT data can support a reduced scope of qualification activities, provided that traceability to URS and QRM principles is maintained and justified [11, 12, 85].

7.3.3.4 Installation, Operational, and Performance Qualification

Installation Qualification

IQ verifies that the equipment has been correctly delivered, installed, and assembled in accordance with the manufacturer's specifications and installation guidelines. However, IQ also plays a broader role in confirming that the equipment has been fabricated to meet the URS and includes any design features or control elements identified through QRM to mitigate unacceptable risks. This ensures that the equipment, as installed, is suitable for its intended use in the manufacturing process.

For equipment incorporating software, it is essential to verify that software versions, configurations, and parametrizations are correctly implemented and aligned with process-specific requirements. Many types of product equipment are nonconfigurable but offer flexible parameter settings, which can be stored as process-specific "recipes." In such cases, a risk-based approach can be used to define a testing strategy focused on verifying critical software functionality on a representative unit.

If equivalence can be demonstrated—through documented risk assessment criteria such as component matching, version control, and manufacturing lot traceability—testing does not need to be repeated across all identical units. This integrated IQ approach ensures both installation accuracy and compliance with user-driven, risk-mitigated design and performance expectations.

Operational Qualification

The primary objective of OQ is to provide documented evidence that the equipment and its critical components operate consistently and reliably across the full range of conditions defined in the URS. This includes verifying that the equipment can support the control of CPPs and maintain those parameters within acceptable limits throughout the intended duration of operation, as determined through risk assessment. This targeted approach enables a more efficient and risk-based validation strategy by focusing testing efforts on functions directly linked to product quality, safety, and efficacy.

For example, in an automated fill and finish system, the accurate dispensation of reagents is directly tied to the dosage and therefore impacts a CQA. In this case, OQ would rigorously assess the system's ability to fill product containers consistently and accurately across various operating ranges. Testing would typically include evaluation of key components such as peristaltic pumps, load cells, and cooling systems. In addition, computerized logic used to control fill volumes, particularly algorithms that adjust for fluid density or viscosity, should be verified to ensure consistent performance. Based on the risk assessment, OQ should also include testing of alarms, interlocks, and error responses for components critical to maintaining control over CPPs.

From a Good Engineering Practice (GEP) and contractual standpoint, equipment vendors are expected to supply complete as-built specifications, including component-level information and functional descriptions. These datasets provide the foundation for developing OQ protocols. Where appropriate, data from the vendor's standard qualification package may be leveraged, provided the qualification package is verified and traceable to the specific equipment in use. By applying QRM principles, manufacturers can place greater reliance on vendor-supplied verification for noncritical features—such as Graphical User Interface (GUI) elements, utility connections, and standard startup functions—while directing internal qualification resources toward process-critical functionalities. This approach improves efficiency and strengthens assurance that equipment performance supports product quality and regulatory compliance.

Performance Qualification

It is recognized in ATMP manufacturing that PQ, the final stage of equipment qualification, can be strategically combined with PPQ or PV. PPQ is a demonstration that the entire manufacturing process, including the connected systems, can consistently produce a product that meets quality attributes and specifications. A PQ focuses solely on the equipment's performance as intended in the actual production environment.

Due to the high cost of healthy donor material and simulated product runs in ATMP manufacturing, it is beneficial to demonstrate the equipment's performance in concert with other process components in a PPQ that ensures the final product meets all CQAs. Therefore, a risk assessment that links the interaction between equipment functionality and CPPs allows for the simultaneous validation of equipment functionality and process performance that provides a comprehensive view of the process's capability to meet quality requirements. This reduces duplication of effort and ensures consistent production of quality product that can consistently meet predefined attributes.

The manufacturing process should be assessed as per the PV plan. The PV batches will be executed to determine if the process is capable of reproducible commercial manufacturing, by combining the facility, utilities, equipment, and trained personnel with the commercial manufacturing process, control procedures, raw materials, and components required to produce commercial batches. Successful PV batches are required to confirm the process design and demonstrate that the commercial manufacturing process performs as expected.

7.3.3.5 Equipment Equivalence

Equipment equivalence refers to the principle that different units or models of manufacturing equipment, despite variations in design manufacturer, can achieve the same operational outcomes, meet the same quality standards, and adhere to the same regulatory requirements for the production of ATMPs. It encompasses the comprehensive evaluation and demonstration that these different pieces of equipment can consistently produce products with identical CQAs and operate within the defined CPPs, thereby ensuring product consistency, safety, and efficacy.

Equivalence is essential to expand production capacity, replace legacy equipment that has reached the end of life, or introduce flexibility into manufacturing operations without compromising product quality or compliance. It provides a structured approach to demonstrate that different equipment can be used interchangeably, thereby supporting operational efficiency, scalability, and continuity in manufacturing across multiple sites.

Equivalence can be demonstrated through a systemic comparison that includes the steps outlined next.

Technical Assessment

A technical assessment is a detailed comparison of the design specifications, functionalities, and operational capabilities of the equipment to ensure it will achieve the same process outcomes. This may involve reviewing the operational control of the equipment and demonstrating that the critical design elements identified will satisfy the process outputs required. In cases where equipment inherently functions differently (i.e., perfusion washing versus centrifugation), then it is recommended to identify the equipment recipe parameters to produce product within predetermined quality specifications to demonstrate equivalency.

Equipment Qualification and Process Comparability

Comparability studies and characterization runs should empirically demonstrate the equipment can operate within the established CPPs and produce products meeting predefined CQAs. A robust IQ and OQ of the equipment will demonstrate clearly where the similarities and differences lie between the equipment. In addition, the expected performance in the manufacturing process can be directly compared when assessing the qualified operating ranges.

For further details on comparability, see the *ISPE Guide: Advanced Therapy Medicinal Products – Recombinant AAV Comparability and Lifecycle Management* [86].

Risk Management

Risk management identifies and assesses any potential risks associated with using different equipment, including variations in process control, monitoring capabilities, and maintenance requirements. Appropriate mitigation strategies should be developed to ensure these risks do not impact product quality or patient safety.

Documentation and Traceability

Maintaining comprehensive records that document the equivalence assessment process—including technical evaluations, comparability data, risk assessments, and mitigation implementation—is crucial for submissions and subsequent regulatory inspections. Documentation and a clear statement of equipment equivalence should follow a structured approach that is detailed in a standard operating procedure or other enforceable training document.

7.4 Process

The validation of the manufacturing process for ATMPs necessitates a tailored approach that accommodates the unique characteristics and requirements of these innovative products. In addition, because terminal sterilization of the final product is not possible, a robust APS strategy is essential. Therefore, the PV of an ATMP needs to include validation of the process to consistently manufacture products that meet quality attributes, as well as the ability to consistently manufacture product that is free from contamination. A well-defined process that has undergone successful PV will facilitate scale-out capabilities, which are especially beneficial for the rapid production of consistent quality individualized batches for an autologous process. Key elements include a comprehensive technology transfer that demonstrates equivalence in equipment and environmental conditions, a documented CCS, and a process control strategy. When combined, these elements ensure consistent quality product manufactured at multiple sites.

Through integration of these components—operations, equipment, and process—into the validation framework, a holistic risk-based approach is achieved that not only adheres to the highest standards of manufacturing practice but also ensures the delivery of safe, effective, and high-quality ATMPs to patients. A risk-based approach facilitates compliance, but it does not serve as a justification for noncompliance with regulatory requirements nor for a lack of documented evidence in managing the identified risks.

7.4.1 Process Factors for Consideration

As pertaining to ATMPs, quality risks are dependent upon the biological characteristics of the starting material (i.e., patient versus healthy donor), characteristics of the vector (i.e., lentiviral versus reverse transcription), properties of the noncellular components, and the manufacturing process. Additionally, safety risks pertaining to ATMPs should consider health and environmental risks, not only to the patient but also to the operators responsible for DP manufacturing. The risks related to product quality, safety, and efficacy pose challenges for the ATMP manufacturer, as the finished product will have inherent variability due to the characteristics of the starting material. Therefore, the control strategy for a product should include testing of key in-process controls as relevant to the final product CQAs, incorporate critical processing times, and identify CPPs that will produce quality product when starting from a variable material that meets minimum established requirements (e.g., total cell count, viability, White Blood Count (WBC) population).

7.4.1.1 Defining Critical Quality Attributes and Critical Process Parameters

The linkage of a QTPP to a product's CQAs and identified CPPs is detailed in PV guidelines as discussed in Chapter 4 and will not be repeated here. Instead, this section highlights some key risks common to ATMPs that should be considered to help assess process parameters and quality attributes. It is expected that the finished product will have variability due to starting material characteristics, but the final product should be within a given range.

For example, consider an autologous product. Some attributes that may be identified through a risk assessment are based on the vector component of the process. For instance, CAR T expression, which is related to the percentage of cells expressing the gene of interest, may be determined through clinical characterization work. Furthermore, the vector copy number is an important safety attribute and should have ranges determined to limit impact of tumorigenicity in patients.

Other attributes that may be identified through a risk assessment include the means of determining the identity of the product, especially for a multiproduct facility. In cases where a product contains multiple antigens or relies on additional cytokine transgenes for enhanced CAR T activity, then there should be mechanisms to assess these attributes accordingly. Purity is a common quality attribute that should be assessed for the unique autologous process, and it may be appropriate to demonstrate impurity clearance through validating the functionality of cell processing equipment. In these cases, a manufacturer may assess that it does not need to test for impurity results at batch release, given that the equipment operated as expected and within the validated range.

This example demonstrates the need to determine the CPPs for a given manufacturing process that, when controlled, will consistently produce quality products. It is essential for a manufacturer to identify the process parameters that control the product within specified ranges of these identified CQAs. Identifying these process parameters will establish appropriate points for in-process testing that demonstrate process performance.

Considering the example of an autologous product again, one potential mitigation to reducing starting material variability is through establishing a minimum required number of cells present. This number of cells would coincide with the number of cells required for downstream manufacturing steps (i.e., activation, transduction/transfection, expansion, and harvest). If the cells are assessed to be under ideal conditions, regarding viability and total number, then a manufacturer will have a better understanding of expected performance prior to initiating the process. Furthermore, related to the number of cells present in the starting material, a manufacturer should identify methods for ensuring consistent isolation from the material.

A fixed cell-to-bead ratio may be required to ensure consistent isolation of T cells, for instance. If equipment is used to perform the isolation step, then it is recommended to use fixed settings. However, changes to autologous starting material specifications need to be thoroughly vetted by all stakeholders, as this may change the treatable patient population and/or indication.

As related to the viral vector attributes, parameters that may be identified during the risk assessment include the use of a fixed multiplicity of infection via controlling the amount of vector introduced to the product for a fixed duration of time. Similar to the isolation step, in cases where equipment is used for the transduction or transfection step, fixed settings should be used. Following transduction/transfection, other parameters may consist of ensuring that proper cell density is maintained through the expansion days and until a sufficient number of cells are available to harvest.

The identification of these CPPs will aid the documented evidence to be provided that a process can constantly produce product within specified quality attributes, when considering inherent donor variability of the starting material.

7.4.1.2 Noncellular Components for Consideration

A risk-based approach should be applied to the properties of the noncellular components, or the raw materials, to develop appropriate specifications that will ensure the quality of the raw material and overall final product. Raw materials for ATMPs have separate risk levels due to the properties of the raw material in addition to the use within the process and as such should be considered during a risk assessment. For instance, cell culture media containing growth factors will have a different risk profile than basic media that may require additional components to be added prior to introduction to the manufacturing process.

Furthermore, a raw material that directly contacts starting and/or in-process material will have a higher risk profile and may include more stringent specifications prior to being released by the Quality unit. The manufacturer should also consider direct product contact and contact with other materials that may interact with the DP. Specific factors include contamination risk as related to viral, microbiological, and TSE. In addition, it is important to understand that mycoplasma and endotoxin are not detected with sterility testing and based on the risk of the raw material, may require specific testing. Therefore, these specific microbial contamination tests should also be assessed for a given raw material.

Specifications for raw materials should include transport conditions, storage conditions, sampling, and testing requirements. For critical raw materials, there is an expectation to assess the quality requirements with established acceptance criteria that ensure suitability for the intended use in each manufacturing process. The manufacturer is responsible for determining the criticality of a raw material in their specific process based on assessed risk. Therefore, the manufacturer needs to agree with the supplier that the specifications contain the appropriate testing, control, and handling of the material as identified through the risk assessment.

If the risk of a raw material is thoroughly documented and understood, then it may be appropriate to qualify the supplier and rely solely on the CoA provided to release the material for use in manufacturing. In contrast, where there may be gaps, or higher level of risk that can only be mitigated through further testing by the manufacturer, this should be documented as such in the risk assessment.

7.4.1.3 Contamination Control Strategy

A robust CCS is critical to ensuring the DP is free from microbial contamination. Aside from microbial contamination, the risk of cross-contamination is prevalent in ATMP manufacturing where multiple products are produced within the same facility. Therefore, the CCS encompasses a comprehensive approach, addressing potential contamination sources from facilities, equipment, personnel, starting material, and raw materials. The objective is to ensure that the measures in place collectively guarantee the absence of contamination in the finished product. This comprehensive approach is imperative, as reliance on final product testing to ascertain product purity and/or sterility is not acceptable.

ATMPs are unique in that there are various points of potential cross-contamination. For instance, the starting material, gene of insertion, and type of vector each introduce unique risks of cross-contamination that must be considered by the manufacturer. Segregation of lots that use unique starting material are most common in autologous facilities or facilities that have been designed with a scale-out, ballroom approach. The cleaning and segregation required between the different lots manufactured should be determined by the assessed risk. In addition, there are separate considerations to consider when designing a facility that will manufacture products that use different vector types—Retroviral Vector (RVV) or Lentiviral Vector (LVV)—compared to the manufacture of separate products that use the same viral vector (i.e., differing in the gene of insertion).

In a multiproduct facility, the potential for cross-contamination between different products, especially when handling sensitive materials such as viral vectors, requires stringent controls. There should be well-defined and strictly enforced segregation policies that address the various risks related to cross-contamination between equivalent products. For a facility where the gene of insertion and viral vector are identical, cross-contamination measures should be focused on ensuring the starting material remains segregated from other lots.

This consists of ensuring the unique lots are never incubated in the same area, unless clearly segregated through the use of closed and uniquely labeled culture vessels (e.g., culture bags, flasks). Furthermore, it is acceptable to manufacture multiple lots in one cleanroom so long as there are defined cleaning procedures and clear demarcations of workstations for each lot being manufactured. Multiple workstations are common in a ballroom-style facility and come with inherent risks that require mitigations such as a higher number of BSCs and isolators, the use of shared processing equipment, and the separation of material and personnel flow.

Processing equipment may be appropriate for multiple lots given the use of a single-use kit can segregate the product from the cleanroom environment and that there is cleaning performed between different lots. The separation of personnel and materials can be achieved with airlocks and unidirectional flow when entering and exiting the cleanroom. These factors must be carefully assessed and planned to ensure concurrent manufacturing of lots within the same cleanroom area to prevent cross-contamination and lot mix-ups.

In addition to routine contamination control measures, accidental releases or spillages—particularly those involving live organisms—must be promptly and safely managed. Facilities should maintain accessible, organism-specific decontamination procedures that are aligned with the risks posed by the biological materials in use. Spill response protocols need to be well-documented, clearly communicated to personnel, and supported by the immediate availability of appropriate containment and neutralization agents. Personnel must be trained in the proper handling of accidental exposures to ensure containment and mitigate the risk of environmental contamination or cross-contamination of other products in production.

Other facilities that manufacture products using different vectors, or the same type of vector but a different gene of insertion, require additional measures regarding cross-contamination. For instance, the cleanroom should be dedicated to one vector and gene of insertion at a time. The space can only be switched over following a validated cleaning process and therefore can never have concurrent manufacturing runs within the same space. There must be 100% air exhaustion from the cleanroom when manufacturing with different vectors to ensure no recirculation of airborne viral particulates. Finally, vector waste must remain closed and segregated from other vector waste to prevent potential vector mix-ups. These additional facility measures help ensure manufacture of multiple types of ATMPs while maintaining segregation of unique lots starting material, gene of insertion, and vector type.

Regardless, if operations have physical segregation or share a common operating space with other products, it is imperative to establish and validate cleaning procedures that intentionally degrade and remove genetic, viral, and cellular residues to acceptable limits, ensuring no cross-contamination occurs between all products. This includes not only product contact surfaces but also non-product contact surfaces. Although these pose lower risk, they must also be effectively decontaminated to prevent inadvertent transfer of bioactive materials.

Cleaning procedures must demonstrate the inactivation or destruction of residual materials to a level that mitigates the risk of carryover, even below the limits of conventional detection. This involves detailed documentation and rigorous testing to confirm the effectiveness of the cleaning agents and the process for all product contact and non-product contact surfaces commensurate with identified risk. Implementing these measures requires a comprehensive understanding of the production process and potential risks associated with each step. Therefore, facilities must conduct detailed risk assessments periodically and update the CCS accordingly to ensure compliance with regulatory standards and maintain product quality and patient safety.

7.4.2 Aseptic Process Validation

Establishing an effective aseptic PV strategy is paramount in ATMP manufacturing, as final product sterilization is not possible when the final product contains live cells or is greater than the 0.2 µm sterilizing filter. Aseptic PV ensures that the manufacturing process, including environment and handling, consistently produces a product that is free from microbial contamination ingress. This process involves a comprehensive strategy encompassing aseptic simulation tests, media fills, monitoring of critical processing steps, and rigorous training and qualification of personnel.

7.4.2.1 Aseptic Simulation Tests

Aseptic simulation, often referred to as media fill, is a critical component of aseptic PV. The media selected for this process must be a sterile microbiological growth medium, typically Tryptic Soy Broth (TSB), but may also include an alternative placebo that supports microbial growth. The media should represent optimal growth conditions for the types of microbial contamination most likely to be encountered in the specific manufacturing environment. Selection of the appropriate media should be based on a documented risk assessment and must be scientifically justified. To confirm the suitability of the selected media, growth promotion testing must be performed in accordance with compendial requirements, such as USP <71>, USP <61> [53], and Ph. Eur. 2.6.1 [52], to demonstrate that it can adequately support microbial growth. This testing validates the ability of the media to detect contamination during the simulation process. Additionally, any mock materials used in the simulation must not compromise the integrity of the process by containing antibacterial or antifungal properties.

Actual process and environmental conditions should be followed as closely as possible to demonstrate the adequacy of the aseptic process to prevent contamination. The simulation, therefore, should use the same equipment, incorporating typical interventions and recovery scenarios (i.e., transferring to a new single-use kit and/or transfer bag), and account for expected operational challenges. These include, but are not limited to, maximum operating personnel in a room and working overnight shifts.

7.4.2.2 Modifications in Aseptic Simulation Test

Certain steps in the process that are inherently longer, such as centrifugation or incubation, may be shortened during simulation if a robust justification supports the modification. Incubation duration for process simulation with growth medium must be scientifically justified and based on the process to ensure it reflects the intended conditions accurately, with an importance placed on covering the ideal conditions for microbial growth. For processes using closed systems, the focus should be on the aseptic connections and transfers to and from the system. Furthermore, all operations with product contact surfaces should ensure that all parts of the container come into contact with the growth medium.

7.4.2.3 Bracketing and Matrix Approaches

The separation of the process into key stages can facilitate a bracketing and matrix approach in aseptic PV that identifies similar operations between key steps. Unlike a filling line, where operations are measured in hours, many ATMP processes take weeks, necessitating step qualification for practical reasons. A bracketing and matrix approach is especially beneficial in cases where two products share a similar piece of manufacturing equipment, a fill and finish system, or a cell processing system, for instance. Where these operations are identical, it may be appropriate to leverage prior aseptic validation work if the risk is assessed accordingly and the rationale to exclude these operations is scientifically supported. When implementing equivalent process closure initiatives across multiple processes, this approach aids in reducing repetitive testing while still capturing the worst-case scenarios within the matrix.

7.4.2.4 Validation and Periodic Revalidation

For initial validation, at least three consecutive successful aseptic simulation tests are required for each production process. To maintain ongoing assurance of aseptic conditions, these simulations should be periodically repeated, with the frequency determined by a documented risk assessment. Regulatory expectations, such as those outlined in EU Annex 1 [35] and FDA guidance [81], generally require that APS be performed at least every six months for each process. Unique accommodations may be justified for processes executed infrequently, such as those occurring fewer than two times per year. However, if more than one year has passed since the last batch was manufactured, it is recommended to perform an additional three aseptic simulation runs beyond the original qualification set to reestablish the validated state of the process.

In addition, significant changes to the aseptic process, such as the introduction of new single-use consumables, modifications to the Heating, Ventilation, and Air Conditioning (HVAC) system, or changes to materials intended to support process closure, require revalidation to confirm that such alterations do not compromise the sterility assurance of the product. In these cases, three consecutive, successful APS runs must be performed to reestablish the validated state of the process. Operators involved in these simulations must already be qualified in aseptic techniques applicable to the process; APS intended for PV must not be used to qualify untrained or newly trained personnel. If operator qualification on new or modified procedures is required due to process change, this should be completed before executing the validation runs, using separate qualification activities such as media fills or simulation exercises specifically designed for training purposes. This approach ensures that the process simulation reflects a validated and controlled state, consistent with regulatory expectations.

7.4.2.5 Sterilization and Filter Validation

Some final products or in-process materials may be sterilized via passage through a 0.2 µm filter into a presterilized container. When a sterile filter is used to ensure product sterility, it is critical to validate the filter to confirm it does not adversely impact product quality. This includes a comprehensive assessment of E&L, as well as evaluating the retention of excipients and ensuring filter integrity can be maintained under the maximum expected process volume and stress conditions.

Validation of filter integrity should include standard pre-use and post-use integrity tests such as bubble point, diffusive flow, or pressure hold. However, in the context of gene therapy products such as rAAV vectors, the implementation of Pre-Use Post-Sterilization Integrity Testing (PUPSIT) may not be technically feasible at early clinical development stages due to small batch volumes and dead volume limitations of PUPSIT assemblies. In such cases, a phase-appropriate approach should be adopted. Waiving PUPSIT may be acceptable if it is justified through a comprehensive risk assessment and supported by appropriate risk mitigation measures. Sponsors are encouraged to engage with regulatory agencies to discuss the appropriateness of such approaches on a case-by-case basis.

The filter's microbial retentivity must be validated using a bacterial challenge test with *Brevundimonas diminuta*, prepared and cultivated in accordance with ASTM F838-20 [87]. The organism shall be inoculated into the product or a suitable surrogate, and the bacterial challenge test must be conducted under actual process conditions to confirm the filter's capacity to retain microorganisms during use. The strategy for filter validation—including microbial retention, E&L, and integrity testing—must be documented in a risk assessment.

Using the same filter across multiple product lots is not permitted. However, extended use of a filter for longer than one day may be acceptable within a single lot or batch, provided this is supported by a validated hold time and performance justification. In ATMP manufacturing, open steps are often associated with the preparation of in-process reagents. Bulk preparation and subsequent sterile filtration of these reagents through validated 0.2 µm filtration is a widely accepted approach to introduce process closure while maintaining aseptic integrity without significant modification to the established open process.

7.4.3 Process Validation Strategy

Establishing robust PV strategies is critical to ensure the consistent production of safe and effective therapies in the rapidly evolving field of ATMPs. Due to the inherent variability of starting materials—such as autologous cells—and the complexity of manufacturing platforms, a structured risk-based approach to PV should be developed for each facility. This strategy should be tailored to the product and process, while also enabling the potential to leverage prior knowledge and process characterization data from similar products within a multiproduct facility. This supports the definition of meaningful acceptance criteria and process parameters that are commensurate with risk.

As outlined in the EMA guideline on quality, non-clinical and clinical requirements for investigational ATMPs [88], it is essential that PV strategies address the control of critical manufacturing steps and intermediates. These steps often represent the highest risk to product quality and consistency. Therefore, clear control strategies, including in-process controls, should be established early and verified through process characterization and risk assessments.

Additionally, establishing hold times and storage conditions for critical intermediates is a key enabler of process robustness. The earlier in development such data are generated and justified, the more reliable and defensible the overall PV strategy will be. Data supporting intermediate stability, process hold conditions, and their impact on product CQAs should be integrated into the control strategy and validation lifecycle framework.

Given the limited clinical and manufacturing datasets that may be available during early development phases, it is recommended that facilities adopt a lifecycle validation framework consistent with ICH Q8(R2) [41], Q9(R2) [1], and Q10 [42]. This allows for adaptive integration of process changes both pre- and post-approval while ensuring sustained product quality and patient safety.

In addition, for ATMPs using viral vectors or components of biological origin, viral safety must be a fundamental consideration within the PV strategy. This includes the implementation of viral clearance studies, inactivation steps, and sourcing controls for animal-derived materials in alignment with ICH Q5A(R2) [89]. Demonstrating effective viral clearance is essential for regulatory acceptance, patient safety, and long-term product success in global markets.

7.4.3.1 Documentation and Validation Plan Structure

A well-structured validation plan is essential to execute a PV. The plan should include detailed documentation of the strategy to justify the defined CPPs, CQAs, and acceptance criteria that were derived from developmental and characterization data. A description of the equipment and facilities, along with qualification for intended use in the manufacturing process, should be outlined. The analytical methods used for the in-process controls shall be qualified at a minimum, with an expectation of method validation by market authorization. Comprehensive testing strategies, which include a sample plan and rationale explaining the selection of batches and testing frequencies, must be documented, especially in case of limited runs. Procedures shall be established and all personnel assisting in the PV campaign should be trained accordingly to represent the release and certification of batches, ensuring each batch has met the predefined specifications, regardless of human use. Finally, there should be a detailed approach to handling nonconformances during the PV and clarification when re-execution would be required versus an additional run to meet the required consecutive runs based on risk. Overall, a comprehensive validation plan forms the backbone of consistent, compliant, and high-quality manufacturing processes, fostering robust adherence to company and regulatory requirements.

7.4.3.2 Defining the Number of Runs

Regulatory guidance [29] suggests that, as a minimum, three consecutive batches under routine manufacturing conditions are used to validate a process. However, the variations in starting material, particularly for autologous therapies, necessitate adjustments to this number. The number of runs determined may be justified by demonstrating equivalence with similar products manufactured on-site or by establishing minimum criteria for starting material. Or they may be limited due to a rare disease with few batches available. The flexibility in the number of validation runs must be documented in a risk assessment and should be supported by additional in-process testing to demonstrate consistent product quality and process control. Despite the inherent variability of starting material and other challenges, the number of runs determined should reasonably support that a product can be manufactured within specified quality attributes given the control of specific parameters.

7.4.3.3 Establishing Acceptance Criteria and Considerations

Acceptance criteria should be rigorously defined to ensure that despite the inherent variability in starting material, the final product consistently falls within an acceptable quality range. These criteria should be established based on a thorough analysis of developmental data and should be documented comprehensively. Furthermore, the acceptance criteria should reflect a deep understanding of the product and process interdependencies, ensuring that they are realistic and achievable while safeguarding product quality. Some ATMPs pose the challenge of limited data available to make statistically meaningful criteria, and therefore it is recommended to use similar products, where possible and appropriate, as determined in a risk assessment. Additionally, when surrogate material is used in characterization runs to offer additional data points for statistical analysis, it is important to consider the impact on these established ranges when using patient material, if any.

The challenge of limited data regarding ATMPs emphasizes the importance of identifying the CQAs, as based on the impact to patient safety and efficacy, and prioritizing control around these attributes. Therefore, control limits should be established, allowing for potential variation while ensuring the final product meets predefined specifications. The monitoring of these process parameters throughout the manufacturing process will substantiate the limits with patient material and thus assist in future validation campaigns with setting appropriate limits. This is ultimately achieved post PV and during the continuous process verification lifecycle of a product. During this phase, the monitoring of identified parameters based on risk provides real-time assurance that the process remains controlled and capable of consistently meeting the quality attributes.

Developing a robust PV strategy for ATMP products is paramount to maintaining the consistent production of high-quality, safe, and effective therapies. Tailoring this approach to a company's specific facility, while leveraging data from similar products for a multiproduct facility, allows for establishing robust acceptance criteria that align with risk-based assessments. Documenting this process in a comprehensive validation plan ensures that CPPs, equipment qualification, analytical methods, and testing strategies are effectively outlined, fostering adherence to company and regulatory requirements. Acceptance criteria should be clearly defined and supported by developmental and characterization data, with consideration for inherent variability in starting materials. This strategy must emphasize continuous monitoring throughout the product's lifecycle, allowing for the early identification of nonconformances and timely process adjustments. Ultimately, by implementing a meticulous, risk-based validation strategy that accommodates limited datasets and evolving process changes, manufacturers can ensure product quality, safety, and regulatory compliance throughout product development and commercialization.

8 Cleaning

8.1 Introduction

Cleaning validation is a requirement for biopharmaceutical operations whenever hazards exist for product contamination from product contact surfaces. Regulatory agencies expect the development and validation of cleaning methods designed to prevent product cross-contamination to safeguard patient safety and product quality. ATMPs are typically manufactured with single-use components, and consequently, do not require product cleaning methods. However, any surface from components or equipment that is reused or any contamination hazards that may impact the ability to execute effective aseptic processing steps may still be part of the process and therefore requires controls to prevent product carryover and cross-contamination.

For readers seeking in-depth understanding of cleaning validation concepts, scope, and practices, further guidance can be obtained from the ISPE Baseline Guide Risk-Based Manufacture (Risk-Mapp) [90] and the ISPE Good Practice Guide on Cleaning Validation [10], which provide a comprehensive framework and more detailed methodologies.

8.2 Cleaning Validation Program

Requirements for creating a cleaning validation program are clearly defined in regulatory guidance documents, such as FDA [91, 92], EMA [6, 93, 94], Health Canada [37], and ICH [95]. Regulatory guidance may vary between agencies, but all follow similar principles that include phases organized as a process lifecycle:

1. **Cleaning process design:** This stage is used to design the cleaning method and typically consists of identifying the residues to be cleaned, selecting the cleaning agents, determining critical parameters for the cleaning method, selecting the cleaning method, and defining the cleaning limits (based on Health-Based Exposure Limits (HBELs) or another cleaning limit scientifically agreed-upon with regulators), sampling plans, and cleaning validation acceptance criteria.
2. **Cleaning process qualification:** The second stage qualifies the cleaning method for use and includes the qualification of equipment, validation of analytical methods used by the cleaning procedure, adoption of visual inspection criteria, creation of written procedures to execute the cleaning method, and execution of qualification protocols to generate the data necessary to qualify the process.
3. **Continued cleaning verification:** The last stage of the cleaning lifecycle consists of monitoring the cleaning method performance, evaluating changes to the method, and periodically assessing method effectiveness.

8.3 Early-Stage ATMP Cleaning

For investigational ATMPs, cleaning verification is acceptable after each batch. Equipment cleaning data must be captured after each batch and evaluated to support the equipment cleaning conclusion. This information, and the use of cleaning records for each piece of equipment, can be used to support the cleaning verification practice due to the lack of identical batches.

Analytical test methods should be validated to analyze rinse and swabbing samples. The selection of sampling sites should be based on hard-to-clean areas (i.e., “worst-case”).

Equipment visual inspection is conducted as part of the routine cleaning verification.

Single-use components may be discarded, and equipment non-product contact surfaces may be wiped with low-lint wipes.

8.4 Equipment and Facility Cleaning and Sanitization

The cleaning/decontamination procedures applied to multi-use equipment coming into contact with the product should be validated. It is preferable to use single-use disposable technology to avoid the need for cleaning validation because many ATMPs are generated in limited batches and multiple lots are not always available to study or to plan for equipment cleaning studies to perform validation.

A risk assessment should be used to determine the extent of the cleaning required and the procedures necessary to conduct validation activities.

In general, equipment should not be moved from high-risk areas to other areas, or between high-risk areas (e.g., equipment used for the handling of cells from infected donors or the handling of oncolytic viruses). When this happens, appropriate measures need to be applied to avoid the risk of cross-contamination.

Where disinfectants are used, the efficacy thereof should be checked. It is also advisable to use more than one type of disinfectant to avoid the development of resistant strains and to achieve a broader range of bio-decontamination activity. Disinfectants, detergents, and cleaning materials used in clean areas of Grades A and B should be sterile.

8.5 Equipment and Process Platforms

Manufacturing ATMPs can require a variety of equipment, systems, and platforms. These can vary from entirely manual processes to fully automated process steps. The unit operations may be conducted in cleanrooms, BSCs, or isolators, or be fully enclosed processes. Therefore, a risk assessment should be performed to determine the level of controls required for each process phase to ensure no product contamination from previous processing steps.

Figure 8.1 illustrates the possible causes of failure for a cleaning process. Risk assessments should consider these potential issues to improve the overall design of the cleaning program.

Figure 8.1: Elements of a Validated Cleaning Process [10]

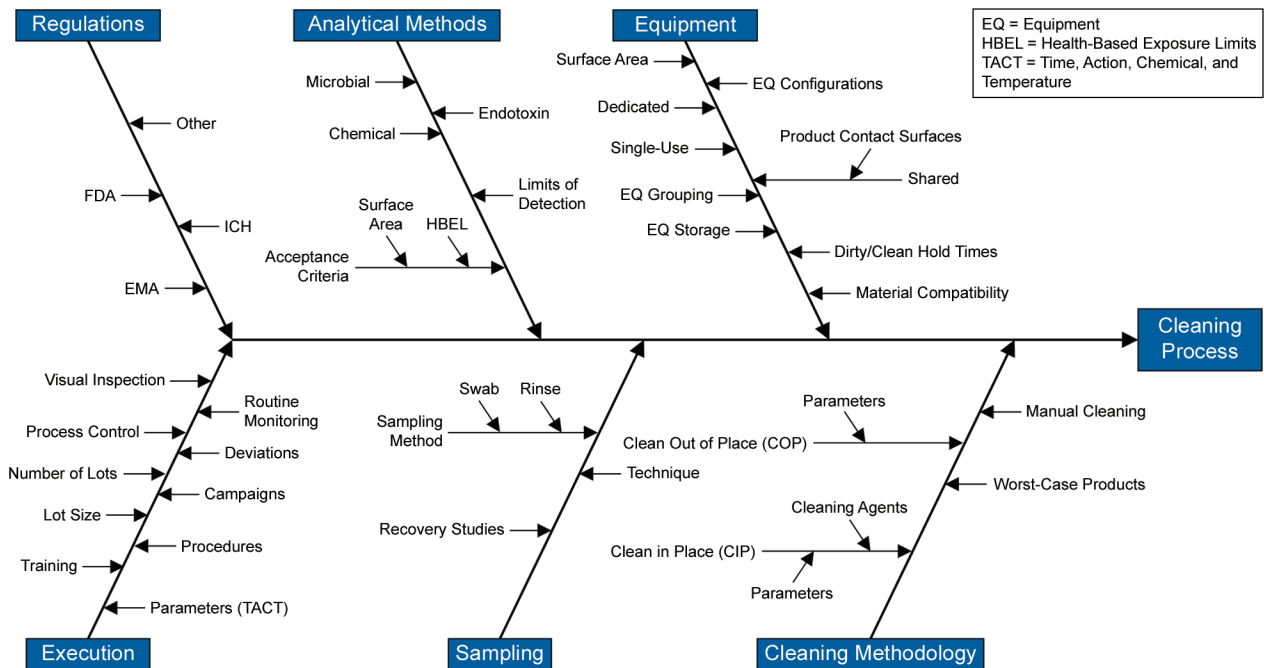
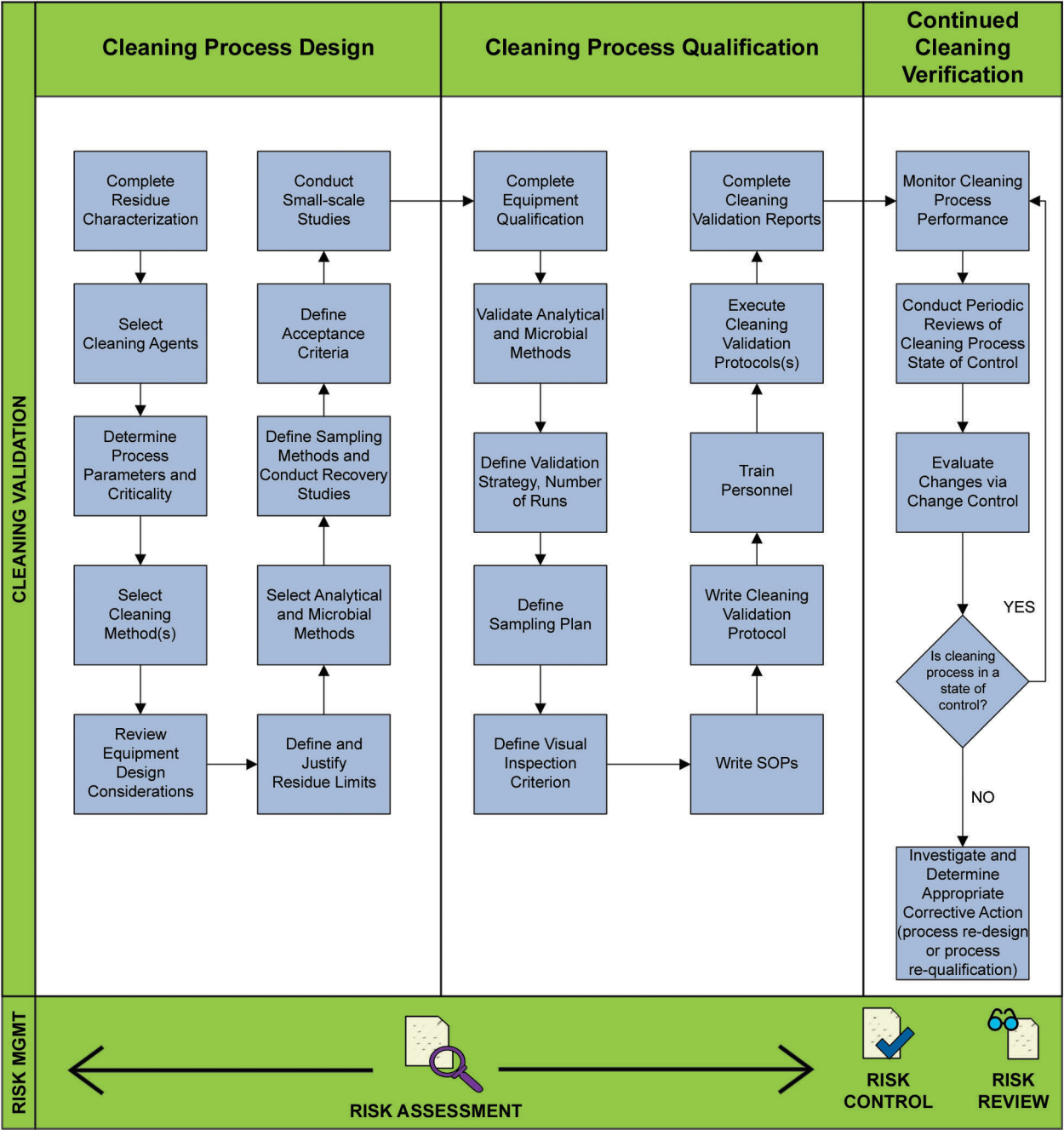


Figure 8.2 describes the typical roadmap to validate a cleaning process, where cleaning design, qualification, and ongoing monitoring is presented. In addition, it illustrates the risk management phases as they relate to the development of cleaning processes.

Figure 8.2: Cleaning Process Validation Road Map [10]



The following considerations are relevant:

- Develop necessary analytical methods to detect impurities or contaminants of interest
- Single validation studies should address worst-case conditions as an acceptable strategy
- Develop detailed cleaning procedures for equipment and tools
- Justify sampling plans (e.g., frequency, location)
- Establish scientifically justified cleaning limits (based on HBEL or equivalent)
- Perform ad-hoc studies to determine acceptable levels of residual VHP for isolator decontamination processes

8.6 Single-Use Equipment and Components

The use of single-use equipment, components, and tools is common in ATMP processes. Cleaning validation is not required for disposables and single-use systems or components. Refer to the *ISPE Good Practice Guide: Single-Use Technology* [15].

Additional controls to consider include the following:

- Confirm cleanliness level of single-use components from the manufacturer prior to use
- Establish effective specifications for all single-use components to ensure consistent quality and performance
- Define clear supplier controls
- Establish procedures to conduct periodic review for changes (from suppliers and manufacturers)
- Assess stability and storage conditions for single-use components
- Determine risks for E&L
- Establish effective controls to decontaminate in place or inactivate potentially biohazardous materials prior to disposal
- Determine risks for particles originating from the single-use components and establish controls to mitigate the risks to meet process requirements

9 Supply Chain

9.1 Introduction

This chapter focuses on the supply chain for ATMPs and the risk-based approach to the validation requirements. As part of a supply chain for any pharmaceutical product, raw materials and finished product storage as well as shipments are part of the topics in this chapter. A unique part of the supply chain for ATMPs is the dependence on donor material. Because of the mostly on-demand nature of these types of products, donor material collection, shipping, and handling require special attention to ensure that the finished product meets the quality and efficacy requirements for the patient. The distributed nature of the supply chain—from the hospital to the logistics and storage to the manufacturing site and then back to the hospital/clinical site for patient administration—requires all organizations involved to qualify their processes and systems to meet certain requirements.

9.2 Cell Donor Collection

Gene-modified cell therapy—also known as *ex vivo* gene therapy or sometimes referred to as simply cell therapy—includes genetic modification of the cells from donors. If the donor cells also originate from the patient receiving the resulting DP, the therapy is referred to as autologous. If the cells originate from one or more donors different than the patient receiving the resulting DP, the therapy is referred to as allogeneic. There may be increased checks for infectious diseases and other safety factors for cells that are used for allogeneic therapies. Regardless of cell type (blood cells, bone marrow, stem cells) or the autologous versus allogeneic nature of the therapy, from a supply chain perspective, the considerations are similar.

Starting with apheresis, the first consideration is the transportation of cells to the manufacturing location. This transportation can be achieved with the cells fresh or frozen. In both cases, the impact of the transportation conditions and duration of transport prior to the start of manufacturing on cell viability and quality should be studied. Early in clinical studies, vendor qualification of shipping containers can be leveraged. In later phases and prior to commercial launch, a product-specific study is expected. Depending on the distances and the transportation options between donor cell collection centers and the manufacturing location, freezing may be necessary to ensure manufacturing starting material quality. In this case, freezing protocols may need to be defined and staff training needs to be considered. Qualification of freezing may be needed in later clinical phases.

Clinical materials that are used for collection (e.g., IV bags, vials, syringes, catheters, tubing, gauze pads, alcohol swabs) may be provided to the collection centers as kits. The use of kits provides greater control of the materials used in the clinical process. The primary packaging of starting material is one of the critical aspects of the supply chain and may impact cell quality. Qualification studies should include the ability to support material temperature requirements, impact of E&L for materials that will come in contact with the cells, container closure integrity, etc. Primary labels should also be considered as part of the starting material. Further, label stability at the shipment temperature should be confirmed and label content (e.g., ISBT 128 Standard [96]) should be decided prior to initiation of clinical studies.

9.3 Other Raw Materials

Numerous raw materials other than the donor cells are required to ensure supply continuity. General principles for starting and source materials and for materials management as described in ICH Q7 [95] and Q11 [97] apply to ATMPs. Although the required raw materials for manufacturing are main considerations, materials related to DP QC release testing are also critical to consider as part of the supply chain continuity. As for any raw material risk, the manufacturing process and quality grade (research grade or GMP) of raw materials should be considered during the development phases. Using GMP-grade materials early on in the development, when possible, can prevent any changes or comparability studies being required at later phases.

A phase-appropriate testing strategy for raw materials should be deployed. During the early phases of clinical trials, the release of raw materials can rely on the vendor CoA. However, a raw material testing program is expected to be implemented before commercial licensure. If regulatory filing in multiple regions is expected, sourcing and testing per the most stringent compendia should be instituted. Supplier audit and qualification programs are important to establish early during the product lifecycle to verify GMP status of suppliers and as a complementary initiative for QC and supply continuity. The following are further related considerations.

9.3.1 *Vectors and Plasmids*

As one of the complex raw materials, vectors that are used in the production of ATMPs or vectors that are used directly as gene therapies require attention and advanced planning. Although vectors used in gene therapies can be deemed bulk DS and DP, regulatory agencies recognizing the complexity in manufacturing vectors assign DS status even to those vectors used as part of cell therapies, where vectors are added to cell culture to achieve the desired gene editing. Because of this elevated importance, a QbD framework should be implemented to vector manufacturing early on. This begins with defining the CQAs, having appropriate process characterization to define CPPs and a control strategy, and performing PPQ before applying for market authorization. Because this work can take a number of years, it is important to plan and start as early as possible. The cell lines and plasmids used during vector production would require additional attention before the start of process characterization. To avoid variability, it is advisable to create an MCB and rely on WCBs created from the MCB for the duration of product lifecycle.

Vectors produced with transient transfections require various plasmids, including transfer plasmids. These need to be added to the supply planning and risk assessments. USP <1043> [53] considers plasmids an ancillary raw material. The FDA guidance on CMC information for human gene therapy INDs [32] considers plasmids as intermediates. The EU Directive 2001/83/EC [98] considers plasmids as starting materials. Therefore, the regions where regulatory filing will be submitted should be considered as part of the selection, risk assessment, and qualification activities from an early point in development. The most stringent guidelines based on the target licensure regions should be followed. Although plasmid manufacturing and characterization may not be as complex according to regulatory requirements, traditional aspects of forecasting and stock building are required to avoid delays.

9.3.2 *Critical Raw Materials*

It is important to define critical raw materials early in process development to be able to identify alternative sources, if possible. Linking CQAs to raw materials can help determine criticality and differentiate raw materials. Many raw materials such as cell culture media, scaffolds, matrices, reagents, buffers, and excipients can also be defined as critical due to their complex nature and/or their availability from a single supplier. A Raw Material Risk Assessment (RMRA)—which considers various attributes such as impact to patient safety, product quality and/or potency, and supply considerations—will help prioritize studies when necessary. The RMRA document can be used as a living knowledge depository for additional information learned during development. Although manufacturing-related raw materials are the main focus of an RMRA, it is also important to define the risks related to DP testing-related materials, especially those related to quality and supply chain continuity.

In the early stages of development, the main focus is to ensure that the raw materials have the appropriate functionality to support their intended role in the DS or DP. It is important to source the highest-grade material that meets the criteria to avoid comparability studies at later phases. Understanding of the raw materials will increase along with the usage of multiple lots as a project progresses through the product lifecycle. At the later stages of development, quality and compliance considerations will become more important, especially toward product registration, launch, and commercial supply.

For raw materials used in early-phase projects, it may be sufficient to check the supplier CoA. As the project moves closer to commercial production, more in-depth testing may be needed, including identity verification and compendial testing. Especially for excipients, market registration strategies should be aligned with the type of compendial raw materials chosen [99].

9.3.3 Single-Use Systems

Single-use systems are widely used during ATMP processes, especially during manufacturing of autologous therapies. These components (e.g., bags, tubing sets, connection kits) are generally accepted as essential to product sterility and safety. Although these components have defined materials of construction and are widely used, documentation and, if required, further testing related to particulates and E&L may need to be considered. Risk assessment can be implemented early in the clinical phases, and studies related to single-use systems should be planned before commercial licensure.

9.3.4 Sourcing Strategy and Additional Aspects

A supplier qualification program is important to address the risks of raw material sourcing. Vendor audits can be implemented in a risk-appropriate and clinical phase-appropriate manner. Early in the clinical phases, relying on reputable vendors or vendor relationships can suffice; not having supplier qualification is mainly a business risk. However, as the supply chain matures and the product advances to later clinical phases, diligence in vendor assessments is expected. Not having supplier qualification may pose a compliance risk.

Supplier qualification is a crucial step before a commercial license is granted for the product. A supplier qualification program should address the ability of the supplier to meet the necessary standards and regulatory requirements for the materials or services they provide. It also addresses the storage, logistics, and transportation requirements of each material. Various aspects of the supply chain, including temperature requirements and packaging, can be addressed during a supplier qualification program, as these can impact the stability and CCS. USP <1043> [53] provides risk categorization guidance that can be used to prioritize the qualification/validation activities.

9.4 Cryopreservation

Cryopreservation is an important part of the ATMP process to be qualified to support the supply chain. Early-stage developers may often avoid freezing cells for transport to avoid introducing manipulations that could impact cell viability or potency. Cost or time to clinic can also add to pressures during development that may result in the need to use fresh donor material or DP. Scaling up a supply chain based on fresh cells will pose additional challenges at later stages, as cryopreservation is the only sustainable path to scaling.

The main aspects related to cryopreserved material that need to be defined include the following:

- Controlled freezing
- Hold time during storage or transport
- Controlled thawing prior to use

Justification of the practices or qualification of these aspects is needed before starting clinical trials. During process development, widely accepted controlled rate freezing and thawing parameters can be used prior to clinical studies. Similarly, early studies for DP stability may justify the hold time at cryogenic or freezing temperatures. However, these processes should be characterized further and should be included in PV before commercial license application.

Based on whether the process starts with fresh or frozen donor material, the need for cryopreservation may start at the clinic. The controls around cryopreservation can impact the quality of the starting material. Thus, it is important to define the critical parameters and how they are controlled during the process. If freezing of the donor material occurs in the clinic, it is possible for various clinics and technicians to be involved with this process. Defined and clear procedures, training of involved personnel, and a possible periodic qualification of the freezing process at each site may reduce the variability in the starting material due to this important step.

Developers also may opt for fresh donor material shipments because many hospitals and clinical sites that draw patient/donor tissues or blood lack the infrastructure and/or knowledge for cryopreservation. However, starting with fresh donor material necessitates validation of the duration of the storage at non-frozen conditions. This could result in more regional manufacturing or cryopreservation centers during clinical and commercial phases.

The final DP may need to be cryopreserved to extend the shelf life and support the delivery of product to the clinic for administration. This is essential for maintaining the integrity and functionality of ATMP products during storage and transport. Given the complexity of scheduling patients, especially if the product is autologous and has a multiple-week manufacturing turnaround time, it would be a necessity to cryopreserve the DP.

Introducing cryopreservation and related qualifications/validations early in process development will make supply chain planning for later phases easier and could avoid costly comparability studies. Long-term impacts should be considered when deciding between fresh versus frozen options, as early changes may only require an amendment rather than an entirely new regulatory filing.

9.5 Transportation, Logistics, and Shipping

There are unique challenges to the ATMP supply chain. Although a cold chain supply is typical for various biopharmaceuticals and vaccines, the just-in-time, on-demand nature of the ATMP supply chain and cryogenic transportation is complex. Therefore, there is a need to address various elements of logistics and shipping between the hospitals (where donations of human cells and tissues are made), the manufacturing facilities, and the location of the final product administration (which could be the same donation facility or a different one focusing on administration). In cases where the donor material or DP will be outside of well-managed hospitals and manufacturing sites, there is also an increased risk of damaging the product and ultimately not being able to administer it to the patient.

Because environmental and packaging conditions can impact the quality of the ATMP products, there is a need for greater knowledge and control during the transportation of these products. The first step is to ensure delivery of consistently high-quality products to patients. This is product-related knowledge. Product stability as a function of temperature and storage time should be studied during development and QC stability studies using appropriate assays. The approach to these studies is outside the scope of this chapter; however, data generated during these studies are used to set time and temperature ranges appropriate for the product during transit and storage. Shippers should be qualified to maintain the product within the specified temperature ranges up to a qualified shipping time. Considerations for the validation of transportation and logistics include:

- **Pack-in and pack-out procedures:** The packing configuration of minimum and maximum loads and the impact of these on temperature control should be considered while designing the studies.
- **Temperature control:** These ATMP products require strict temperature management. The validation process must ensure that the packaging can maintain the required temperature range for the duration of the shipment. For refrigerated shipping, the range is typically between 2°C and 8°C. For cryogenic shipping, a temperature between -150°C and -196°C is maintained. During early clinical phases, shipping qualification can be based on vendor-provided studies for shippers (e.g., liquid nitrogen shippers). Product-specific validation of the shippers and qualification of the shipment lanes providing temperature versus time data during transportation can be performed during transport simulations. Worst-case product-specific shipping validation is preferred by health authorities.
- **Transport simulation:** Simulating transportation conditions should be considered early on during clinical studies to support licensure. These simulations should include testing various stresses such as vibrations, drops and shocks, and changes in pressure and temperature on the packaging and the simulated product. Transportation industry standard test methods such as ASTM D4169 [100] can be consulted while designing the studies. If the product is transported via an airplane, pressure studies with different relative atmospheric conditions (such as in an airplane at 30,000 ft) will be important to conduct early in the clinical lifecycle to avoid any unforeseen impact.

- **Packaging integrity:** ATMP products are packed in various layers other than just primary: secondary packaging (overwraps and/or cassettes) and tertiary packaging (cassettes and/or shipping containers). The packaging must be robust enough to withstand wide temperature ranges and to protect the product from physical damage and contamination. Testing the packaging for durability, seal integrity, and sterility is important. Transport or stress simulations can provide important information about failure points and prevent product loss.
- **Labeling integrity:** As part of the integrity of packaging, labeling integrity (including secondary and tertiary packaging) should be considered. Labels that are specifically designed for cryogenic or wide temperature range conditions should be chosen early during development to avoid any late-stage changes.
- **Monitoring and tracking:** Monitoring systems are crucial for ATMP logistics. These systems track the location and/or condition of the product throughout the shipping process and can provide alerts for any deviations from the set parameters. Real-time access to the tracking data is not necessary but is advisable to implement for greater control, especially as the supply chain grows.

Starting with clinical studies, compliance with regulations from the US FDA [2], EMA [3], and other local regulatory authorities is mandatory for transportation of the ATMP products. A thorough risk assessment should be conducted to identify potential hazards in the logistics process and implement mitigation strategies. Any validation must follow the current GMP guidance. Complete documentation of the validation process is essential for traceability and to provide evidence of compliance in case of audits. An ongoing evaluation is advisable for continual improvement and adaptation to new regulations in this area.

Cleaning and maintenance of the equipment used to transport the primary containers should also be considered as part of risk management. The effectiveness of the procedure should be incorporated into shipping validation studies. The logistics providers used and their quality management systems are important factors when deciding on the supply chain for ATMP products. Providers who handle the product should be familiar with, and capable of, maintaining appropriate storage conditions. Routine follow-up on key performance indicators should be part of vendor management from early clinical stages. Security of the product during transportation through COI and COC is of utmost importance, especially for autologous therapies. These general considerations are covered in Section 9.7.

9.6 Patient Administration

Patient administration is part of clinical or health care provider responsibility. However, due to the complex nature of ATMP products, sponsor and manufacturer company guidance may be necessary to ensure the patient consistently receives a high-quality product.

9.6.1 Kitting

Similar to the kitting requirements on the donor collection side, the materials used to administer the finished product to the patient need to be specified or qualified to ensure a safe, efficacious product. Material compatibility needs to be examined and qualified. In addition, clear directions need to be stated as part of the protocol for administration.

9.6.2 Special Administration Protocols

Although the patient administration portion of the product is covered from a medical perspective with physicians and nurses, the product may have unique requirements that the sponsor organization is responsible to communicate. There can be a cryopreservation thawing process that defines controlled thawing rates and any special equipment needed beyond a water bath. In addition, there could be postproduction processing of the final product such as cryoprotectant removal such as Dimethyl Sulfoxide (DMSO) that could be further defined. Finally, the product may need final formulation adjustments conducted by the hospital pharmacy, which will require a stated process and will need qualification. If there are specific requirements for administration, an initial training and routine follow-up on training may be required to ensure consistency in administration.

9.7 General Requirements

Global regulations impose tight control over ATMP logistics. The following sections outline additional considerations.

9.7.1 Chain of Identity

There is a need to maintain identification of the starting material, manufacturing work in progress, and finished product throughout the supply chain. Although there is no regulatory mandate for what type of identifier to use, the industry standard is defined by the International Council for Commonality in Blood Banking Automation (ICCBBA) [96]. A new IG-050 ISBT 128 COI Identifier implementation guide document [101] was published in December 2023. The document supports the ST-028 ISBT 128 Standard COI Identifier document [102] published in December 2022.

In terms of the unique materials that must be identified for use within a manufacturing batch or material sourcing such as autologous cells, a COI should be established for each entity in the supply chain. It should be shared across all partners and applied to the material's label to identify it. The process should be qualified and requires all organizations in the supply chain to follow it and show that they have qualified their use of it. If there are information systems that support the lifecycle of the COI, they would also need to be qualified and validated.

9.7.2 Labeling

Labels are an important part of products, and all aspects of the label should be considered before finalizing the design. Important considerations for labels include container type, label application procedure, desired label type, final storage temperature, packaging configuration during shipment, and target countries for clinical activities.

Regulatory agencies have certain baseline requirements for DP labeling, including lot number, expiration date, and dosage (e.g., Current Good Practice for Finished Pharmaceuticals: Packaging and Label Control, 21 CFR Part 211 Subparts G and J [91]). The adoption of ISBT 128 [96] for cellular therapy products is increasing steadily worldwide. In addition to specifying the requirements for the electronic coding of information, ISBT 128 provides a standard labeling format that ensures a consistent layout of the bar codes on the product label.

The autologous therapy supply chain starts with the patient. Thus, especially in early clinical phases, manufacturers send preprinted labels to apheresis sites. With this approach, reaching a higher number of patients can be challenging as the therapy scales up. Therefore, cell therapy orchestration systems and on-demand label printing should be considered as an option. Although this option brings its own challenges—such as whether to use printers available at health care provider sites or to provide each site with a label printer—on-demand label printing can be the most compliant option.

9.7.3 Chain of Custody

There is a need to maintain information on identity at every point in the supply chain. Similar to COI requirements, the COC also requires a process to maintain custody of the material throughout the supply chain, and the process must be shared across all partners. The process should be qualified and all organizations in the supply chain required to use it and show that they have qualified their use of it. If there are information systems that support the COI lifecycle, they would also need to be qualified and validated.

9.7.4 Chain of Condition

For any material used in the manufacturing process—including donor material, raw materials, and finished product—the chain of condition must be maintained throughout the supply chain. Due to the risk of changing the properties of the materials or finished product, the condition needs to be maintained. This can include temperature, humidity, and container closure integrity. The supply chain should support the chain of condition with specifications on each relevant parameter. The supply chain should also be designed and qualified to support those specifications as well as to document results of logistics, transport, and storage. In addition, the conditions should be documented for review by the responsible quality organization.

9.7.5 Chain of Compliance

There is a need to maintain GMP status of all equipment and systems supporting the transportation and storage of the starting material and the finished product. Like the COI, the chain of compliance also requires a process to maintain all equipment, instrumentation, and controls used throughout the supply chain. Examples include tertiary containers used for shipping the material or finished product and temperature monitoring devices within them. This should be shared across all partners. As with the COC, the process should be qualified and all organizations in the supply chain required to use it and show that they have qualified their use of it.

If reusable containers are used, they may need to be cleaned prior to the next use. The cleaning process should also be qualified. Although it will not be sterilized, the intent is to ensure there is no contamination from previous uses that might contaminate the next material.

If there are information systems that support the chain of compliance lifecycle, they would also need to be qualified and validated.

9.7.6 Data Sharing Across the Supply Chain

There is a significant level of data shared across the supply chain on the starting material and the finished product. Organizations need to be concerned with manual and automated systems that support information sharing. The processes and associated systems will typically be outside of individual organizations. The data may be shared in documents or more granularly in database systems.

The term “cell orchestration” has been used to describe commercial information systems that support the sharing of data from each organization collaborating in the supply chain. Whether using an orchestration system or not, the process (including security measures) should be qualified to ensure it is executed properly and no data is shared inappropriately.

9.7.7 Clinical Considerations

Although the supply chain does not formally manage product clinical trials, it must support the parameters and controls that the clinical trial requires. This includes steps, such as supporting the placebo and final product shipment for the clinical trial sites, that may need to be qualified separately or as part of the clinical trials process or information. Data on the product labeling to indicate clinical trials and ensure any special requirements defined for the clinical materials may also be qualified to support the clinical trial program.

9.7.8 Commercial Considerations

Additional considerations for commercial product include capacity requirements and the CoA process. The sponsor of the product in the supply chain should evaluate, define, and ensure all organizations participating in the supply chain have sufficient capacity in terms of units, timing, and other metrics. This helps ensure proper handling and delivery to the patient receiving the product.

In many situations, the CoA may be used in the supply chain to simplify and ensure proper documentation is created and reviewed. This helps ensure donor material and raw materials can be used by the manufacturing organization for the product batch. This process should be qualified. Sponsor organizations may have a CoA process that can be applied.

10 Phase Appropriate

10.1 Introduction

Due to their transformative nature, ATMPs are often on an accelerated track for clinical trial programs. A phase-appropriate approach within ATMPs should consider the following phases of development.

- Preclinical is the stage in development prior to clinical trials. For the purpose of this Guide, this stage encompasses R&D as well as toxicology studies.
- Early phase is the stage in development that represents Phase I and Phase II (or Phase I/II) clinical trials. An application must be submitted with the appropriate regulatory authority in order to enter this phase (e.g., an IND application is filed with the FDA [2] or a Clinical Trial Application (CTA) is filed with the EMA [3]).
- Late phase is the stage in development that represents Phase III (pivotal) clinical trials. Following successful clinical trials, an application for market approval is submitted to the appropriate regulatory authority (e.g., a BLA is filed with the FDA, or an MAA is filed with the EMA).
- Commercial is the stage in development after licensure approval.

QRM should be applied throughout the product development lifecycle to ensure that there is appropriate focus on product quality and patient safety. Throughout the ATMP product development lifecycle, additional knowledge is gained and technology is improved. Any changes that are considered should be evaluated for risk and impact to the phase-appropriate validated state. The appropriate level of impact should be assessed and actions taken to maintain the validation and to ensure patient safety.

10.2 Phase-Appropriate Validation

The example in Table 10.1 includes recommendations for what should be in place to successfully implement each phase. Specific ATMP product considerations, risks, and regulatory commitments should be considered when companies design their phase-appropriate validation approach.

Table 10.1: Example of a Phase-Appropriate Validation Strategy

	Preclinical	Early Phase	Late Phase	Commercial
Analytical	Method development	• Sterility and microbial assays validated • Patient safety assays validated (e.g., test for replication-competent retrovirus) • Suitability of methods for CQAs	• Potency assay validated • Analytical methods for batch release validated • Analytical methods for stability testing validated	• All analytical methods validated • Continued method monitoring

Table 10.1: Example of a Phase-Appropriate Validation Strategy (continued)

	Preclinical	Early Phase	Late Phase	Commercial
Process	Process development	• Stage 1 – Process design • Aseptic process simulation • Appropriate monitoring and control measures implemented • Equipment IQ/OQ • Potential CQA/CPP	• Stage 2 – Process qualification (PPQ) • Equipment PQ • CQA/CPP	• Stage 3 – Continued process verification
Cleaning	Appropriate for laboratory	• Cleaning and disinfection strategy and rationale	• Qualified facility cleaning and sanitization program	• Ongoing monitoring program
Supply Chain	Shipment tracking (including temperature, if required)	• Packaging configuration, transport conditions, and shipping lanes, all defined in writing • Shipping studies to demonstrate adequacy of defined packaging and transport • Temperature monitoring during shipment • COI/COC established and maintained	• Formal shipping validation for intended shipping lanes	• Continued supply chain monitoring

11 Appendix 1 – References

1. ICH Harmonised Guideline: *Quality Risk Management – Q9(R1)*, Step 4, 18 January 2023, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.
2. US Food and Drug Administration (FDA), www.fda.gov.
3. European Medicines Agency (EMA), www.ema.europa.eu/en.
4. FDA Draft Guidance for Industry: Manufacturing Changes and Comparability for Human Cellular and Gene Therapy Products, July 2023, US Food and Drug Administration (FDA), www.fda.gov.
5. PIC/S Guide to Good Manufacturing Practice for Medicinal Products, Annex 2A Manufacture of Advanced Therapy Medicinal Products for Human Use, 25 August 2023, Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.
6. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Part IV: Guidelines on Good Manufacturing Practice specific to Advanced Therapy Medicinal Products, 22 November 2017, https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en.
7. *ISPE Guide: Advanced Therapy Medicinal Products – Autologous Cell Therapy*, International Society for Pharmaceutical Engineering (ISPE), First Edition, November 2021, www.ispe.org.
8. *ISPE Guide: Advanced Therapy Medicinal Products – Allogeneic Cell Therapy*, International Society for Pharmaceutical Engineering (ISPE), First Edition, January 2024, www.ispe.org.
9. *ISPE Good Practice Guide: Practical Application of the Lifecycle Approach to Process Validation*, International Society for Pharmaceutical Engineering (ISPE), First Edition, March 2019, www.ispe.org.
10. *ISPE Guide: Cleaning Validation Lifecycle – Applications, Methods, and Controls*, International Society for Pharmaceutical Engineering (ISPE), First Edition, August 2020, www.ispe.org.
11. *ISPE GAMP® 5: A Risk-Based Approach to Compliant GxP Computerized Systems*, International Society for Pharmaceutical Engineering (ISPE), Second Edition, July 2022, www.ispe.org.
12. *ISPE Baseline® Pharmaceutical Engineering Guide, Volume 5 – Commissioning and Qualification*, International Society for Pharmaceutical Engineering (ISPE), Second Edition, June 2019, www.ispe.org.
13. *ISPE Baseline® Pharmaceutical Engineering Guide, Volume 6 – Biopharmaceutical Manufacturing Facilities*, International Society for Pharmaceutical Engineering (ISPE), Third Edition, November 2023, www.ispe.org.
14. *ISPE Good Practice Guide: Packaging, Labeling, and Warehousing Facilities*, International Society for Pharmaceutical Engineering (ISPE), First Edition, June 2012, www.ispe.org.
15. *ISPE Good Practice Guide: Single-Use Technology*, International Society for Pharmaceutical Engineering (ISPE), First Edition, November 2018, www.ispe.org.
16. *ISPE Good Practice Guide: Validation 4.0*, International Society for Pharmaceutical Engineering (ISPE), First Edition, June 2025, www.ispe.org.
17. FDA Guidance for Industry: Expedited Programs for Regenerative Medicine Therapies for Serious Conditions, February 2019, US Food and Drug Administration (FDA), www.fda.gov.

18. European Medicines Agency (EMA), “PRIME: priority medicines,” 5 August 2025, www.ema.europa.eu/en/human-regulatory-overview/research-development/prime-priority-medicines.
19. US Food and Drug Administration (FDA), “Regenerative Medicine Advanced Therapy Designation,” 18 July 2025, www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/regenerative-medicine-advanced-therapy-designation.
20. PIC/S Guide to Good Manufacturing Practice for Medicinal Products, Annex 1 Manufacture of Sterile Medicinal Products, 25 August 2023, Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.
21. Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.
22. European Commission (EC), https://commission.europa.eu/index_en.
23. International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.
24. World Health Organization (WHO), www.who.int.
25. Health Canada, www.canada.ca/en/health-canada.html.
26. Therapeutic Goods Administration (TGA), www.tga.gov.au.
27. International Organization for Standardization (ISO), www.iso.org.
28. PIC/S Guide to Good Manufacturing Practice for Medicinal Products, Annex 2B Manufacture of Biological Medicinal Substances and Products for Human Use, 25 August 2023, Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.
29. PIC/S Guide to Good Manufacturing Practice for Medicinal Products, Annex 15 Qualification and Validation, 25 August 2023, Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.
30. EMA Questions and answers on the principles of GMP for the manufacturing of starting materials of biological origin used to transfer genetic material for the manufacturing of ATMPs, 24 February 2021, European Medicines Agency (EMA), EMA/246400/2021, www.ema.europa.eu/en.
31. ICH Harmonised Guideline: *Analytical Procedure Development – Q14*, Step 4, 1 November 2023, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.
32. FDA Guidance for Industry: Chemistry, Manufacturing, and Control (CMC) Information for Human Gene Therapy Investigational New Drug Applications (INDs), January 2020, US Food and Drug Administration (FDA), www.fda.gov.
33. BioPhorum, “Industry recommendation for phase appropriate approach to assay validation in cell and gene therapies (CGTs),” 17 January 2024, www.biophorum.com/download/phase-appropriate-approach-to-assay-validation.
34. ICH Harmonised Guideline: *Validation of Analytical Procedures – Q2(R2)*, Step 4, 1 November 2023, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.
35. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 1: Manufacture of Sterile Medicinal Products, 22 August 2022, https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en.

36. FDA Draft Guidance for Industry: Platform Technology Designation Program for Drug Development Guidance for Industry, May 2024, www.fda.gov.
37. Health Canada Cleaning validation guide (GUI-0028), 29 June 2021, www.canada.ca/en/health-canada/services/drugs-health-products/compliance-enforcement/good-manufacturing-practices/validation/cleaning-validation-guidelines-guide-0028/document.html.
38. EU Guidelines of 5 November 2013 on Good Distribution Practice of medicinal products for human use Text with EEA relevance, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=oj:JOC_2013_343_R_0001_01.
39. FDA Guidance for Industry: Process Validation – General Principles and Practices, January 2011, US Food and Drug Administration (FDA), www.fda.gov.
40. The Committee for Advanced Therapies (CAT) and the CAT Scientific Secretariat, “Challenges with Advanced Therapy Medicinal Products and How to Meet Them,” *Nature Reviews Drug Discovery*, Vol. 9, No. 3, March 2010, <https://doi.org/10.1038/nrd3052>.
41. ICH Harmonised Guideline: *Pharmaceutical Development – Q8(R2)*, Step 4, 1 August 2009, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.
42. ICH Harmonised Guideline: *Pharmaceutical Quality System – Q10*, Step 4, 4 June 2008, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.
43. Yu, L.X., “Pharmaceutical Quality by Design: Product and Process Development, Understanding, and Control,” *Pharmaceutical Research*, Vol. 25, pp. 781–791, January 2008, <https://doi.org/10.1007/s11095-007-9511-1>.
44. Pugh, Keith, “Prior Knowledge in Product Development/Design,” Regulators’ Perspective – Session 2: European Medicines Agency (EMA), 7 May 2018, www.ema.europa.eu/en/documents/presentation/presentation-regulators-perspective-session-2-keith-pugh_en.pdf.
45. FDA Guidance for Industry: Q8, Q9, and Q10 Questions and Answers(R4), November 2011, US Food and Drug Administration (FDA), www.fda.gov.
46. Millili, P.G., A. Chang, S. Finnie, and M. Amaya, “Post-Approval Change Management for Cell and Gene Therapy Products,” *Pharmaceutical Engineering*, Vol. 44, No. 3, May/June 2024, <https://ispe.org/pharmaceutical-engineering/may-june-2024/post-approval-change-management-cell-and-gene-therapy>.
47. 21 CFR Part 312 Subpart I – Expanded Access to Investigational Drugs for Treatment Use, Code of Federal Regulations, US Food and Drug Administration (FDA), www.ecfr.gov.
48. Challener, C.A., “Process Analytical Technologies for Manufacturing Cell and Gene Therapies,” *BioPharm International*, Vol. 34, No. 12, December 2021, www.biopharminternational.com/view/process-analytical-technologies-for-manufacturing-cell-and-gene-therapies
49. Szarzynski, A., O. Spadiut, M. Reisbeck, G. Jobst, R. L. Paterson, A. Kamenskaya, E. Gateau, H. P. Lesch, L. Henry, and B. Kozma, “CGT 4.0: A Distant Dream or Inevitable Future? Smart Process Automation is Critical to Make Efficient Scalability of CGT Manufacturing a Reality,” *Frontiers in Bioengineering Biotechnology*, Vol. 13, 19 March 2025, <https://doi.org/10.3389/fbioe.2025.1563878>.
50. Hewitt, M.M., Trainor, N., Ostrout, N., and E. Abraham, “Cell Therapy Manufacturing: Process Analytic Technologies Needed to Achieve Flexible, Feedback Driven Automation,” *Current Opinion in Biomedical Engineering*, Vol. 20, December 2021, <https://doi.org/10.1016/j.cobme.2021.100358>.

51. Cell and Gene Therapy Catapult, “Cell and Gene Therapy Catapult Brings Together Over 20 Organisations with the Aim to Accelerate Technology Development and Potentially Lower Cost in Cell and Gene Therapy Manufacturing,” 8 June 2021, <https://ct.catapult.org.uk/news/press-release-cell-and-gene-therapy-catapult-brings-together-over-20-organisations-with-the-aim-to-accelerate-technology-development-and-potentially-lower-cost-in-cell-and-gene-therapy-manufacturing>.
52. European Pharmacopoeia (Ph. Eur.), EDQM Council of Europe, www.edqm.eu/en/.
53. United States Pharmacopeial Convention (USP), www.usp.org.
54. Smith, M., and S. Kassim. “‘The Process Is the Product’ Is No Longer Enough. Now, Data Is the Deliverable,” *Bio-IT World*. 1 June 2023. www.bio-itworld.com/news/2023/06/01/the-process-is-the-product-is-no-longer-enough.-now-data-is-the-deliverable.
55. Dream, R., and B. Loxley. “Process Hold Times: What is it and Why is it Important?” PDA Letter. 28 August 2024. www.pda.org/pda-letter-portal/home/full-article/process-hold-times-what-is-it-and-why-is-it-important.
56. Rathore, A.S., and H.S. Winkle. “Quality by Design for Biopharmaceuticals,” *Nature Biotechnology*, Vol. 27, No. 1, January 2009, <https://doi.org/10.1038/nbt0109-26>.
57. ICH Harmonised Guideline: Specifications: *Test Procedures and Acceptance Criteria for Biotechnology/Biological Products – Q6B*, Step 4, 10 March 1999, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.
58. Wiwi, C., “Analytical Considerations for Cellular Therapy Manufacturing,” *Cell and Gene Therapy Insights*, Vol. 2, No. 6, December 2016, <https://doi.org/10.18609/cgti.2016.071>
59. Karanu, F., Ott, L., Webster, D.A., and L. Stehno-Bittel, “Improved Harmonization of Critical Characterization Assays Across Cell Therapies,” *Regenerative Medicine*, Vol. 15, No. 5, June 2020, <https://doi.org/10.2217/rme-2020-0003>
60. FDA Guidance for Industry: Potency Tests for Cellular and Gene Therapy Products, January 2011, US Food and Drug Administration (FDA), www.fda.gov.
61. Stroncek, D.F., Jin, P., Ren, J., Feng, J., Castiello, L., Civini, S., Wang, E., Marincola, F.M., and M. Sabatino, “Quality Assessment of Cellular Therapies: The Emerging Role of Molecular Assays,” *Korean Journal of Hematology*, Vol. 45, No. 1, 2010, <https://doi.org/10.5045/kjh.2010.45.1.14>.
62. Kazi, A., “Why Are Lentivirus Vector Safety Tests So Crucial?” Eureka Blog, Charles River Laboratories, 27 February 2023, www.criver.com/eureka/why-are-lentivirus-vector-safety-tests-so-crucial.
63. BioPhorum, “A Practical Proposal for Standardizing Traceability of Cell and Gene Therapies,” *Cell & Gene*, 21 July 2023, www.cellandgene.com/doc/a-practical-proposal-for-standardizing-traceability-of-cell-and-gene-therapies-0001.
64. FDA Draft Guidance for Industry: Safety Testing of Human Allogeneic Cells Expanded for Use in Cell-Based Medicinal Products, April 2024, US Food and Drug Administration (FDA), www.fda.gov.
65. Baraniak, P., “Getting Off to a Good Start: Why finding high-quality donor cells is so important in cell therapy development,” *The Medicine Maker*, 2 February 2024, <https://themedicinemaker.com/issues/2024/articles/feb/getting-off-to-a-good-start>.
66. King, D., Schwartz, C., Pincus, S., and N. Forsberg, “Viral Vector Characterization: A Look at Analytical Tools,” *Cell Culture Dish*, October 2018, <https://cellculturedish.com/viral-vector-characterization-analytical-tools>.

67. FDA Guidance for Industry: Testing of Retroviral Vector-Based Human Gene Therapy Products for Replication Competent Retrovirus During Product Manufacture and Patient Follow-up, January 2020, US Food and Drug Administration (FDA), www.fda.gov.
68. Charles River Laboratories, "Process and Product-Related Impurities," Accessed 26 August 2025, www.criver.com/products-services/biologics-testing-solutions/contamination-and-impurity-testing/process-and-product-related-impurities.
69. Blay, E., Hardyman, E., and W. Morovic, "PCR Based Analytics of Gene Therapies Using Adeno Associated Virus Vectors: Considerations for cGMP Method Development," *Molecular Therapy – Methods & Clinical Development*, Vol. 31, December 2013, <https://doi.org/10.1016/j.omtm.2023.101132>.
70. Kang, D.H., et al, "A Study on the Application of Recombinant Factor C (rFC) Assay Using Biopharmaceuticals," *Microorganisms*, Vol. 12, No. 3, March 2024, <https://doi.org/10.3390/microorganisms12030516>.
71. Alliance for Regenerative Medicine, "Addressing Potency-Assay Related Development Delays for Cell and Gene Therapies," March 2023, <https://alliancerm.org/potency-assay-report>.
72. Amaya, M., et al, "Potency Measurements for Cellular and Gene Therapy Products," *Pharmaceutical Engineering*, Vol. 44, No. 5, September/October 2024, <https://ispe.org/pharmaceutical-engineering/september-october-2024/potency-measurements-cellular-and-gene-therapy>.
73. Salmikangas, P., Carlsson, B., Klumb, C., Reimer, T., and S. Thurstup, "Potency Testing of Cell and Gene Therapy Products," *Frontiers in Medicine*, Vol. 10, May 2023, <https://doi.org/10.3389/fmed.2023.1190016>.
74. Charles River Microbial Solutions, "Rapid Microbial Methods for Short-Shelf-Life and ATMP Testing," 10 October 2023, www.rapidmicrobiology.com/news/rapid-microbial-methods-for-short-shelf-life-and-atmp-testing.
75. Alvergnas, E. and L. Jallow, "Using Rapid Microbiological Methods Flexibly to Ensure Safe Cell and Gene Therapeutics," *American Pharmaceutical Review*, Vol. 27, No. 3, April 2024, www.americanpharmaceuticalreview.com/Featured-Articles/612552-Using-Rapid-Microbiological-Methods-Flexibly-to-Ensure-Safe-Cell-and-Gene-Therapeutics.
76. FDA Guidance for Industry: Considerations for the Development of Chimeric Antigen Receptor (CAR) T Cell Products, January 2024, US Food and Drug Administration (FDA), www.fda.gov.
77. Pierce, L., Sumona, S., Chan, L. L.-Y., Lin, B., and J. Qui, "Outcomes from a Cell Viability Workshop: Fit-for-Purpose Considerations for Cell Viability Measurements for Cellular Therapeutic Products," *Cell & Gene Therapy Insights*, Vol. 7, No. 4, May 2021, <https://doi.org/10.18609/cgti.2021.076>.
78. Nieuwenhuizen, P., and C. Rogers, "Challenges of Analytical Method Validation for ATMPs," *American Pharmaceutical Review*, 1 October 2023, www.americanpharmaceuticalreview.com/Specialty/Chromatography/Featured-Articles/607964-Challenges-of-Analytical-Validation-for-ATMPs.
79. Capelli, C., et al, "Potency Assays and Biomarkers for Cell-Based Advanced Therapy Medicinal Products," *Frontiers in Immunology*, Vol. 14, June 2023, <https://doi.org/10.3389/fimmu.2023.1186224>.
80. Abdellatif, A., Jaoudeh, M.B., Zwiers, A., and G. Breda, "Advancing Potency Assay Development for Advanced Therapy Medicinal Products: A Comprehensive Approach and Regulatory Insights," *Human Gene Therapy*, Vol. 36, No. 4, April 2025, <https://doi.org/10.1089/hum.2024.249>.
81. FDA Guidance for Industry: Sterile Drug Products Produced by Aseptic Processing – Current Good Manufacturing Practice, September 2004, US Food and Drug Administration (FDA), www.fda.gov.

82. US Food and Drug Administration (FDA), "Transcript for the FDA CBER OTP Town Hall: CMC Readiness for Gene Therapy BLAs," 4 June 2024, www.fda.gov/news-events/otp-town-hall-cmc-readiness-gene-therapy-blas-06042024.
83. 21 CFR Part 11 – Electronic Records; Electronic Signatures, Code of Federal Regulations, US Food and Drug Administration (FDA), www.ecfr.gov.
84. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 11: Computerised Systems, Manufacture of Sterile Medicinal Products, January 2011, https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en.
85. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 15: Qualification and Validation, 30 March 2015, https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en.
86. *ISPE Guide: Advanced Therapy Medicinal Products – Recombinant AAV Comparability and Lifecycle Management*, International Society for Pharmaceutical Engineering (ISPE), First Edition, January 2024, www.ispe.org.
87. ASTM Standard F838-20, "Standard Test Method for Determining Bacterial Retention of Membrane Filters Utilized for Liquid Filtration," ASTM International, www.astm.org.
88. EMA Guideline on quality, non-clinical and clinical requirements for investigational advanced therapy medicinal products in clinical trials, 20 January 2025, European Medicines Agency (EMA), EMA/CAT/22473/2025, www.ema.europa.eu/en.
89. ICH Harmonised Guideline: *Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin – Q5A(R2)*, Step 4, 1 November 2023, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.
90. *ISPE Baseline® Pharmaceutical Engineering Guide, Volume 7 – Risk-Based Manufacture of Pharmaceutical Products*, International Society for Pharmaceutical Engineering (ISPE), Second Edition, July 2017, www.ispe.org.
91. 21 CFR Part 211 – Current Good Manufacturing Practice for Finished Pharmaceuticals, Code of Federal Regulations, US Food and Drug Administration (FDA), www.ecfr.gov.
92. US Food and Drug Administration (FDA), "Validation of Cleaning Processes (7/93) – Guide to Inspections Validation of Cleaning Processes," July 1993, www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/inspection-guides/validation-cleaning-processes-793.
93. EMA Guideline on setting health based exposure limits for use in risk identification in the manufacture of different medicinal products in shared facilities, 20 November 2014, European Medicines Agency (EMA), EMA/CHMP/CVMP/SWP/169430/2012, www.ema.europa.eu/en.
94. EMA Questions and answers on implementation of risk-based prevention of cross-contamination in production and 'Guideline on setting health-based exposure limits for use in risk identification in the manufacture of different medicinal products in shared facilities' (EMA/CHMP/CVMP/SWP/169430/2012), 19 April 2018, European Medicines Agency (EMA), EMA/CHMP/CVMP/SWP/246844/2018, www.ema.europa.eu/en.
95. ICH Harmonised Guideline: *Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients – Q7*, Step 4, 10 November 2000, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.

96. International Council for Commonality in Blood Banking Automation (ICCBBA), www.isbt128.org/about-iccbba.
97. ICH Harmonised Guideline: *Development and Manufacture of Drug Substances (Chemical Entities and Biotechnological/Biological Entities)* – Q11, Step 4, 1 May 2012, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.
98. EU Directive 2001/83/EC of the European Parliament and of the Council of 6 November 2001 on the Community code relating to medicinal products for human use, <https://data.europa.eu/eli/dir/2001/83/oj>.
99. Milne, J., et al, "Regulatory Landscape for Raw Materials: CMC Considerations," *Pharmaceutical Engineering*, Vol. 42, No. 5, September/October 2022, <https://ispe.org/pharmaceutical-engineering/september-october-2022/regulatory-landscape-raw-materials-cmc>.
100. ASTM Standard D4169-22, "Standard Practice for Performance Testing of Shipping Containers and Systems," ASTM International, www.astm.org.
101. ISBT 128 Implementation Guide ICCBBA IG-050, "Using the ISBT 128 Chain of Identity (Col) Identifier," International Council for Commonality in Blood Banking Automation (ICCBBA), www.isbt128.org.
102. ISBT 128 Standard ICCBBA ST-028, "Chain of Identity (Col) Identifier," International Council for Commonality in Blood Banking Automation (ICCBBA), www.isbt128.org.
103. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 2: Manufacture of Biological Active Substances and Medicinal Products for Human Use, June 2018, https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en.

12 Appendix 2 – Glossary

12.1 Acronyms and Abbreviations

AAV	Adeno-Associated Virus
AI	Artificial Intelligence
APS	Aseptic Process Simulations
ATMP	Advanced Therapy Medicinal Product
ATP	Adenosine Triphosphate
AUC	Aggregate Ultracentrifugation
BLA	Biologics License Application
BSC	Biosafety Cabinet
BSE	Bovine Spongiform Encephalopathy
C&Q	Commissioning and Qualification
CAL	Cells at the Limit
CAR	Chimeric Antigen Receptor
CBER	Center for Biologics Evaluation and Research (FDA)
CBMP	Cell-Based Medicinal Product
CCS	Contamination Control Strategy
CFR	Code of Federal Regulations (US)
CMA	Critical Material Attribute
CMC	Chemistry, Manufacturing, and Control
CoA	Certificate of Analysis
COC	Chain of Custody
COI	Chain of Identity
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
CTA	Clinical Trial Application
ddPCR	Droplet Digital PCR
DLS	Dynamic Light Scattering
DMSO	Dimethyl Sulfoxide
DNA	Deoxyribonucleic Acid
DOE	Design of Experiments
DP	Drug Product
DQ	Design Qualification
DS	Drug Substance

E&L	Extractables and Leachables
EC	European Commission
EDQM	European Directorate for the Quality of Medicines & HealthCare
ELISA	Enzyme-Linked Immunosorbent Assay
EM	Environmental Monitoring
EMA	European Medicines Agency
EPC	End-of-Production Cell
FAT	Factory Acceptance Testing
FDA	Food and Drug Administration (US)
FFF-MALS	Field-Flow Fractionation–Multi-Angle Light Scattering
GEP	Good Engineering Practice
GMP	Good Manufacturing Practice
GOI	Gene of Interest
HBEL	Health-Based Exposure Limits
HCP	Host Cell Protein
HIV	Human Immunodeficiency Virus
HVAC	Heating, Ventilation, and Air Conditioning
ICCBBA	International Council for Commonality in Blood Banking Automation
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
IMP	Investigational Medicinal Product
IND	Investigational New Drug
IPC	In-Process Control
IQ	Installation Qualification
ISO	International Organization for Standardization
IV	Intravenous
LAL	Limulus Amebocyte Lysate
LVV	Lentiviral Vector
MAA	Marketing Authorization Application
MAH	Marketing Authorization Holder
MAP/EAP	Managed Access Program/Expanded Access Program
MCB	Master Cell Bank
ML	Machine Learning
MS	Mass Spectrometry
NGS	Next-Generation Sequencing
NIR	Near-Infrared Spectroscopy
OOS	Out-of-Specification

OQ	Operational Qualification
OTP	Office of Therapeutic Products (FDA)
OTS	Off-The-Shelf
PAT	Process Analytical Technology
PCR	Polymerase Chain Reaction
PDS	Product Design Specification
PIC/S	Pharmaceutical Inspection Co-operation Scheme
PPQ	Process Performance Qualification
PQ	Performance Qualification
PRIME	PRiority Medicines (EU)
PUPSIT	Pre-Use Post-Sterilization Integrity Testing
PV	Process Validation
Q&A	Questions and Answers
QbD	Quality by Design
QC	Quality Control
qPCR	Quantitative PCR
QRM	Quality Risk Management
QTPP	Quality Target Product Profile
R&D	Research and Development
rAAV	Recombinant Adeno-Associated Virus
RCL	Replication Competent Lentivirus
rFC	Recombinant Factor C
RMAT	Regenerative Medicine Advanced Therapy (FDA)
RMRA	Raw Material Risk Assessment
RVV	Retroviral Vector
SARS	Severe Acute Respiratory Syndrome
SAT	Site Acceptance Testing
SDS-PAGE	Sodium Dodecyl Sulfate Polyacrylamide Gel Electrophoresis
SEC-HPLC	Size Exclusion Chromatography–High-Performance Liquid Chromatography
SEC-MALS	Size Exclusion Chromatography with Multi-Angle Light Scattering
sIND	Single-Subject Investigational New Drug
SOP	Standard Operating Procedure
STR	Short Tandem Repeat
TEM	Transmission Electron Microscopy
TGA	Therapeutic Goods Administration (Australia)
TPP	Target Product Profile

TSB	Tryptic Soy Broth
TSE	Transmissible Spongiform Encephalopathy
URS	User Requirement Specification
USP	United States Pharmacopeia
VHP	Vaporized Hydrogen Peroxide
WBC	White Blood Count
WCB	Working Cell Bank
WFI	Water for Injection
WHO	World Health Organization



3001 N. Rocky Point Dr. E., Suite 200-242, Tampa, Florida 33607 USA
Tel: +1-813-960-2105, Fax: +1-813-264-2816

www.ISPE.org